Performance Before Outcomes

When the Human Brain Grew, Why It Stalled, and Why Primary Devotion Outranks Secondary Mating Criteria

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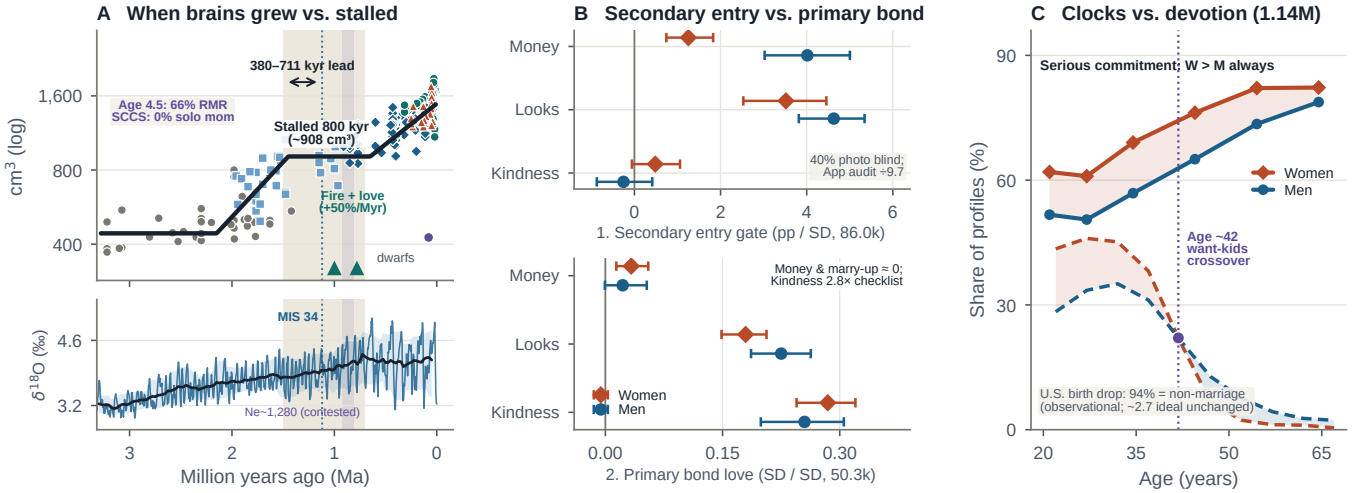


Fig. 1. **When the human brain grew versus stalled, and why secondary entry filters reverse into primary pair-bond devotion. (A)** Across 233 dated hominin crania [1–3] synchronized with the 57-core LR04 δ^{18}O stack [4], endocranial volume rose to $\sim$908 cm^3 by 1.50 Ma, stalled for 800 kyr (380–711 kyr before late-plateau crises), and surged $+50\%$/Myr once 1.0–0.78 Ma cooking [5, 6] and biparental/alloparental care (66.3% age-4.5 brain RMR [7]; 0.0% solo-mother care across 136 societies) lifted the juvenile caloric ceiling. **(B)** Across 175 countries [8], 43 longitudinal couples studies [9], and Stanford HCMST [10], *secondary criteria* (money, photo rank, $6'0''$ cutoffs, checklist tags) govern stranger entry while apps compress witness audits 9.7×, whereas *primary criteria*—kindness ($+0.29$ SD, 2.8× checklist fit, 5.1–8.9× homophily) and lived commitment (44.2% vs. 7.5% trait R^2)—govern union love and survival. **(C)** Across 1,137,309 profiles, female serious commitment stays above men's at every age even as childless "Want children" crosses near age 41.8, while 94% of the U.S. birth decline sits in non-marriage [11].

***Abstract*—Human partner and ally choice must evaluate *performance before outcomes*—committing years before juvenile survival or lifelong fidelity can be observed. Across twenty-three propositions and fifteen datasets spanning 233 hominin crania, cross-cultural ethnography, laboratory encounters, longitudinal couples, multimodal facial attribution, and 7,996,528 platform records, we resolve two linked gaps: when the human brain grew versus stalled, and why *primary* pair-bond devotion outranks *secondary* mating criteria. First, synchronizing fossil crania with the 57-core LR04 δ^{18}O stack shows brain volume climbed to ~ 908 cm^3 by 1.50 Ma, stalled for 800 kyr (380–711 kyr before 1.12 Ma cooling, the contested 930–813 ka bottleneck, or 789 ka impact ejecta), and surged $+50\%$/Myr across Africa and Eurasia only after 1.0–0.78 Ma cooking and biparental/alloparental provisioning overcame the age-4.5 brain peak (66.3% resting energy; 0.0% solo-mother care across 136 societies). Second, single-context stranger markets select on *secondary criteria* with high instant legibility—portrait rank (39.9% consensus gap beyond HoloFace-L1), $6'0''$ cutoffs (null fertility $r = +0.006$), income brackets, hobby tags ($r = +0.014$), and demographic sorting—while misreading opposite-sex priorities and compressing pre-romance witness audits 9.7× (tripling breakup when cohabitation rushes ≤ 3 months). Inside unions across 175 countries and 43 longitudinal studies, secondary criteria collapse near zero while *primary criteria*—kindness ($+0.29$ SD, beating checklist fit 2.8× and politics/class/degree homophily 5.1–8.9×), live conversational play, nine-domain mind breadth (4.9× validity), mutual reverence, and lifelong hearth devotion—govern love, marital stability, and the 94% non-marriage share of U.S. fertility decline.**

***Index Terms*—Human brain evolution, paleoclimate, primary and secondary mating criteria, pair-bond devotion, general intelligence, facial attribution, stature, digital platforms, fertility, AI evaluation.**

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I. Introduction

HOMININ endocranial volume roughly tripled over the past three million years, rising from Australopithecine means near 450 cm^3 to late Pleistocene means near 1,350–1,500 cm^3 (Fig. 1A) [1–3]. Across $\sim 10^5$ generations, that net trajectory requires a long-run directional response of only $\sim 10^{-4}$ phenotypic standard deviations (SD) per generation—two to three orders of magnitude smaller than selection differentials routinely measured in living populations [12]. Yet brain tissue is metabolically expensive, consuming roughly a fifth of resting adult energy and peaking at 66.3% of resting metabolism in early childhood [7, 13]. Meanwhile, human children require nearly two decades of biparental and alloparental provisioning [14, 15], and coalitionary cooperation unfolds across years of reciprocal exchange [16, 17]. Prospective partners and band members must therefore commit *before* completed family size, crisis provisioning, or lifelong fidelity can be observed—selecting on *performance before outcomes*.

This structural timing problem links two major unresolved gaps across evolutionary anthropology and modern partner choice:

1) **When the human brain grew, when it stalled, and what released it (Section II):** Why did endocranial volume expand between 2.1 and 1.5 Ma, stall near $\sim$908 cm^3 for 800 kyr (1.50–0.70 Ma) hundreds of millennia *before* Mid-Pleistocene climate crises or proposed bottlenecks, remain small in insular or calorie-constrained *Homo* lineages, and then surge $+50\%$/Myr simultaneously across Africa and Eurasia after 0.7 Ma?
2) **Secondary entry filters versus primary pair-bond criteria (Sections III–V):** Why do single-context stranger markets select heavily on *secondary mating criteria*—attributes with high zero-acquaintance legibility (ρ_X) such as static portrait rank, $6'0''$ stature cutoffs, income brackets, hobby tags, and demographic sorting labels that contribute near zero to union satisfaction or completed fertility ($\alpha_X \approx 0$)—whereas *primary mating criteria* ($\rho_{K,0} \ll \rho_X$: everyday kindness, conversational play, general intelligence g, mutual reverence, emotional fidelity, and unstaged hearth devotion) require multi-context witness breadth yet govern pair-bond love, survival, and juvenile encephalization (α_K,$\alpha_D \gg |\alpha_X|$)?

Appendix A formalizes this architecture in twenty-three propositions across three stages: (i) *what observation reveals* across domains and contexts (Propositions 1–7, 20), proving that K-domain residential breadth multiplies general-capacity aim K-fold, taxes single-domain gaming as $1/K$, and breaks the single-context validity ceiling $R^2_{\infty,1} < 1$; (ii) *what sustains delivery, love, and marital survival* (Propositions 15–17, 21, 23), proving that staged tests ($d \to 1$) erase delivery validity $R_p \to 0$, stranger legibility inverts secondary entry weights against primary pair-bond welfare, and compressed pre-cohabitation witness audits inflate lock-in failure and asymmetric marital exit; and (iii) *how juvenile metabolism and pairing gates allocate evolution* (Propositions 9–14, 18–19, 22), proving that cooking efficiency and pair-bond devotion break the solitary-mother childhood brain ceiling supermodularly while modern reproduction is gated by union entry rather than marital birth ideals.

Rather than estimating unobservable Pleistocene selection gradients, we test the observable comparative statics of these equations across fifteen open and archival datasets spanning deep time, cross-cultural ethnography and song corpora, laboratory encounters, longitudinal and national couples cohorts, deep-learning facial attribution, and 7,996,528 digital platform records (including 484,387 multi-domain `OkCupid` essays and 1,052,790 Plenty of Fish self-descriptions). Table I summarizes the eight macro selection key performance indicators (KPIs) and scope boundaries.

II. When the Human Brain Grew, Why It Stalled for 800 kyr, and What Released It

A. Four Regimes of Growth and Stasis—and the Onset–Crisis Timing Constraint

Across the three open hominin cranial compilations—233 specimens over 3.19 Myr in Gingerich [3], 166 Pleistocene *Homo* crania in Will et al. [2], and 94 specimens in Du et al. [1]—a constant log-linear expansion is decisively rejected (ΔAICc $= 43.2$; Fig. 2A). Instead, hominin brain evolution alternated sharply between stasis ceilings and rapid releases across four regimes: (i) an Australopithecine ceiling near 443 cm^3 prior to $\sim$2.1 Ma (Phase I, when brains did not grow despite bipedality); (ii) an early-*Homo*/*H. erectus* expansion doubling volume to $\sim$ 908 cm^3 between 2.1 and 1.45 Ma (Phase II); (iii) an $\sim$800 kyr Middle/Early Pleistocene stasis plateau between $\sim$ 1.50 and 0.70 Ma (Phase III, within-window slope $+0.02 \pm 0.11$ ln /Myr, indistinguishable from zero; within-clade Monte Carlo rate dropping -58% from 0.70 to 0.30 ln /Myr [1]); and (iv) a renewed $+50\%$/Myr multi-continental surge after $\sim$0.65 Ma (Phase IV), even as calorie- or island-constrained *H. naledi* (465–560 cm^3) and *H. floresiensis* (426 cm^3) retained small brains into the Late Pleistocene.

Synchronizing this cranial record with the 57-core `LR04` benthic foraminiferal δ^{18}O stack [4] establishes a decisive chronological constraint on catastrophe-based theories of why the brain stalled (Fig. 2A,B; Appendix Table B2). Phase III stasis began near 1.50 Ma (fitted 1.45 Ma), under mild 41 kyr obliquity cycles (`LR04` P90–P10 swing 0.80‰). Every major environmental catastrophe invoked in recent debates occurred hundreds of millennia *after* stasis was already established: $+250$ kyr for Mid-Pleistocene Transition onset (1.25 Ma), $+380$ kyr for MIS 34 European cooling (1.12 Ma), $+570$ kyr for the contested 930–813 ka coalescent bottleneck, and $+711$ kyr for the 789 ka Australasian impact. Under Proposition 10 and Corollary 6, early stasis onset without catastrophe reflects energetic and juvenile co-limitation: once raw-foraging gut allocation and *H. erectus* body size hit pre-cooking caloric limits ($\kappa_B \bar{B} \approx a\beta_{\text{eff}}$) [13, 27, 28], brain volume stalled near 908 cm^3 despite ongoing social selection.

B. Separating Late-Plateau Crises from the Cooking, Childhood, and Hearth Release

Evaluating overlapping 1.25–0.70 Ma global events by epistemic tier separates processes that constrained late-plateau habitats or triggered the 0.7 Ma release from contested or unlinked hypotheses (Fig. 2B; Table B2):

- **Measured paleoclimatic cooling and regional depopulation:** Across the Mid-Pleistocene Transition (1.25–0.70 Ma) [29], glacial periodicity shifted from 41 kyr to high-amplitude 100 kyr cycles (`LR04` P90–P10 glacial–interglacial swings rising 51% from 0.80‰ to 1.21‰), accompanied by atmospheric CO_2 drawdown [30, 31] and a severe 950–860 ka (MIS 24–22) ocean-circulation crisis [32]. Earlier, at MIS 34 ($\sim$1.12 Ma, `LR04` peak 4.50‰), climate-envelope modeling and Iberian marine pollen cores indicate extreme cooling that plausibly drove regional European hominin depopulation [18] prior to Po Valley/southern European recolonization near 0.9 Ma [33]. Crucially, this depopulation was *regional* to western Eurasia rather than a global extinction, while climate-driven habitat fragmentation later accelerated allopatric diversification [34, 35].
- **Contested coalescent bottleneck (930–813 ka):** Using `FitCoal` on modern site-frequency spectra, Hu et al. [19] and Zhou et al. [36] inferred a prolonged ancestral con-

TABLE I
EXECUTIVE SCORECARD: EIGHT MACRO SELECTION KPIS ACROSS BRAIN EVOLUTION, SECONDARY VS. PRIMARY MATING CRITERIA, AND MODERN PLATFORMS.

Macro Selection KPI	Core Selection Question	Empirical Scale	Headline Quantitative Contrast	Boundary Check & Formal Link
1. When Brains Grew vs. Stalled	When did hominin brains expand, why did they stall for 800 kyr (1.50–0.70 Ma), and what released the surge?	233 crania + `LR04` + 136 SCCS societies	Grew to $\sim 908\,\mathrm{cm}^3$ by 1.5 Ma, stalled **800 kyr** (**380–711 kyr** before crises), then surged $+50\%/\mathrm{Myr}$ post-cooking; age-4.5 brain uses **66.3% RMR**	SCCS solo-mother care is **0/136 (0.0%)**; cooking w/o pair-bond love stalls at $\sim 1{,}055\,\mathrm{cm}^3$ vs. 1,323 (Props. 10, 22)
2. Multi-Modal Legibility Wall	Why do zero-context strangers over-weight secondary cues while missing primary mind and devotion?	20,000 `HoloFace-L1` portraits + 8,378 dates + 484k essays + 118 songs	**39.9%** photo rank-variance gap; 4-min stranger explains **0.36%** of partner SAT vs. **4.9**× across 9 essay domains	Blind raters ($N = 29{,}357$) decode lullabies ($d = +1.84$) **5.0**× over love songs ($+0.37$; Props. 3, 20)
3. Mind-Reading Error & Play	How badly do men and women misread opposite-sex priorities, and what creates live romantic spark?	8,128 ratings + 7,116 hobby dyads + 690 matches	Both sexes overestimate opposite-sex looks weight ($+8.6$ **W** / $+7.0$ **M pts**) & underestimate intelligence (-6.9 / -4.6); men call **4.5**× more	17-hobby overlap is **null** ($r = +0.014$) vs. live play $r = +0.400$ (**27.6**×); **57.6%** unique SRM spark (Prop. 21(ii))
4. Voice vs. Secondary Staging	What do men and women seek in their own words vs. how secondary platform filters trigger strategic staging?	1,052,790 POV + 484k `OkCupid` essays + 65.9k effort logs	Women write family (**122**× $6'0''$); "message me if" asks humor **12.4**× tall & **22.7**× wealth; courtship effort drops -14%/-20% in marriage	Straight men surge $+17.1\%$ at $6'0''$ (gay men -1.3%) & show income **1.70**× more (Props. 16, 23(i))
5. Audit Collapse & Rush Hazard	What happens when digital apps compress pre-romantic witness audits and couples rush into lock-in?	3,510 Stanford `HCMST` adults (2,578 cohabited)	Apps compress pre-romance acquaintance **9.7**× (0.27 vs. 2.58 yr); rushing to cohabit in $\leq$ 3**mo triples breakup** (9.3% vs. 3.0%)	Non-marital breakups are near-symmetric, whereas **marital divorces flip to 64.3% female initiation** (85.7% in husbands' reports; Prop. 23(ii–iii))
6. Primary Love vs. Homophily	Do secondary sorting tags (politics, class, degrees, money) or primary kindness and reverence sustain love?	117,293 adults (47,932 couples) + 43 longit. studies	Primary kindness ($+0.29$ **SD**) beats politics/class/degree homophily (**5.1–8.9**×), checklist fit (**2.8**×), money, & marry-up (≈ 0)	Settling on kindness penalizes love **1.8**× steeper than surplus adds; 43 longit. studies: devotion **44.2%** vs. traits 7.5% R^2 (Prop. 21)
7. Clocks, Verbs & Second Acts	How do male and female desires, hearth verbs, and remarriage choices diverge or converge from 18 to 79?	1.14M POV + 1.05M essays + 85.0k jealousy + 160 SCCS	Childless "Want kids" flips at **Age** ~ 42, casual SOI widens ($1.46 \rightarrow 1.77\times$ M>W), yet hearth cooking verbs rise 4.2–4.9× & commitment stays W>M	40–49 divorced W cut marriage intent -24% (keeping **80.8%** serious bond), while widowed M exceed W (Prop. 23(iv))
8. The 94% Modern Pairing Gate	Did modern births fall because adults stopped wanting families or because stranger markets stall union entry?	34,092 U.S. GSS adults (incl. `WORDSUM`) + 232 meta-effects	**94%** of U.S. birth drop sits in non-marriage; male pairing is positive on income ($+0.046$), schooling ($+0.015$), & `WORDSUM` ($+0.011$)	Within-marriage births invert all four (-0.034 to -0.081); height fertility null ($r = +0.006$; Props. 14, 19)

Every cell is generated programmatically from SHA-256-verified outputs (`scripts/run.py -study performance-before-outcomes`). Observational and contested tiers are labeled explicitly; pre-committed falsifiers F1–F15 appear in Appendix Table D3.

traction to coalescent effective size $N_e \approx 1{,}280$, proposing links to chromosome 2 fusion and Phase IV speciation. Two qualifications are essential. First, $N_e \approx 1{,}280$ is a harmonic-mean coalescent parameter shaped by population structure and selective sweeps, not a literal headcount of living individuals (Proposition 18). Second, independent statistical reanalyses found that alternative demographic histories without a severe bottleneck fit modern SFS and linkage patterns as well or better, attributing the `FitCoal` signal to numerical smoothing artifacts or weakly structured African stem populations [20, 37–40], consistent with continuous 930–813 ka Acheulean occupation across Africa, the Levant, and Spain [41, 42]. We therefore mark the severe bottleneck as a *contested genomic hypothesis*.

- **Impacts, super-eruptions, and magnetic reversal without hominin die-off:** Five dated geophysical events overlap Phase III—the $\sim$ 1.07 Ma Bosumtwi impact (10.5 km crater, Ghana) [43], $\sim$ 815 ka Pantasma impact (14 km, Nicaragua) [44], $\sim$800 ka Oldest Toba Tuff super-eruption [45], $\sim$789 ka Australasian tektite impact scattered across 10–30% of Earth's surface (linked to a buried Bolaven Plateau source in Laos [21, 46, 47], though crater identification remains debated [48]), and the $\sim$773 ka Matuyama–Brunhes geomagnetic reversal [49]. None establishes a global hominin die-off; notably, 789 ka Bose Basin biface assemblages in southern China and Vietnam show behavioral continuity directly across the tektite horizon [50].
- **Campfire energetic release, childhood pair-bond ceiling, and hearth song legibility:** Controlled fire at Wonderwerk Cave ($\sim$1.0 Ma) [5] and structured fish cooking at Gesher Benot Ya'aqov ($\sim$ 780 ka) [6, 51] align directly with the $\sim$ 0.65 Ma termination of Phase III. Crucially, the sharpest metabolic bottleneck sits in childhood rather than adulthood: PET/MRI glucose data show that the human brain peaks at 66.3% of resting metabolic rate (43.2% of daily energy expenditure) at age 4.5—3.2× the adult share (20.5%) [7], imposing the $\sim 650\,\mathrm{cm}^3$ primate "Gray Ceiling" under solitary maternal foraging [22, 23] (Fig. 2C). Across 136 coded Standard Cross-Cultural Sample societies [52], mothers provide exclusive early-childhood care in 0.0% of societies (0/136; non-mothers provide majority care in 73.5%), paternal closeness rises from 30.5% in infancy to 54.7% in early childhood (85.3% at least occasional), and community food sharing operates in 92.1%. Under Proposition 22 (`E22`, verified across 20,000 simulated agents within $\leq 0.75\,\mathrm{cm}^3$), Lande equilibria step from $650\,\mathrm{cm}^3$ (solitary ape mother) to $912\,\mathrm{cm}^3$ (raw alloparental plateau), stall at $1{,}055\,\mathrm{cm}^3$ under cooking *without* durable pair-bond commitment, and reach $1{,}323\,\mathrm{cm}^3$ only when cooking combines supermodularly with pair-bond devotion ($+268\,\mathrm{cm}^3$ love gain). Evening hearths simultaneously extended multi-domain social auditing ($K \uparrow$, $(1-d)n \uparrow$): across 118 field recordings from 86 societies rated blind by 29,357 naive listeners in the Harvard `Natural History`

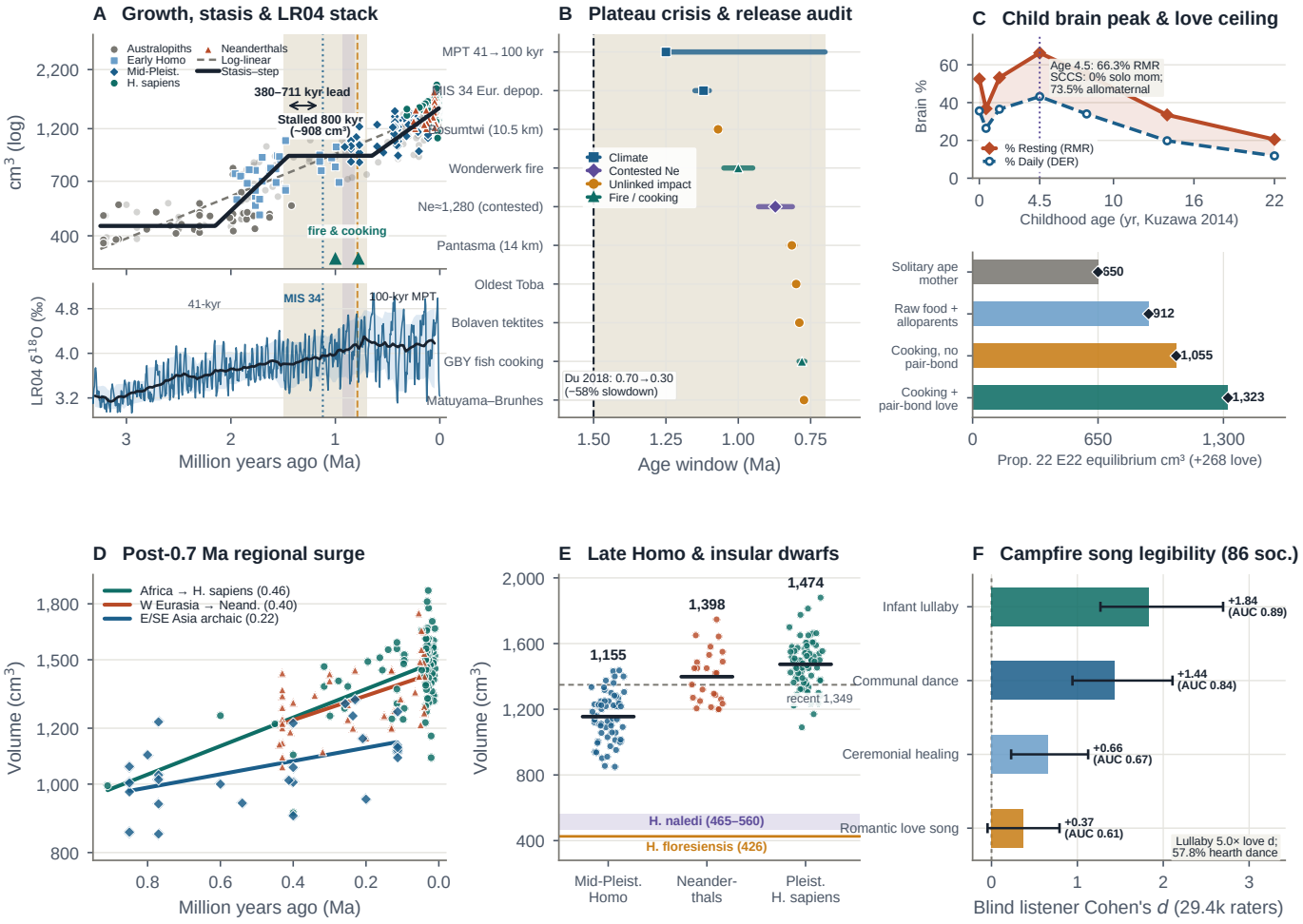


Fig. 2. **When the hominin brain grew, why it stalled for** 800 kyr**, and how cooking, childhood pair-bond care, and hearth song legibility released it. (A)** Endocranial volume across 233 crania [1–3] (ΔAICc = 0 stasis–step vs. 43.2 log-linear) synchronized with the 57-core LR04 δ^{18}O stack [4]: 1.50–0.70 Ma stasis ($\sim$908 cm^3) preceded 1.12 Ma cooling [18], the contested 930–813 ka bottleneck [19, 20], and 789 ka tektites [21] by 380–711 kyr before 1.0–0.78 Ma fire and cooking [5, 6]. **(B)** Epistemic audit of ten 1.25–0.70 Ma events and within-clade slowdown (0.70 $\rightarrow$ 0.30 ln /Myr, -58% [1]). **(C)** Age-4.5 brain metabolic peak (66.3% RMR, 3.2$\times$ adult [7]), 0.0% solo-mother care across 136 SCCS societies [22, 23], and Proposition 22 Lande equilibria (650 $\rightarrow$ 912 $\rightarrow$ 1,055 $\rightarrow$ 1,323 cm^3; +268 cm^3 pair-bond gain). **(D, E)** Post-0.7 Ma $+50\%$/Myr multi-regional surge [2] alongside small-brained *H. naledi* [24] and *H. floresiensis* [25]. **(F)** Harvard Natural History of Song (118 recordings, 86 societies, 29,357 blind listeners [26]): lullabies ($d = +1.84$) and communal dance ($d = +1.44$) are 5.0$\times$ and 3.9$\times$ more stranger-legible than love songs ($d = +0.37$).

of Song corpus [26] (4,709 ethnographic performances), zero-context strangers decode infant lullabies ($d = +1.84$, AUROC 0.89) and communal motor dance ($d = +1.44$, AUROC 0.84; 57.8% of coded hearth performances) 5.0$\times$ and 3.9$\times$ more strongly than romantic love songs ($d = +0.37$, AUROC 0.61; Fig. 2F)—confirming that alloparental soothing and coalitionary synchrony are species-legible public broadcasts whereas romantic devotion requires lived relational proof. Immediately post-0.7 Ma, African *Homo* into *H. sapiens* (0.46 ln /Myr) [53], western Eurasian *Homo* into Neanderthals (0.40 ln /Myr), and East Asian archaic *Homo* (0.22 ln /Myr) [54] expanded in parallel at $+50\%$/Myr (Fig. 2D), whereas small-brained *H. naledi* (465–560 cm^3 at $\sim$285 ka) [24] and insular *H. floresiensis* (426 cm^3 at $\sim$80 ka) [25] persisted where local ecological budgets kept κ_B high (Fig. 2E).

III. Stranger Encounters: Mind-Reading Blindness, Multi-Context Juries, and Secondary Gaming

A. Stranger Legibility, Cross-Sex Mind-Reading Blindness, and Post-Match Initiation

Why do solitary first encounters overweight secondary surface cues while missing primary mind and devotion? Reanalyzing 8,378 four-minute Columbia speed-dating encounters among 551 participants [55] exposes a sharp legibility wedge alongside systematic cross-sex misperception (Fig. 3A,B; Fig. 9A). Before meeting, women allocate nearly equal stated weights across intelligence, sincerity, fun, ambition, and looks (17.9 of 100 points to looks; men allocate 27.0). Yet when asked to predict what the *opposite sex* seeks, both sexes severely overestimate opposite-sex preoccupation with secondary looks while underestimating primary mind and character: women overestimate men's looks weight by +8.6 points (95% CI $[+5.9, +11.1]$) while underestimating men's valuation of intelligence (-6.9 points, $[-7.9, -5.9]$) and sincerity (-5.1 points),

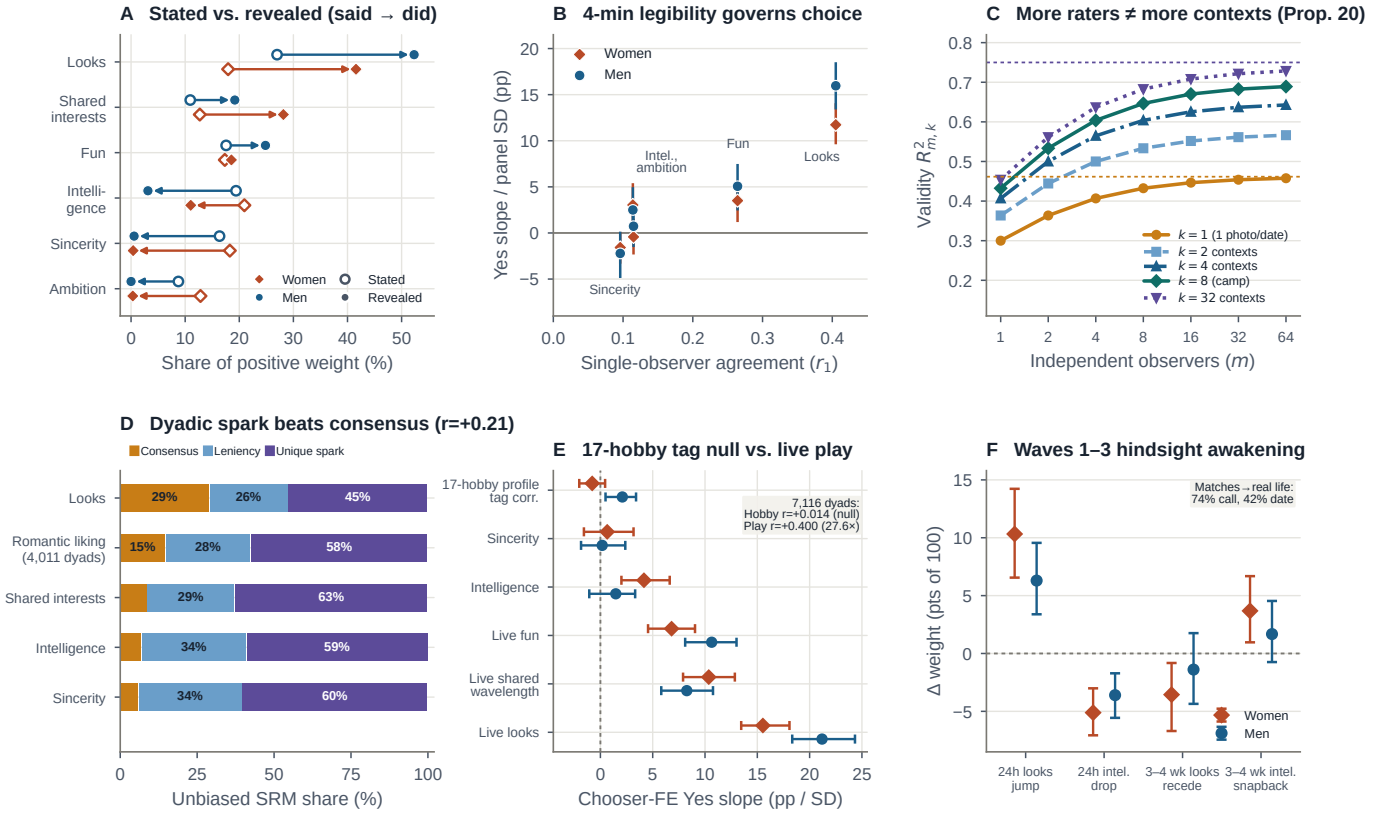


Fig. 3. **Four-minute stranger legibility, the two-way jury law, unique dyadic spark, the 17-hobby checklist null, and longitudinal hindsight. (A)** Stated pre-event weights (open) versus revealed fixed-effects choice weights (filled) across 8,378 speed dates of 551 participants [55]. **(B)** Single-observer agreement (r_1) governs revealed choice slope: looks ($r_1 = 0.40$) are immediately legible, whereas intelligence ($r_1 = 0.11$) and sincerity ($r_1 = 0.10$) are nearly opaque to one stranger. **(C)** Two-way validity $R^2_{m,k}$ across m observers and k contexts (Proposition 20): single-context crowds ($k = 1$) stall at $R^2_{\infty,1} = 0.46$, whereas a residential band ($m = 8$,$k = 8$) reaches $R^2_{8,8} = 0.65$. **(D)** Bipartite Social Relations Model ($n = 8{,}128$ ratings, 4,011 dyads [56]): partner consensus explains only 14.6% of liking (6.9% intelligence, 5.9% sincerity), whereas unique pair-specific spark accounts for 57.6% ($r = +0.21$ reciprocity). **(E)** Across 7,116 dyads, objective 17-hobby tag overlap (`int_corr`) is null for "Yes" ($r = +0.014$), whereas live conversational wavelength (`shar`) reaches $r = +0.400$ (27.6× stronger [57]). **(F)** Waves 1–3 hindsight [55, 58]: 24 h post-event, women's looks weight jumps +10.3 points before intelligence rebounds +3.7 points 3–4 weeks later.

and men mirror the error about women (+7.0 points on looks, −4.6 on intelligence, −3.1 on sincerity), reinforced by same-sex pluralistic ignorance (+9.1 points among women, +5.4 among men; Fig. 9A). In actual four-minute "Yes" decisions with chooser fixed effects, both sexes act heavily on physical attractiveness (+11.8 pp/ SD in women, +15.8 pp/ SD in men; bottom-to-top quintile 15.0%→57.5% and 24.4%→71.3%).

Inter-observer reliability explains why without invoking hypocrisy (Fig. 3B). Two strangers meeting a candidate for four minutes agree at $r_1 = 0.40$ on looks and $r_1 = 0.26$ on fun, but only $r_1 = 0.11$ on intelligence and $r_1 = 0.10$ on sincerity—requiring ~ 6 raters for looks versus ~ 25 for intelligence and > 60 for sincerity (Eq. (A31)). Proposition 20 further proves that pooling $m \to \infty$ raters within a *single* curated context ($k = 1$) never averages out person-by-context state σ^2_C, stalling at $R^2_{\infty,1} = 0.46$ whereas a residential camp ($m = 8$,$k = 8$) reaches $R^2_{8,8} = 0.65$ (Fig. 3C). Moreover, a finite-sample-unbiased bipartite Social Relations Model across $n = 8{,}128$ ratings and 4,011 dyads [56] proves that attraction is relational rather than scalar "trophy value" (Fig. 3D; Proposition 21(ii)): partner consensus explains only 14.6% of liking (28.9% looks, 8.6% shared interests, 6.9% intelligence, 5.9% sincerity), whereas chooser leniency explains 27.8% and unique pair-specific spark accounts for 57.6% ($r = +0.21$ mutual residual reciprocity), lifting women's combined live spark slope (+23.4 pp/ SD; feeling chosen back +9.5 pp W, +9.4 pp M) above physical attractiveness (+15.2 pp/ SD).

Speed dating likewise refutes profile-tag matching [57, 58] (Fig. 3E,F). Across 7,116 dyads, exact objective correlation across 17 pre-scored leisure hobbies (`int_corr`) has a null effect on saying "Yes" ($r = +0.014$, $[-0.019, +0.043]$; women's FE slope −0.8 pp/ SD), whereas live four-minute conversational wavelength (`shar`) correlates at $r = +0.400$ (27.6× stronger; +10.4 pp/ SD). Tracking Waves 1–3 reveals both an initiation bottleneck and a primary-trait rebound: across 690 mutual matches 3–4 weeks post-event, men place 4.5× as many follow-up calls per match as women (49.7% vs. 11.1%; 77.9% of matched women place zero outbound calls vs. 32.7% of men), and once real follow-up conversations occur (74.3% of matched participants call; 41.6% date), women's temporary 24-hour surge on looks (+10.3 points) reverses as intelligence rebounds +3.7 points ($[+0.8, +6.8]$) and looks recede −3.6 points.

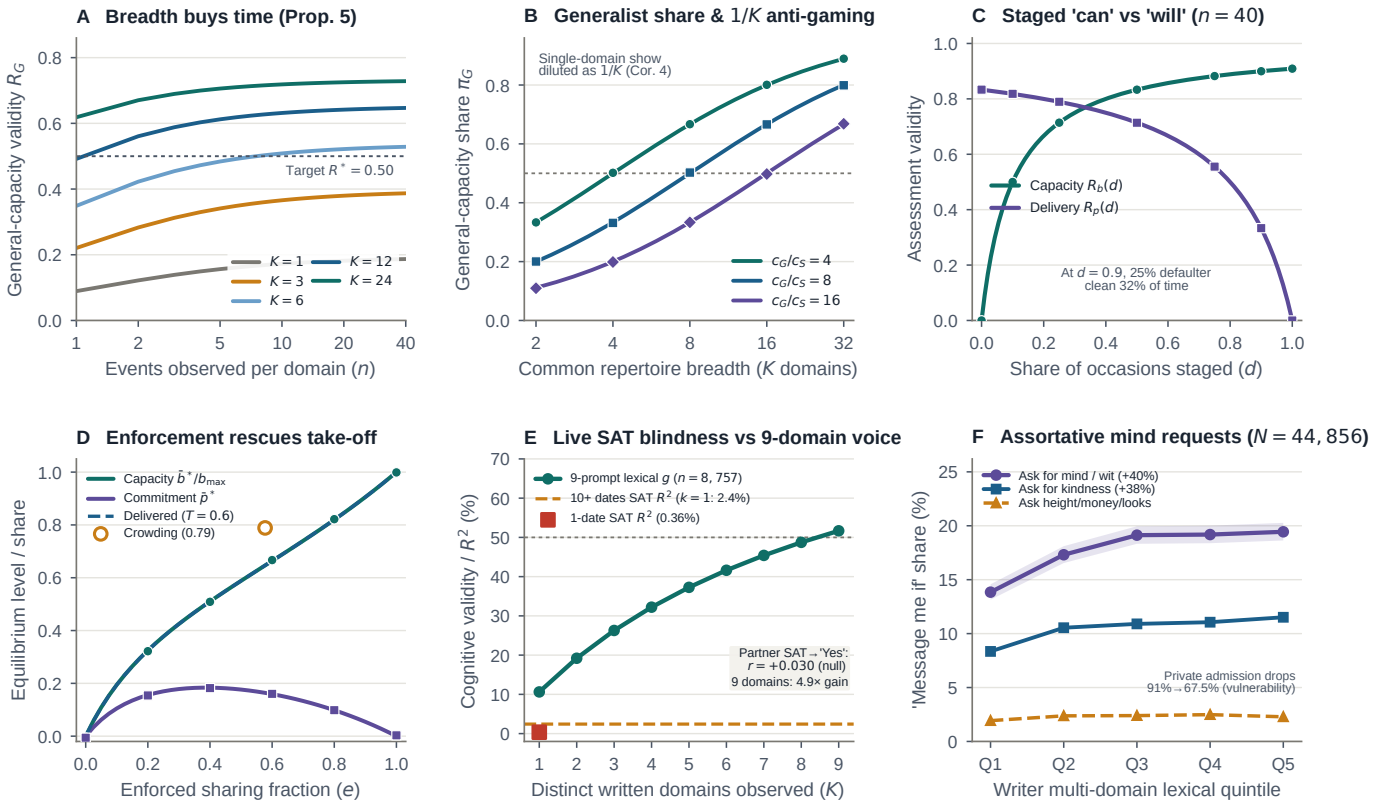


Fig. 4. **Generalist breadth, anti-gaming dilution, staged versus unstaged delivery (E15–E16), and nine-domain mind selection.** **(A)** Events per domain $n^*(K)$ to reach $R^* = 0.5$ across $K = 1 \ldots 24$ domains (Proposition 5): impossible for $K \leq 5$, 7.5 at $K = 6$, and ≈ 1 at $K = 12$. **(B)** Equilibrium general-capacity investment share $\pi_G = Kc_S/(Kc_S + c_G)$ (Proposition 6) and $1/K$ dilution of single-domain gaming (Corollary 4). **(C)** Opposite response of capacity validity $R_b(d)$ and delivery validity $R_p(d)$ as staged share d rises across $n = 40$ occasions (Proposition 16). **(D)** Equilibrium capacity and voluntary commitment across third-party enforced sharing $e \in [0,1]$ below ($T = 0.6$) versus above ($T = 1.5$) threshold (Proposition 17). **(E)** Across 2,989 speed dates (202 participants [55]), one stranger explains 0.36% of partner SAT ($r = +0.030$ for "Yes", 10+ raters stall at 2.42%), whereas observing 8,757 `OkCupid-JSE` writers across $K = 1 \rightarrow 9$ prompts [59] lifts lexical reliability 4.9× (0.106 → 0.517). **(F)** Across writer lexical quintiles ($N = 44,856$ [59]), explicit requests for curiosity, intelligence, and wit climb +40% (13.8%→19.4%) alongside kindness (+38%), while height/money/looks tags stay flat ($\sim$2.3%).

B. Generalist Breadth, Staged Capacity, Enforced Delivery, and Empirical Mind Selection

When a residential band observes every adult across K common domains—foraging, toolmaking, childcare, argument, navigation, and conflict mediation—four theoretical and empirical properties emerge (Fig. 4):

1) **Breadth aims selection at g and buys time:** Under Corollary 3, comparing candidates on K shared domains multiplies the selection differential on general capacity relative to any specialty by K (simulated ratios 1.00,4.01,15.97 at $K = 1,4,16$), whereas occupational division of labor where each person is judged in their best specialty weakens generalist aim 3.4-fold at $K = 10$ (Proposition 7). Breadth also makes pre-reproductive assessment feasible: reaching $R^* = 0.5$ is impossible for $K \leq 5$ domains, takes 7.5 events per domain at $K = 6$, and requires roughly one event per domain at $K = 12$ (Proposition 5; Fig. 4A). Evolutionarily, general-capacity investment dominates total expenditure ($\pi_G > \frac{1}{2}$) once repertoire breadth K exceeds the tissue-to-skill cost ratio c_G/c_S, while single-domain exaggeration is taxed as $1/K$ (Proposition 6, Corollary 4; Fig. 4B).
2) **Staged tests dissociate from unstaged delivery:** Raising the share of announced, staged assessments d from 0.1 to 0.9 across $n = 40$ occasions lifts capacity validity R_b from 0.50 to 0.90 while collapsing delivery validity R_p from 0.82 to 0.33, allowing a candidate who defaults on 25% of everyday obligations to maintain a clean staged record 32% of the time (Proposition 16; Fig. 4C).
3) **Third-party enforcement breaks the take-off trap:** Below the co-evolutionary threshold ($T = 0.6 < 1$), neither capacity nor commitment evolves under dyadic choice alone ($e = 0$). Enforcing camp-wide sharing of $e = 0.2$ or 0.5 of output [16, 60] immediately lifts equilibrium capacity to 0.32 and 0.59 of its maximum while peaking voluntary commitment at partial enforcement ($e^* = 0.39$), whereas full enforcement ($e \rightarrow 1$) crowds out selection on internal commitment above threshold ($T = 1.5$; Proposition 17; Fig. 4D).
4) **Four-minute SAT blindness versus nine-domain written portfolios:** Pairing speed-dating ratings with participants' objective college SAT scores (2,989 encounters across 202 participants [55]) and 484,387 multi-prompt `OkCupid-JSE` essays ($N = 59,946$ San Francisco profiles [59]) tests Propositions 3, 5, and 20 directly (Fig. 4E,F). One stranger in four minutes explains only



Fig. 5. **Male height percentile versus attraction, the 6′0″ platform cliff, the straight-versus-gay audience-staging natural experiment, and null completed fertility. (A)** Male height percentile versus self-rated attraction percentile across $N = 4{,}375$ Australian men [61]: attraction rises from the 38.4th percentile in the shortest bracket to a plateau near the 51.9th at the 72.2th height percentile before dipping to 47.6th at the 95.9th percentile ($\beta_2 = -1.02$, $[-1.64, -0.43]$). **(B)** Reported male stature across $N = 781{,}680$ POV profiles versus Gaussian expectation, exposing a $1.20\times$ heap at 6′0″ ($n = 99{,}330$) and a -39.1% cliff at 6′1″ ($n = 60{,}483$). **(C)** Audience-targeted staging on the identical San Francisco `OkCupid-JSE` interface [59]: straight men ($n = 31{,}073$) surge $+17.1\%$ from 5′11″ to 6′0″ before a -46.8% cliff at 6′1″, whereas gay men ($n = 3{,}985$) show no 6′0″ surge (-1.3%), and straight men disclose positive income at 24.4%—$1.70\times$ gay men (14.4%) and $2.03\times$ straight women (12.1%). **(D)** Prime-age POV outcomes (ages 30–49) across stature percentiles: $>$ 3-year union history rises across the lower-tail floor ($62.3\% \rightarrow 79.9\%$) and flattens above median height, while casual-intent share ticks upward in the tall tail ($15.3\% \rightarrow 21.2\%$). **(E)** Cross-population meta-analysis [62]: male height predicts short-term mating ($r = +0.050$, $k = 62$) but shows null association with completed fertility ($r = +0.006$, $[-0.052, 0.064]$, $k = 35$, $N = 22{,}984$), whereas functional bodily strength predicts both mating and reproduction ($r = +0.139$). **(F)** Across 47,293 `OkCupid-JSE` "message me if" essays [59], humor (17.0%), kindness (16.7%), and curiosity/mind (13.7%) outnumber 6′0″/tall (1.36%) by $12.4\times$ and wealth (0.75%) by $22.7\times$, while in 48,413 "first noticed" essays smile (27.2%) and eyes (20.5%) outrank height (10.2%).

0.36% of a partner's objective SAT ($r = +0.060$, 95% CI $[+0.022, +0.096]$), objective SAT has a null effect on saying "Yes" ($r = +0.030$, $[-0.006, +0.065]$; matched couples $r = -0.022$), and even pooling 10+ four-minute raters stalls at 2.42% ($r = +0.156$, $6.8\times$ single-rater R^2 yet trapped at $k = 1$). Conversely, across $K = 9$ topically distinct `OkCupid` prompts (within-person cross-prompt engagement $r = +0.467$; $n = 8{,}757$ complete 9-domain writers), single-prompt lexical-sophistication reliability ($r_1 = 0.106$) rises $4.9\times$ to $R_G(9) = 0.517$. Across writer lexical quintiles ($N = 44{,}856$), explicit partner requests for curiosity, intelligence, and witty conversation in `essay9` ("You should message me if…") climb $+40\%$ from 13.8% (Q_1) to 19.4% (Q_5; $1.40\times$) and kindness climbs $+38\%$ ($8.3\% \rightarrow 11.5\%$), whereas height/money/looks demands stay flat near 2.3% and prompt completion drops sharpest on `essay8` ("Most private thing I'm willing to admit," 90.7%→67.5%).

C. Male Height Percentiles: Threshold Floor, 6′0″ Staging, and Null Fertility

Male stature illustrates how a highly legible static cue behaves across perception, strategic online presentation, and completed reproduction (Fig. 5; Table II). In Australian microdata ($N = 4{,}375$ men aged 18–64 [61]), mapping ordinal stature into cumulative height percentiles reveals a concave floor-and-plateau rather than runaway linearity: mean attraction percentile climbs steeply out of the shortest decile (38.4th percentile) to 51.9th near the 72.2th height percentile, then flattens and dips slightly to 47.6th at the 95.9th percentile (standardized quadratic term $\beta_2 = -1.02$, 95% CI $[-1.64, -0.43]$; Fig. 5A) [63, 64]. Standardized multivariate regressions confirm that health (+12.6 attraction pct/SD) outweighs stature (+0.83), and income (+9.3 pp/ SD) outweighs stature (+0.91 pp/ SD) for current partnering.

On digital platforms ($N = 781{,}680$ men with explicit stature on Plenty of Fish), lexical filtering around round-number cutoffs induces Campbell–Goodhart gaming [65] (Fig. 5B): profiles reporting 6′0″ (183 cm, $n = 99{,}330$) reach $1.20\times$ Gaussian expectation before plunging -39.1% at 6′1″ ($n = 60{,}483$).

TABLE II
MALE STATURE PERCENTILES ACROSS ATTRACTION, 6′0″ PLATFORM STAGING, UNION HISTORY, AND META-ANALYTIC FERTILITY.

Source & male stature contrast	Sample (n)	Height pct.	Primary attraction or union metric	Secondary strategy or fertility metric
Panel A: Australian self-rated attraction & partnering by male height bracket (Whyte et al. 2021, ages 18–64)				
Ordinal band 2 (0.0–0.8th pct.)	33	0.4	Attraction 38.4th pct. $[28.0, 49.0]$	Partnered 48.5%; parent 36.4%
Ordinal band 3 (0.8–10.6th pct.)	430	5.7	Attraction 46.7th pct. $[44.1, 49.5]$	Partnered 49.3%; parent 51.6%
Ordinal band 4 (10.6–52.3th pct.)	1,825	31.5	Attraction 49.7th pct. $[48.4, 51.0]$	Partnered 51.1%; parent 55.5%
Ordinal band 5 (52.3–92.1th pct.)	1,738	72.2	Attraction 51.9th pct. $[50.5, 53.2]$	Partnered 54.0%; parent 52.1%
Ordinal band 6 (92.1–99.8th pct.)	337	95.9	Attraction 47.6th pct. $[44.7, 50.7]$	Partnered 51.3%; parent 44.8%
Panel B: POV dating platform male stature distribution, 6'0" rounding cliff & prime-age outcomes (ages 30–49)				
Reported 5'1" (155 cm; 1.43× Gauss.)	2,159	0.4	$>$ 3-yr union 62.3%; athletic 16.7%	Casual 15.3%; parent 49.6%
Reported 5'4" (163 cm; 0.78× Gauss.)	11,636	2.9	$>$ 3-yr union 72.0%; athletic 23.4%	Casual 16.4%; parent 52.3%
Reported 5'7" (170 cm; 1.11× Gauss.)	66,794	16.5	$>$ 3-yr union 76.3%; athletic 31.1%	Casual 17.3%; parent 52.4%
Reported 5'10" (178 cm; 0.96× Gauss.)	94,768	46.8	$>$ 3-yr union 78.7%; athletic 38.3%	Casual 19.5%; parent 49.8%
Reported 5'11" (180 cm; 1.05× Gauss.)	99,662	59.3	$>$ 3-yr union 79.0%; athletic 38.5%	Casual 19.8%; parent 49.7%
Reported 6'0" (183 cm; 1.20× Gauss.)	99,330	72.0	$>$ 3-yr union 79.4%; athletic 42.2%	Casual 20.0%; parent 50.2%
Reported 6'1" (185 cm; 0.92× Gauss.)	60,483	82.2	$>$ 3-yr union 79.9%; athletic 41.9%	Casual 20.0%; parent 49.8%
Reported 6'4" (193 cm; 0.95× Gauss.)	16,836	97.2	$>$ 3-yr union 79.3%; athletic 41.4%	Casual 21.2%; parent 49.7%
Reported 6'6" (198 cm; 0.73× Gauss.)	3,263	99.4	$>$ 3-yr union 77.0%; athletic 38.6%	Casual 19.8%; parent 49.4%
Panel C: Cross-population meta-analysis of male stature vs. functional bodily strength (Lidborg et al. 2022)				
Male height → mating outcomes ($k = 62$)	92,466	–	Random-effects $r = +0.050$ $[+0.027, +0.072]$	Median $r = +0.046$; N-weighted $r = +0.061$
Male height → reproduction / fertility ($k = 35$)	22,984	–	Random-effects $r = +0.006$ $[-0.052, +0.064]$	Median $r = +0.042$; N-weighted $r = -0.052$
Male strength/muscularity → mating ($k = 121$)	38,330	–	Random-effects $r = +0.122$ $[+0.100, +0.143]$	Median $r = +0.112$; N-weighted $r = +0.078$
Male strength/muscularity → reproduction ($k = 14$)	1,355	–	Random-effects $r = +0.139$ $[+0.077, +0.201]$	Median $r = +0.139$; N-weighted $r = +0.133$

Comparing straight ($n = 31{,}073$) and gay ($n = 3{,}985$) men on the identical San Francisco `OkCupid-JSE` interface [59] proves that this cliff reflects *audience-targeted staging* rather than generic measurement rounding (Fig. 5C): straight men surge $+17.1\%$ from $5'11''$ (3,893) to $6'0''$ (4,559) before a -46.8% cliff at $6'1''$ (2,427), whereas gay men exhibit *no* $5'11'' \to 6'0''$ surge ($471 \to 465$, -1.3%), and straight men simultaneously disclose positive income at 24.4%—1.70× gay men (14.4%) and 2.03× straight women (12.1%, $n = 20{,}533$). Across ages 30–49 on POV, prior $>$ 3-year union attainment exhibits a lower-tail floor (62.3% at $5'1''$ rising to 79.9% at $6'1''$ and 77.0% at $6'6''$), while casual-intent share ticks upward in the tall tail ($15.3\% \to 21.2\%$; Fig. 5D). Synthesizing $k = 35$ populations ($N = 22{,}984$) in Lidborg et al. [62] shows that while male height positively predicts mating access ($r = +0.050$, $[0.027, 0.072]$, $k = 62$), its net correlation with completed fertility is statistically indistinguishable from zero ($r = +0.006$, $[-0.052, 0.064]$), whereas functional muscular strength predicts both mating and reproduction ($r = +0.139$, $[0.077, 0.201]$; Fig. 5E; Proposition 11). Meanwhile, across 47,293 `OkCupid` "You should message me if…" essays [59], explicit invitations for humor (17.0%), kindness (16.7%), and curiosity/mind (13.7%) outnumber $6'0''$/height cutoffs (1.36%) by 10.0×–12.4× and wealth (0.75%) by 22.7× (Fig. 5F).

IV. PRIMARY VERSUS SECONDARY MATING CRITERIA ACROSS CULTURES AND LONG-TERM UNIONS

A. Kin Juries, Forager Waiting Windows, and Effective Ancestors

In small-scale societies, partner selection is rarely an unmediated dyadic transaction between strangers (Fig. 6). Across the Standard Cross-Cultural Sample (SCCS, 186 societies [52, 66]), daughters choose spouses autonomously without parental approval in only 8% of societies (12% among hunter-gatherers), compared with 31% for sons (42% among hunter-gatherers), while parents and kin arrange or co-approve the overwhelming majority (Fig. 6A) [72, 73]. Ethnographic records also document substantial coercion and constraint—bride consent is non-essential in 67% of 57 coded societies, war captivity of women occurs in 37% of 158, and wife-beating in 80% of 70—which dilutes assessment-based selection linearly by $(1 - \kappa)$ while redirecting κ toward force (Eq. (A30)). Yet exit remains widespread: divorce is equally accessible to both spouses in 77% of coded societies (68% in foragers; Fig. 6B) [74, 75].

Across 171 hunter-gatherer groups in Binford's compilation [67], first marriage occurs at a mean age of 20.8 yr for men versus 14.1 yr for women—a +6.7 yr male waiting window (95% CI 5.9 to 7.4 yr; men older in 98% of groups; Fig. 6C). That multi-year interval, reinforced by brideservice in 10% of societies and bridewealth in 57%, places prospective husbands under continuous unstaged observation ($(1 - d)n \gg 1$) by parents-in-law and campmates before reproductive commitment stabilizes [76, 77]. Proposition 18 also clarifies a widely misinterpreted genomic statistic: Y-chromosome versus mitochondrial effective-size ratios near $N_{e,f}/N_{e,m} \approx 2$ [78, 79] do not imply that half of ancestral men left zero descendants; identical ratios arise from modest father-to-father family-size dispersion α_m or patrilineal kin-group drift [80], while completed U.S. cohorts (1900–1979, $n = 34{,}092$) sit near 1.01 (1.00 to 1.03; Fig. 6D).

B. Stated Trust, Shared National Culture, and Cross-National Platform Gradients

Modern cross-national microdata confirm that when people evaluate long-run partners rather than four-minute strangers, delivery, emotional connection, and mind sit at the top of the

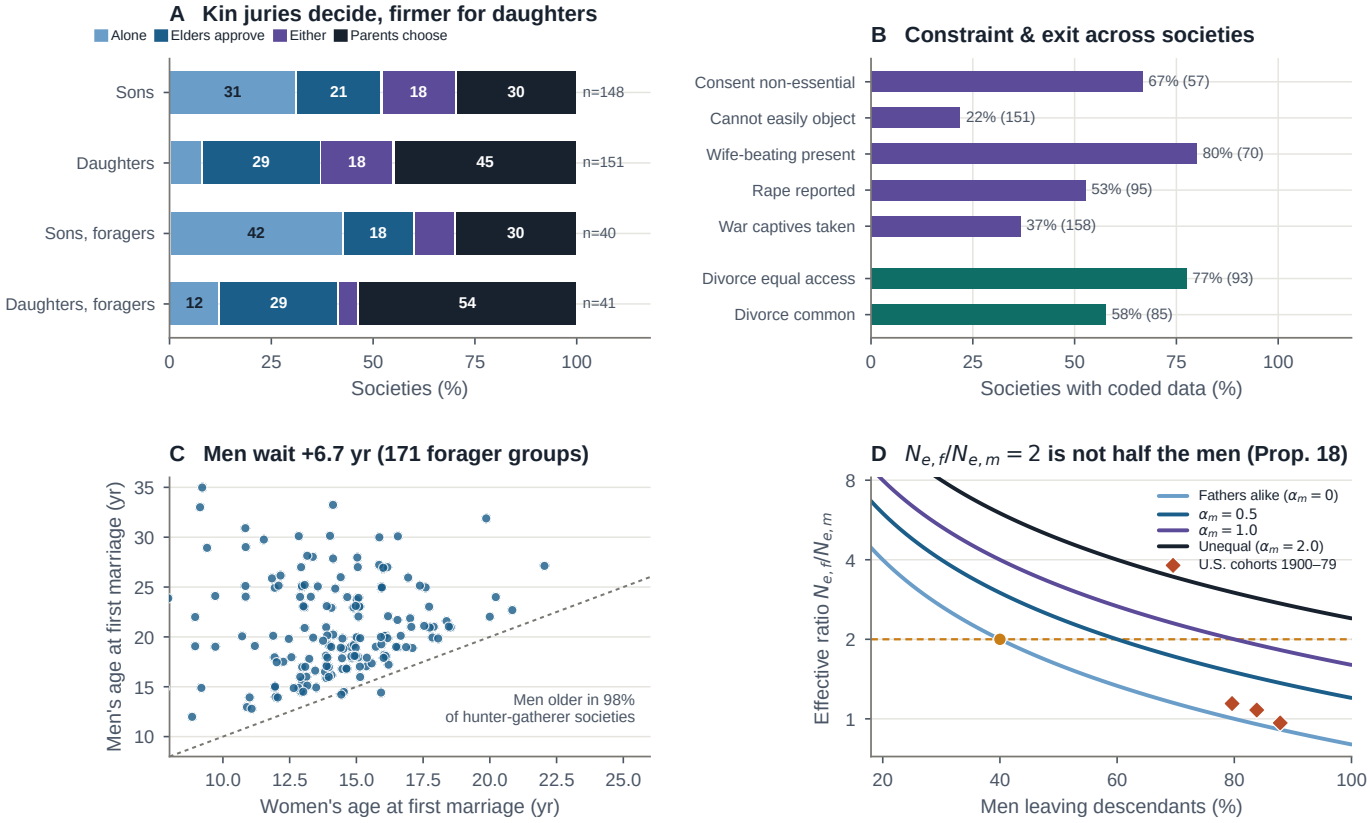


Fig. 6. **Ethnographic marriage juries, constraint versus exit, male waiting times, and effective ancestor ratios. (A)** Who selects a spouse across the Standard Cross-Cultural Sample (SCCS, 186 societies) and hunter-gatherer subset [52, 66]: kin and elders participate in the majority of unions, and autonomous choice is rarer for daughters (8% overall, 12% in foragers) than sons (31%, 42%). **(B)** Prevalence of marriage constraint and coercion alongside symmetric divorce access (77%) and frequent exit (58%) in coded societies. **(C)** Mean age at first marriage across 171 hunter-gatherer groups [67]: men marry 6.7 yr later than women (20.8 vs. 14.1 yr), creating a multi-year brideservice and foraging audit window. **(D)** Female-to-male variance effective population ratio $N_{e,f}/N_{e,m} = [(1-q_f)/(1-q_m)][(1+\alpha_m)/(1+\alpha_f)]$ (Proposition 18): a uniparental genomic ratio of 2.0 does not imply half of men were excluded, and completed U.S. cohorts (1900–1979, $n = 34{,}092$) sit near 1.01.

TABLE III
TAXONOMY AND EMPIRICAL SYNTHESIS OF SECONDARY ENTRY CRITERIA VERSUS PRIMARY PAIR-BOND CRITERIA ACROSS FIFTEEN DATASETS.

Mating Criterion	Empirical Scale	Stranger Legibility (ρ_0)	Entry / Assortative Sorting Effect	Realized Pair-Bond, Divorce & Fertility Return
Panel A: Secondary Mating Criteria (high stranger legibility ρ_X, strong entry/sorting, near-zero bond or fertility welfare $\alpha_X \approx 0$)				
Single-photo facial rank	6.68M HON; 20k `HoloFace-L1`	$\rho = 0.660$ (27.3% 500+ gap)	Entry +4.6/+3.5 pp/SD	Bond +0.22 SD ($\approx \frac{1}{2}$ kindness); aging divorce 5.6%
Male stature & $6'0''$ threshold	781.7k POV; 59.9k `OkCupid`; $k = 97$	Instant box; +17.1% straight surge	Mating $r = +0.050$	$>$ 3-yr union flat (85.6%); fertility $r = +0.006$ (null)
Income & "marrying-up" surplus	90.5k Kowal; 34.1k U.S. GSS	Instant bracket; 1.66× straight M show	Entry +4.0 pp/SD; GSS +0.046	Bond +0.03 SD; marry-up −0.02 SD; marital births −0.034
17-hobby & 5-trait checklist tags	7,941 speed-date dyads; 47.9k couples	Instant profile tags	17-hobby "Yes" $r = +0.014$ (null)	5-trait checklist fit +0.10 SD (5.3× below kindness)
Politics, class & degree sorting	47.4k–50.0k couples (150 countries)	Instant credential / label	Assortative $r = +0.61$ / +0.49 / +0.48	Bond +0.08 / +0.07 / +0.04 SD (5.1–8.9× below kindness)
Panel B: Primary Mating Criteria (low zero-context legibility $\rho_{K,0} \ll \rho_X$, multi-context witness audit, dominant love & survival $\alpha_K \gg \lvert\alpha_X\rvert$)				
Everyday kindness & warmth	47.9k couples; 1.05M POV essays	4-min $r_1 = 0.10$ (near-blind)	Entry −0.25/+0.48 pp/SD (≈ 0)	#1 **bond driver** (+0.29 **SD**); settling kink −0.18 SD/pt
Live play, humor & dyadic spark	8,378 speed dates; 47.3k `OkCupid`	60.5% unique dyadic SRM variance	Live wavelength $r = +0.400$ (20× tags)	#1 "Message me if" cue (25.4× tall, 74.9× wealth)
General intelligence (g) & curiosity	SAT dates ($n = 6{,}822$); 9-domain essays	1-date SAT $R^2 = 0.36\%$ (blind)	$K = 9$ domains: 5.1× lexical validity; GSS +0.011	Couple SAT $r = -0.02$; hindsight snaps back +3.7 pts
Unstaged devotion & witness patience	43 longit. studies; 3,510 `HCMST`	App pre-romance 0.27 yr (÷9.7 kin)	$\leq$ 3 mo rush = 3.1× **breakup** vs. 2.5+ yr	Dyadic devotion explains **44.2%** vs. traits 7.5% R^2
Emotional fidelity & hearth care	160 SCCS soc.; 136 childcare soc.	Lullaby 5.0× love-song d (29.4k raters)	Hearth cooking verbs rise 4.2–4.9× (18–79)	Betrayal cited 14.3× aging looks; +268 cm^3 love gain

Secondary criteria govern low-acquaintance market filtering and assortative sorting (ρ_X high) while contributing near zero inside established unions ($\alpha_X \approx 0$); primary criteria are noisy in single snapshots ($\rho_{K,0} \ll \rho_X$) yet dominate long-term satisfaction, fidelity, and juvenile provisioning ($\alpha_K \gg \lvert\alpha_X\rvert$; Propositions 21–23).

hierarchy (Fig. 7). Across $N = 7{,}060$ Australians [61], interpersonal trust (78.1/100 women, 68.9/100 men), emotional connection (77.5,65.1), openness, and intelligence (69.7,55.8) outrank physical build and income in both sexes (Fig. 7A).

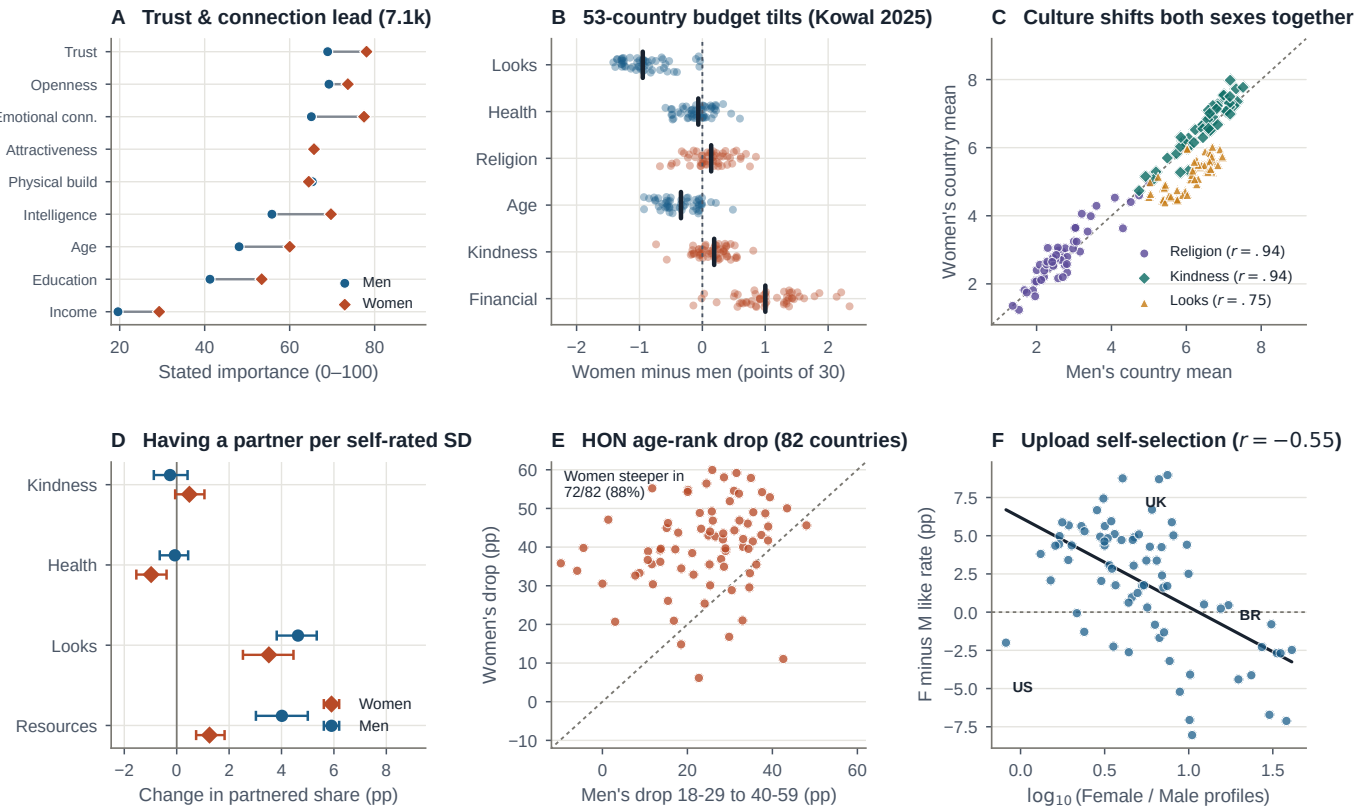


Fig. 7. **Cross-national partner priorities, shared national culture, and global digital platform gradients. (A)** Nine-trait stated partner importance dumbbell across $N = 7{,}060$ Australians [61]: trust (78.1,68.9/100), emotional connection (77.5,65.1), openness, and intelligence (69.7,55.8) outrank physique and income in both sexes. **(B)** Within-country sex differences (women minus men, points of 30) across 53 countries ($n = 67{,}743$ [8]), showing shared directional tilts alongside wide national spread. **(C)** Cross-national co-variation of men's and women's 30-point mate budgets: national culture shifts both sexes in lockstep ($r = 0.94$ religion, $r = 0.94$ kindness). **(D)** Union-entry partnering slopes per self-rated SD across 86,033 adults [8]: financial resources (+4.01 pp) and looks (+4.63 pp) predict men's union entry, whereas self-rated kindness is near zero (-0.25, + 0.48 pp). **(E)** Within-gender HON beauty-percentile drop from ages 18–29 to 40–59 across 82 strict countries ($n \geq 500$ per sex): steeper for women in 72/82 nations (median -40.8 vs. -25.3 pp). **(F)** National platform participation scale versus female-minus-male like-rate advantage across 75 countries ($r = -0.55$), reflecting upload selectivity in smaller markets.

Across 67,743 constrained 30-point mate budgets from 53 countries in Kowal et al. [8] (verified across 32 age-standardized nations), women allocate slightly more to financial prospects and kindness while men allocate more to looks (Fig. 7B), yet national institutional culture shifts men's and women's priorities almost in unison: country-mean allocations correlate across sexes at $r = 0.94$ for religion and $r = 0.94$ for kindness, dwarfing within-country sex gaps (Fig. 7C; assortative partner matching likewise peaks on shared religion $r = 0.70$ and kindness $r = 0.31$). At the union-entry margin across 86,033 adults [8] and 45–70-country surveys ($n = 117{,}293$ [81]), men's self-rated financial standing (+4.01 pp/ SD) and looks (+4.63 pp) strongly predict having a partner, whereas self-rated kindness barely moves entry (-0.25 pp in men, +0.48 pp in women; Fig. 7D).

Comparing 75 countries with ≥ 500 Hot-or-Not profiles per sex (173 countries at ≥ 100 per sex) reveals two global regularities of single-photograph evaluation. First, among 82 countries meeting strict age-cell thresholds, within-gender beauty percentiles fall more steeply across adulthood for women than men in 72/82 nations (median -40.8 vs. -25.3 pp; Fig. 7E; Corollary 7). Second, log national platform participation correlates negatively with the female-minus-male like-rate gap ($r = -0.55$; Fig. 7F), showing that thin-market adopter self-selection inflates observed ratings unless upload selectivity is audited.

C. Primary Devotion Over Secondary Homophily, the Settling Kink, Rushed Lock-In, and Asymmetric Marital Exit

Why do searchable dating markets look obsessed with income, youth, $6'0''$ cutoffs, and political or educational credentials when lived marriages revolve around kindness, mutual reverence, and commitment? Propositions 21 and 23 unify these facts by separating *secondary entry criteria* (ρ_X high, $\alpha_X \approx 0$) from *primary pair-bond criteria* ($\rho_{K,0} \ll \rho_X$, $\alpha_K, \alpha_D \gg |\alpha_X|$; Table III; Figs. 8–9):

1) **Kindness outruns checklist fit, "marrying up," and demographic homophily:** Across $N = 50{,}320$ partnered adults in 175 countries [8], financial resources predict men's union entry (+4.01 pp/ SD) yet collapse near zero inside unions (+0.03, + 0.02 SD; infidelity +0.2 pp, $N = 49{,}668$), whereas partner kindness dominates love (+0.28, + 0.25 SD) and cuts infidelity (-1.8, -3.4 pp/ SD; women even rank kindness #1 for short-term mates, 8.42/11 vs. 8.07 looks). In a country-FE horse race ($N = 47{,}932$ couples across 148 countries; Fig. 8A), kindness (+0.29 SD) beats Conroy-Beam 5-trait idiographic checklist fit (+0.10 SD [68]) by 2.8×,

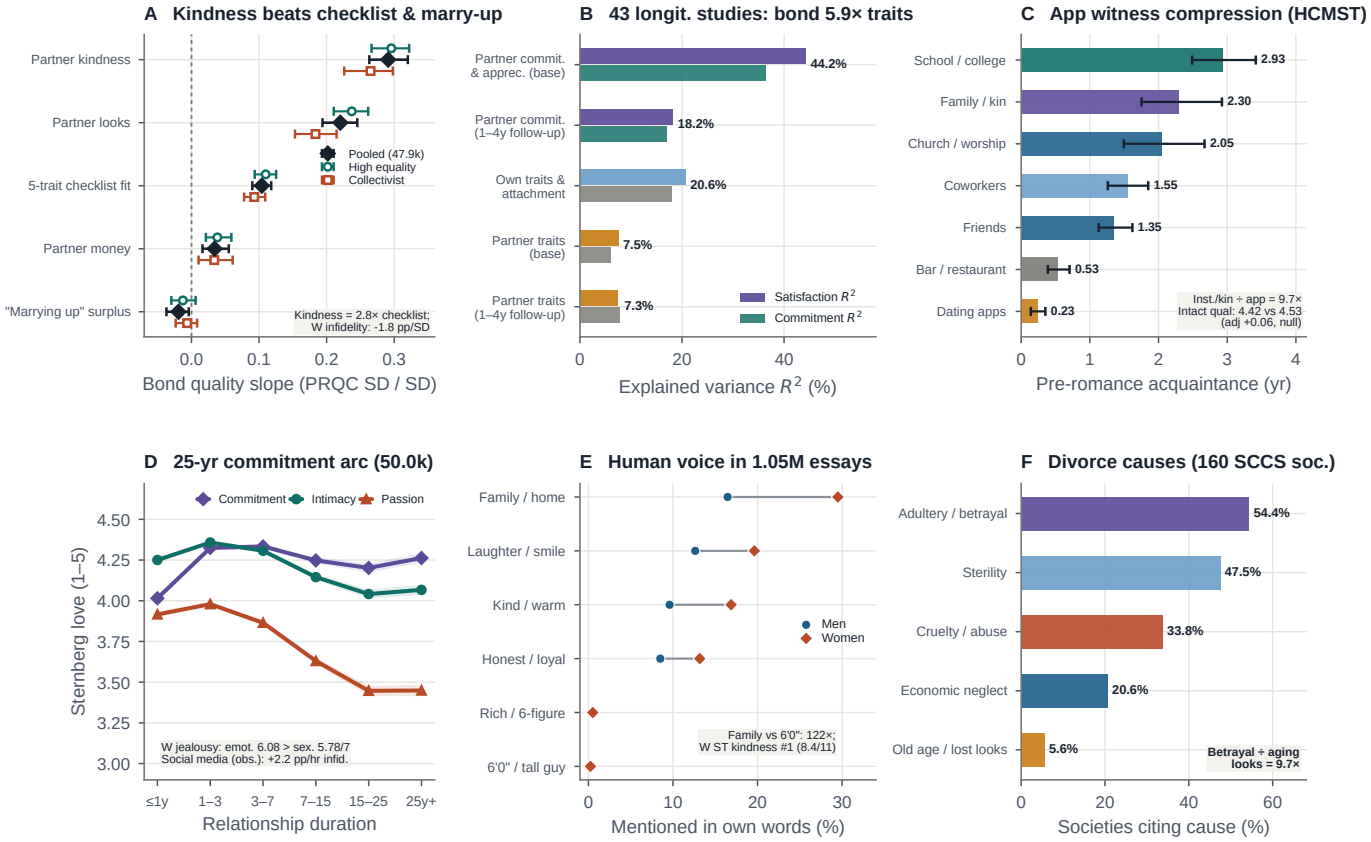


Fig. 8. **Primary pair-bond devotion versus secondary entry filters across 175 countries, 43 longitudinal couples studies, Stanford HCMST, and 1.05M essays (Proposition 21). (A)** Country-FE relationship-quality horse race ($N = 47{,}932$ couples, 148 countries [8, 68]): partner kindness (+0.29 SD) outruns 5-trait checklist fit (+0.10 SD) by 2.8×, partner money (+0.03 SD), and "marrying up" (−0.02 SD) across all four cultural regimes. **(B)** Random-forest synthesis across 43 longitudinal couples datasets (11,196 couples [9]): perceived commitment and dyadic conduct explain 44.2% of baseline satisfaction (18.2% follow-up), 5.9× static partner traits (7.5%). **(C)** Stanford HCMST ($N = 3{,}510$ [10]): apps compress pre-romance acquaintance 9.7× (0.23 vs. 2.26 yr institutional/kin), while surviving intact unions show equal quality (4.42 vs. 4.53/5). **(D)** 25-year commitment arc ($N = 48{,}446$ [8, 69]; 4.02 → 4.26/5), female emotional-over-sexual jealousy (6.08 vs. 5.78/7 [70]), and observational social-media infidelity slope (+2.19 pp/h). **(E)** Self-written themes across 1,052,790 POV essays (29.5% W write family/home, 122× 6′0″ mentions). **(F)** Coded divorce causes across 160 SCCS societies [71]: betrayal (54.4%) and cruelty (33.8%) occur at 9.7× aging looks (5.6%).

partner money (+0.03 SD) by 8.5×, and "marrying out of your league" (−0.02 SD, [−0.03, − 0.01]; −0.02 SD 5-trait controlled) across egalitarian, low-equality, individualist, and collectivist regimes. Crucially, across 149–150 countries ($N = 47{,}410$–$49{,}984$), couples sort heavily on secondary social labels—political ideology ($r = +0.61$, 66.7% exact match), social class ($r = +0.49$, 65.5%), and education ($r = +0.48$, 52.0%)—yet inside-union country-FE quality slopes for similarity are tiny (+0.08, +0.07, and +0.04 SD), so primary partner kindness beats secondary demographic homophily by 5.1×, 5.2×, and 8.9× (Fig. 9B).

2) **Partner reverence versus the "settling" kink, and 43 longitudinal couples studies:** Estimating piecewise country-FE splines in (mate − self) ratings controlling for own trait ($N = 50{,}180$; Fig. 9C) explains why mean "marrying up" looks flat on money while partner admiration anchors love: settling below oneself on primary kindness penalizes love by −0.18 SD /pt in women and −0.16 SD /pt in men—1.8× and 1.7× steeper than positive reverence surplus (+0.10 and +0.09 SD /pt)—whereas reverence surplus on financial resources is null (−0.01 W, −0.04 M), and perceiving one's partner as more physically attractive than oneself cuts reported infidelity by −8.9 pp in women and −15.9 pp in men. Independently, across 43 longitudinal couples datasets (11,196 couples [9]), perceived partner commitment and dyadic responsiveness explain 44.2% of baseline satisfaction (18.2% at 1–4 yr follow-up), 5.9× all static partner traits combined (7.5%, 7.3%; Fig. 8B).

3) **Courtship staging drop, app witness compression, and the ≤ 3-month cohabitation rush hazard:** Consistent with Proposition 23(i) (E23 closed form −17.0%), daily appearance and physical effort across $N = 65{,}916$ adults [8] peaks during active dating (229 min/d women, 212 min/d men) and drops −13.7% (to 198 min/d) in married women and −19.5% (to 171 min/d) in married men (observational cross-section; Fig. 9D). In Stanford HCMST ($N = 3{,}510$, 3,295 meeting-coded couples [10]), dating apps compress pre-romance acquaintance 9.7× (0.23 yr, median 0.00, 50.5% zero years, 89.8% zero prior ties vs. 2.26 yr institutional/kin and 2.93 yr school, 12.5×; surviving intact-union quality null 4.42 vs. 4.53/5, adjusted +0.06; Fig. 8C). Directly confirming Proposition 23(ii), among 2,578 ever-cohabited HCMST couples, rushing from romance onset into cohabita-

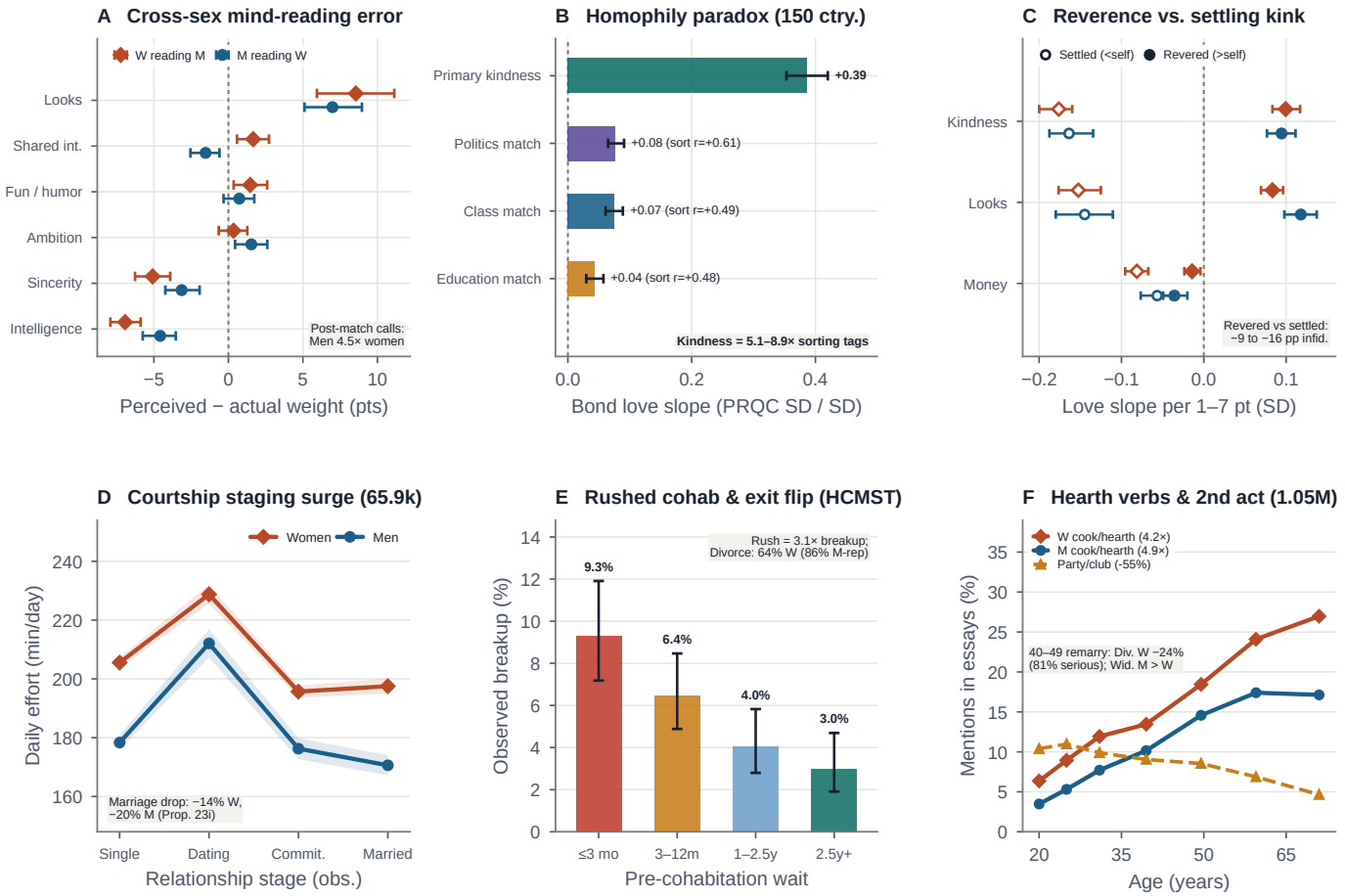


Fig. 9. **Cross-source male–female mating dynamics separating secondary sorting from primary devotion (Proposition 23).** **(A)** Columbia speed-dating cross-sex mind-reading error ($N = 551$ [55]): both sexes overestimate opposite-sex looks weight (+8.6 W, +7.0 M pts) and underestimate intelligence ($-6.9, -4.6$ pts), while men place 4.5× more post-match calls (77.9% W zero calls). **(B)** The homophily paradox across 149–150 countries ($N = 47.4$k–50.0k [8]): couples sort strongly on politics ($r = +0.61$), social class ($r = +0.49$), and education ($r = +0.48$), yet primary kindness (+0.39 SD) beats secondary homophily slopes ($+0.08, +0.07, +0.04$ SD) by 5.1–8.9×. **(C)** Country-FE partner reverence versus "settling" spline ($N = 50{,}180$ [8]): settling below self on kindness penalizes love 1.8× (W) and 1.7× (M) more steeply than surplus adds, while money reverence is null and mutual looks reverence cuts infidelity $-8.9/-15.9$ pp. **(D)** Daily appearance/physical courtship staging peaks in dating (229/212 min/d) and drops $-13.7\%/-19.5\%$ in marriage ($N = 65{,}916$ observational), alongside widening male-over-female casual SOI desire ($1.46 \to 1.77\times$). **(E)** Stanford HCMST ($N = 2{,}578$ ever-cohabited [10]): rushing from romance onset to cohabitation in ≤ 3 months triples breakup (9.3% vs. 3.0% at 2.5+ yr, 3.1×), while marital exits flip to 64.3% female unilateral initiation (85.7% in husbands' reports vs. 58.5% non-marital). **(F)** Across 1,052,790 POV essays (ages 18–79), hearth cooking verbs surge 4.2× (W) and 4.9× (M) as party verbs fall −55.3%, while 40–49 divorced women cut marriage intent −24.2% (keeping 80.8% serious bond) and widowed men exceed widowed women.

tion within ≤ 3 months triples subsequent union dissolution (9.3% vs. 3.0% when waiting 2.5+ years, 3.1×; cohort- and age-adjusted penalty +4.0 pp, $[+1.4, +6.6]$; Fig. 9E).

4) **The 25-year commitment arc, voice, and marriage-specific female unilateral exit:** Across 48,446 couples [8, 69], passion moderates ($3.92 \to 3.45/5$) while intimacy stays high (4.07) and commitment rises ($4.02 \to 4.26/5$; love documented in 88.6% of 166 societies [82]; marry-without-love median $27{,}50/100$, $N = 81{,}596$ [83]), women report stronger emotional than sexual jealousy (6.08 vs. $5.78/7$, +0.29 [70]), and daily social-media browsing tracks higher perceived outside options (+0.020 SD /h), lower quality (−0.035 SD /h), and higher infidelity (+2.19 pp/h, 28.4%→44.1%; Fig. 8D [84]). In 1,052,790 self-written POV essays (327,126 W, 725,664 M), women invoke family/home (29.5%; men 16.5%), laughter (19.6%,12.6%), kindness (16.9%,9.6%), and loyalty (13.2%,8.5%) at 54×–122× 6′0″/tall mentions (0.24%) and 33× wealth (0.51%; Fig. 8E). When primary devotion fails under legal and domestic lock-in, exit turns sharply asymmetric (Proposition 23(iii)): in HCMST, non-marital unilateral breakups ($n = 286$) are near-symmetric (58.5% female; 50.0% in male reports), whereas marital divorces ($n = 67$) flip to 64.3% female unilateral initiation (85.7% in husbands' own reports; E23 65.2%; Fig. 9E), paralleling 160 SCCS societies [71] where betrayal (54.4%) and cruelty (33.8%) trigger divorce at 9.7× aging looks (5.6%; Fig. 8F).

V. Digital Platforms at Scale, the HoloFace-L1 Ceiling, Facial Attribution, and Completed Fertility

A. Platform Attention Concentration and Portfolio Decoupling

Auditing all 6,727,811 canonical Hot-or-Not records (6,677,772 eligible profiles aged 18–79 with ≥ 20 valid votes and 3,685,764 inventoried images; Appendix Table C2) exposes three structural properties of single-context crowd evaluation

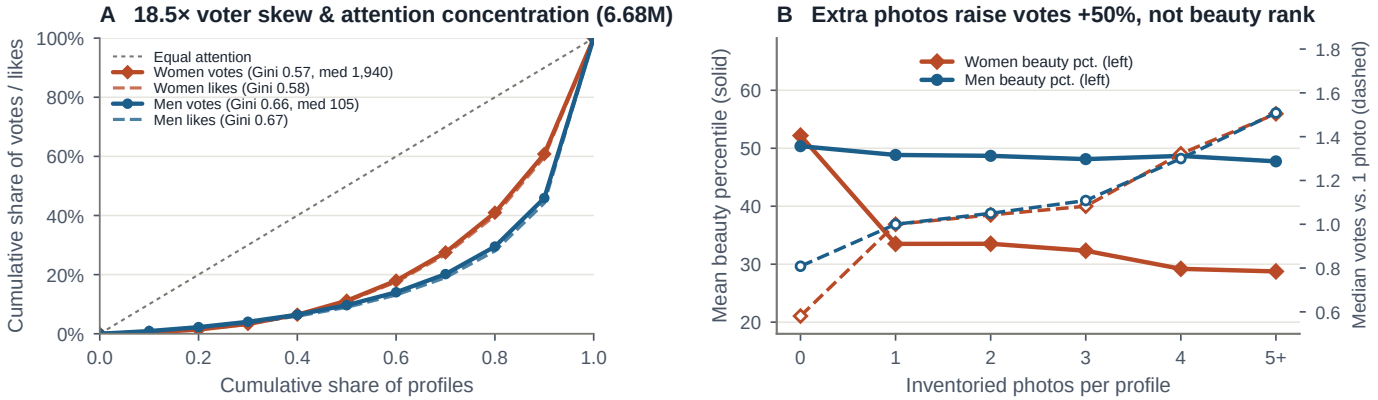


Fig. 10. **Attention concentration and portfolio decoupling across 6,727,811 Hot-or-Not profiles (6,677,772 with ≥ 20 votes). (A)** Lorenz concentration of votes and likes: women receive $18.5\times$ median votes (1,940 vs. 105), while top-decile profiles capture 40.3% (women, Gini 0.580) and 55.8% (men, Gini 0.672) of all positive likes. **(B)** Multi-photo upload portfolio (0 to 5+ inventoried photographs): extra photos raise median vote volume $+50\%$ without lifting mean beauty percentile above $\sim$50th (within-gender age trajectories appear in Fig. 15B).

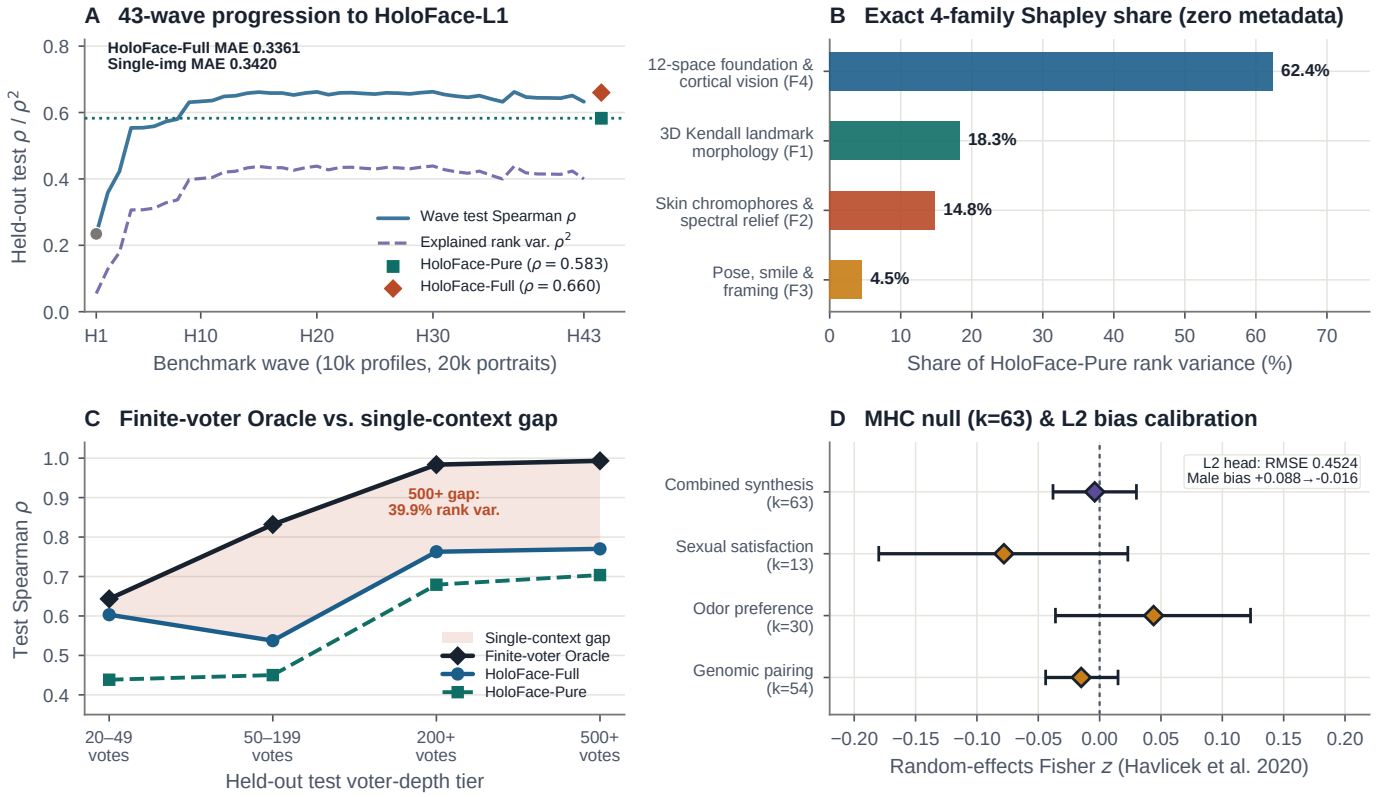


Fig. 11. **The `HoloFace-L1` 43-wave deep-learning benchmark, exact 4-family Shapley attribution, and finite-voter psychometric ceiling** (10,000 **profiles,** 20,000 **portraits). (A)** Held-out test progression from raw UV baseline H1 ($\rho = 0.234$, MAE 0.4831) through strict zero-metadata `HoloFace-Pure` ($\rho = 0.583$, $\rho^2 = 34.0\%$, MAE 0.3776) to multimodal `HoloFace-Full` (`HoloFace-L1`: Test MAE 0.3361, single-image MAE 0.3420, $\rho = 0.660$, $\rho^2 = 43.6\%$). **(B)** Exact $2^4 = 16$ Shapley decomposition of zero-metadata `HoloFace-Pure` rank variance: 95.5% stems from 12-space foundation appearance (62.4%), 3D Kendall Riemannian morphology (18.3%), and skin chromophore/spectral albedo (14.8%), versus only 4.5% from changeable pose, smile, and framing. **(C)** Finite-voter psychometric Oracle ceiling ($\hat{\sigma}^2_{\text{voter}} = 8.23$ on 1–10 rating scale, fitted strictly on 8,000 training OOF residuals) versus `HoloFace-Full` and `HoloFace-Pure` across vote-depth tiers (20–49, 50–199, 200+, 500+), isolating the 39.9% (500+ votes, Oracle $\rho = 0.993$) to 48.3% (all test, Oracle $\rho = 0.918$) irreducible single-context consensus gap. **(D)** Skew-unbiased L_2 calibration eliminating male rating bias ($+0.088 \rightarrow -0.016$, RMSE 0.4524) alongside the $k = 63$ human MHC chemosensory meta-analytic null [85].

(Fig. 10; life-course age curve in Fig. 15B). First, within-gender beauty rank declines much more steeply across adulthood for women than men: women's mean percentile drops -27.8 points from 52.0 at ages 25–29 (top-decile share 8.0%) to 24.2 at ages 45–49 (0.5%), compared with a gentler -20.1-point decline in men from 54.0 to 33.9 after an early-twenties peak (Fig. 15B). Second, voter attention and positive evaluations are heavily concentrated: female profiles receive $18.5\times$ as many median votes as male profiles (1,940 vs. 105), and the top decile captures 40.3% of all female likes (Gini 0.580) and 55.8% of all male likes (Gini 0.672; Fig. 10A). Third, uploading additional photographs (1 to 5+ images) raises median vote

volume $+50\%$ without shifting mean beauty percentile above $\sim$50th (Fig. 10B): repeating static snapshots within the same staged presentation genre raises exposure (m) without adding multi-context trait validity (k).

B. The `HoloFace-L1` 43-Wave Benchmark and Psychometric Voter Ceiling

How much of human consensus rating can be predicted from static portraits and profile metadata, and how much of the residual reflects finite voter noise versus an irreducible single-context ceiling? Across our 43-wave benchmark on 10,000 profiles and 20,000 portraits (8,000 train / 1,000 validation / 1,000 held-out test, zero identity overlap; Fig. 11; Appendix Table C1), progressive nuisance removal and multimodal integration establish four quantitative bounds:

1) **Pure-face predictability and 4-family Shapley anatomy:** Stripping all demographic and tabular metadata, `HoloFace-Pure` lifts held-out pure-face Spearman correlation to $\rho = 0.583$ ($\rho^2 = 34.0\%$, Test MAE 0.3776; rising to $\rho = 0.704$, $\rho^2 = 49.5\%$ among well-voted 500+-vote profiles; Fig. 11A). Exact Shapley attribution across all $2^4 = 16$ visual family subsets proves that 95.5% of pure-face rank variance derives from 12-space foundation and cortical appearance (62.4%), 3D craniofacial and Kendall Riemannian landmark geometry without pixel intensities (18.3%), and skin chromophore/spectral micro-relief (14.8%), whereas changeable camera pose, smile curvature, and framing contribute only 4.5% (Fig. 11B).
2) **Multimodal `HoloFace-Full` optimum and skew calibration:** Combining 29 out-of-fold visual, tabular, and cross-modal predictors via validation cross-validated exact-L_1 linear programming achieves a new benchmark best across all 43 waves: Validation MAE 0.3569, Candidate Test MAE 0.3361, Single-Image Test MAE 0.3420, within-gender percentile MAE 16.6 points, and Test Spearman $\rho = 0.660$ ($\rho^2 = 43.6\%$; $\rho = 0.770$ at 500+ votes). Because male ratings are right-skewed while median-optimal L_1 loss targets conditional medians, pairing the L_1 head with a ridge L_2 calibration head achieves Test RMSE 0.4524 while reducing male mean bias from $+0.088$ to -0.016 (Fig. 11D).
3) **Deattenuated single-context consensus gap:** Fitting heteroscedastic squared OOF residuals $(y_i - \hat{f}_{\text{oof}}(x_i))^2 = \sigma^2_{\text{gap}} + \sigma^2_{\text{voter}}/n_i$ strictly on the 8,000 training profiles yields per-voter rating variance $\hat{\sigma}^2_{\text{voter}} = 8.23$ (SD $= 2.87$ points on the 1–10 rating scale). Finite-voter Oracle ceilings rise monotonically from $\rho = 0.643$ at 20–49 votes to $\rho = 0.832$ at 50–199, $\rho = 0.984$ at 200+, and $\rho = 0.993$ ($\rho^2 = 98.7\%$, Oracle MAE 0.046) at 500+ votes (Fig. 11C). Even where voter sampling error vanishes (500+ votes, median 3,526 raters per portrait), `HoloFace-Full` leaves 39.9% of true human consensus rank variance unexplained (48.3% across the full test cohort, Oracle $\rho = 0.918$). Static photographs and self-reported tags cannot capture dynamic voice, movement, responsiveness, or shared-context conduct ($C_{ic} + H_i$ in Proposition 20).
4) **Chemosensory MHC null:** Nor does unmeasured HLA dissimilarity supply a hidden biological shortcut: across $k = 63$ human effect sizes synthesized from Havlíček et al. [85] and Winternitz et al. [86], random-effects meta-analytic means include zero across genomic pairing ($z = -0.011$), odor preference ($z = +0.044$), and relationship satisfaction ($z = -0.080$; Fig. 11D).

C. Canonical 106-Landmark Facial Attribution Across Sex, Age, and World Regions

To map *where* on the human face visual consensus concentrates without exposing any individual photograph, we registered 5,353 front-facing portraits ($N = 1{,}589$ women, $N = 1{,}977$ men; every composite cell $n \geq 75$) onto a canonical 106-landmark, 222-triangle piecewise-affine Delaunay atlas [87–89] and computed exact 12-zone Shapley attributions [90] from aligned `DINOv3` + `ConvNeXt-V2` + `ArcFace` spatial representations [91, 92] (live spatial models verified at held-out $\rho = 0.557$ in women and $\rho = 0.332$ in men; Figs. 12–14; Table IV).

Three anatomical results emerge. First, beauty drivers and rejectors localize to expressive social-warmth regions—eyes, smile contour, and midface vitality—rather than cranial vault size: in women, eyes, iris & palpebral aperture leads total top-minus-bottom spread (5.42 percentile points: $+1.47$ driver in top 15%, -3.95 rejector in bottom 15%), alongside periorbital eye/iris aperture ($+1.47$ driver), eyebrow arch, vermilion lip and smile contour ($+1.08$), malar cheekbone contour, and tear-trough smoothness; in men, eyebrow arch & spacing (1.52 spread: $+0.46$ driver, -1.06 rejector), jawline/chin definition, nasal bridge/alar proportion, eyes ($+0.13$), and smile contour ($+0.15$) dominate (Fig. 12A–C; Table IV). Second, across five adult age brackets (18–24 to 55+), pure facial anatomy accounts for gradual periorbital tear-trough and nasolabial softening (57.0 $\rightarrow$ 38.8th percentile in women; 62.9 $\rightarrow$ 19.9th in men), whereas observed voter rankings penalize female profile age more steeply (67.4 $\rightarrow$ 29.5th; Fig. 13). Third, across all 7 world regions, spatial beauty driver maps correlate at mean $r = 0.97$ in women (minimum 0.91) and $r = 0.76$ in men (minimum 0.36), even where regional mean voter scores diverge due to upload selectivity (Fig. 14). Independently, across 48,413 self-reported "First thing people notice about me" essays in `OkCupid-JSE` [59], smile/laugh (27.2%, #1) and eyes (20.5%, #2) lead every somatic attribute, outranking stature (10.2%) more than twofold and mirroring the 12-zone `HoloFace-L1` Shapley warmth map.

D. Lifespan Clocks, the Hearth Verb Calendar, Second-Act Re-Entry, and the 94% Pairing Gate

Placing 84,636 realized partner age-gap records [8, 81], 67,733 mate-budget trajectories [8], 6,677,772 rated portraits, and 1,208,771 Plenty of Fish records (1,137,309 opposite-sex profiles aged 18–79, including 1,052,790 self-written essays) onto a shared adulthood axis (ages 18–75; Fig. 15; Fig. 9D,F; Table V; Appendix Tables C3–C4) separates diverging biological clocks from converging hearth priorities. Across the top row of Fig. 15, biological and casual-market

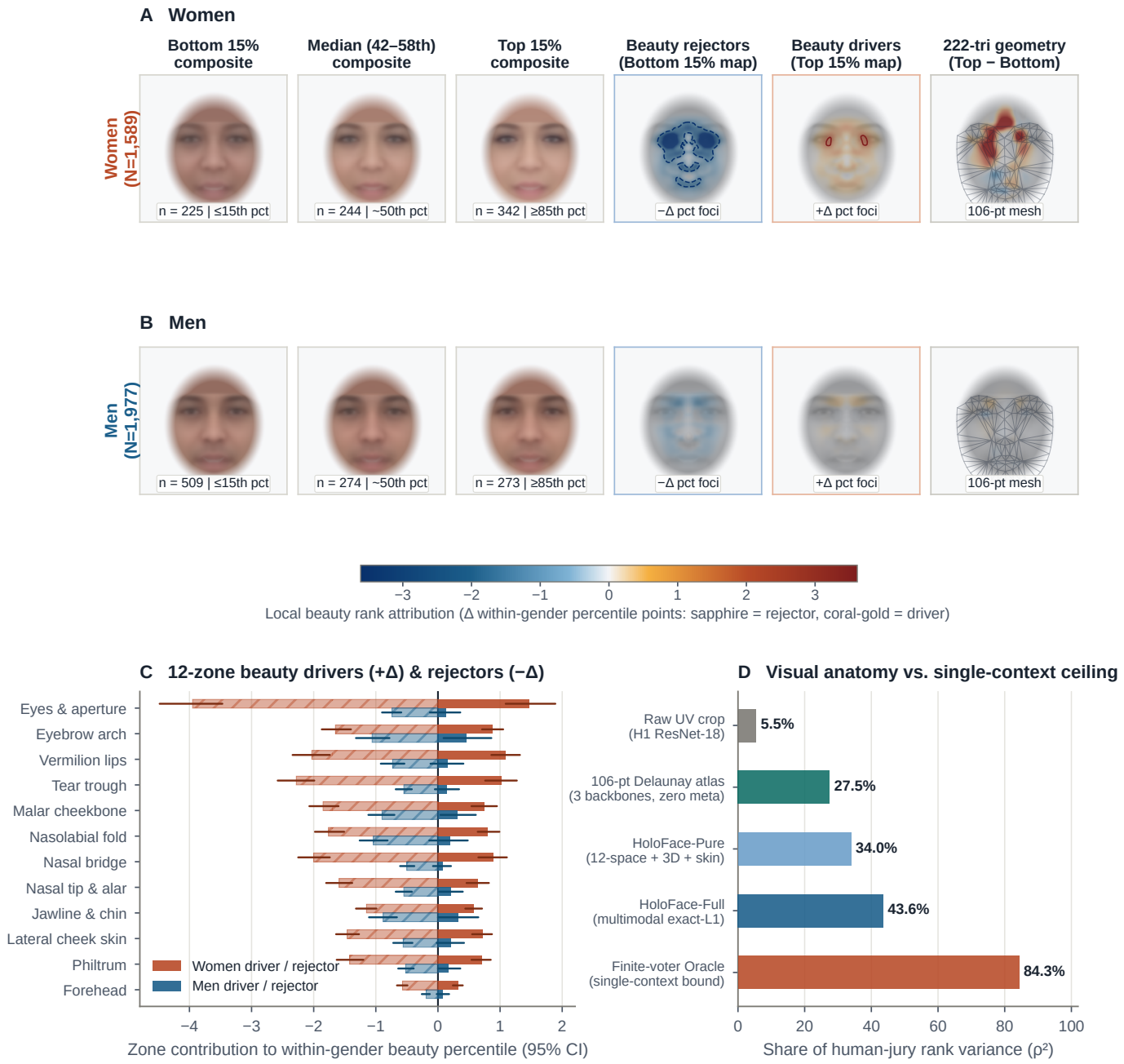


Fig. 12. **Canonical 106-landmark Delaunay facial composites and 12-zone Shapley beauty attribution across 5,353 aligned portraits. (A, B)** Super-resolved (384×384, all cells $n \geq 75$) within-gender bottom-15%, median (42–58th), and top-15% composites alongside signed local percentile driver/rejector maps and 222-triangle Delaunay log-Jacobian geometry differences for women ($N = 1{,}589$) and men ($N = 1{,}977$). **(C)** Exact 12-zone Shapley beauty drivers ($+\Delta$ pct in top 15%) and rejectors ($-\Delta$ pct in bottom 15%) with 95% CIs. **(D)** Progressive share of human-jury rank variance (ρ^2) from raw UV crop (H1), canonical 106-landmark pure-face model, zero-metadata `HoloFace-Pure`, and multimodal `HoloFace-Full` relative to the infinite-voter Oracle ceiling.

clocks diverge by sex: realized partner age gaps scissor apart to $6.5\,\mathrm{yr}$ by ages 60–79 (Fig. 15A), male-over-female casual sociosexual desire widens from $1.46\times$ at ages 18–24 to $1.77\times$ at 60–79 (Fig. 9D), crowd photo ratings penalize female age far beyond pure facial anatomy (Fig. 15B), and explicit "Want children" shares among childless users flip from women higher at 25–29 (46.0% W vs. 33.5% M; $+12.5$ pp) through an interpolated crossover at age 41.8 (exact age 41) to men higher at 45–49 (6.8% W vs. 12.8% M; -6.0 pp; Fig. 15C).

Across the bottom row of Fig. 15 and Fig. 9F, however, primary relational devotion strengthens across life. First, kindness ranks #1 in *both* sexes in every age bracket from 18–24 to 60–79 (Fig. 15D), and across 1,052,790 free-text essays, performative "party/clubbing" action verbs fall -55.3% from ages 18–22 to 65–79 while domestic hearth "cook/bake" verbs surge $4.2\times$ in women ($6.4\%\rightarrow27.0\%$) and $4.9\times$ in men ($3.5\%\rightarrow17.1\%$), directly echoing the Pleistocene hearth (Fig. 9F). Second, women's preference for serious relationship commitment stays strictly above men's in every decade (61.0% vs. 50.5% at 25–29; 76.2% vs. 65.0% at 40–49; 82.3% vs. 78.8% at 60–69; $>$ 10-year prior union 50.1% vs. 38.3%; Fig. 15E), replicating across all five POV country cohorts and

TABLE IV
CANONICAL 12-ZONE FACIAL BEAUTY DRIVERS (TOP 15%), REJECTORS (BOTTOM 15%), AND TOP-MINUS-BOTTOM PERCENTILE SPREAD (5,353 PORTRAITS).

Canonical anatomical zone (106-pt Delaunay atlas)	Women ($N = 1{,}589$)			Men ($N = 1{,}977$)		
	Top 15% driver	Bot. 15% rejector	Spread	Top 15% driver	Bot. 15% rejector	Spread
Eyes, iris & palpebral aperture	$+1.47$ $[+1.09, +1.89]$	-3.95 $[-4.48, -3.47]$	5.42	$+0.13$ $[-0.13, +0.36]$	-0.74 $[-0.90, -0.59]$	0.87
Eyebrow arch & spacing	$+0.88$ $[+0.71, +1.05]$	-1.65 $[-1.87, -1.40]$	2.53	$+0.46$ $[+0.09, +0.86]$	-1.06 $[-1.32, -0.78]$	1.52
Infraorbital / tear trough	$+1.02$ $[+0.76, +1.27]$	-2.28 $[-2.58, -1.99]$	3.30	$+0.14$ $[-0.05, +0.34]$	-0.55 $[-0.68, -0.42]$	0.69
Vermilion lips & smile contour	$+1.08$ $[+0.86, +1.32]$	-2.03 $[-2.34, -1.74]$	3.11	$+0.15$ $[-0.12, +0.41]$	-0.73 $[-0.92, -0.54]$	0.88
Malar cheekbone prominence	$+0.75$ $[+0.54, +0.95]$	-1.85 $[-2.07, -1.60]$	2.61	$+0.31$ $[+0.04, +0.61]$	-0.90 $[-1.12, -0.70]$	1.22
Nasolabial & perioral fold	$+0.80$ $[+0.64, +0.99]$	-1.76 $[-1.98, -1.51]$	2.56	$+0.19$ $[-0.14, +0.48]$	-1.04 $[-1.26, -0.81]$	1.23
Nasal dorsum / bridge	$+0.89$ $[+0.65, +1.11]$	-2.00 $[-2.25, -1.74]$	2.89	$+0.07$ $[-0.08, +0.21]$	-0.50 $[-0.61, -0.38]$	0.56
Nasal tip & alar width	$+0.64$ $[+0.46, +0.82]$	-1.59 $[-1.80, -1.38]$	2.23	$+0.21$ $[+0.01, +0.40]$	-0.55 $[-0.68, -0.42]$	0.76
Lateral cheek skin & tone	$+0.72$ $[+0.55, +0.87]$	-1.46 $[-1.64, -1.27]$	2.17	$+0.20$ $[-0.02, +0.42]$	-0.56 $[-0.72, -0.41]$	0.76
Jawline taper & chin	$+0.57$ $[+0.44, +0.71]$	-1.15 $[-1.32, -0.99]$	1.73	$+0.32$ $[+0.00, +0.65]$	-0.88 $[-1.11, -0.66]$	1.20
Philtrum & upper perioral	$+0.70$ $[+0.54, +0.85]$	-1.42 $[-1.63, -1.20]$	2.11	$+0.17$ $[-0.00, +0.36]$	-0.52 $[-0.64, -0.39]$	0.69
Forehead & glabella	$+0.32$ $[+0.24, +0.40]$	-0.57 $[-0.66, -0.49]$	0.89	$+0.08$ $[-0.02, +0.18]$	-0.19 $[-0.26, -0.13]$	0.28

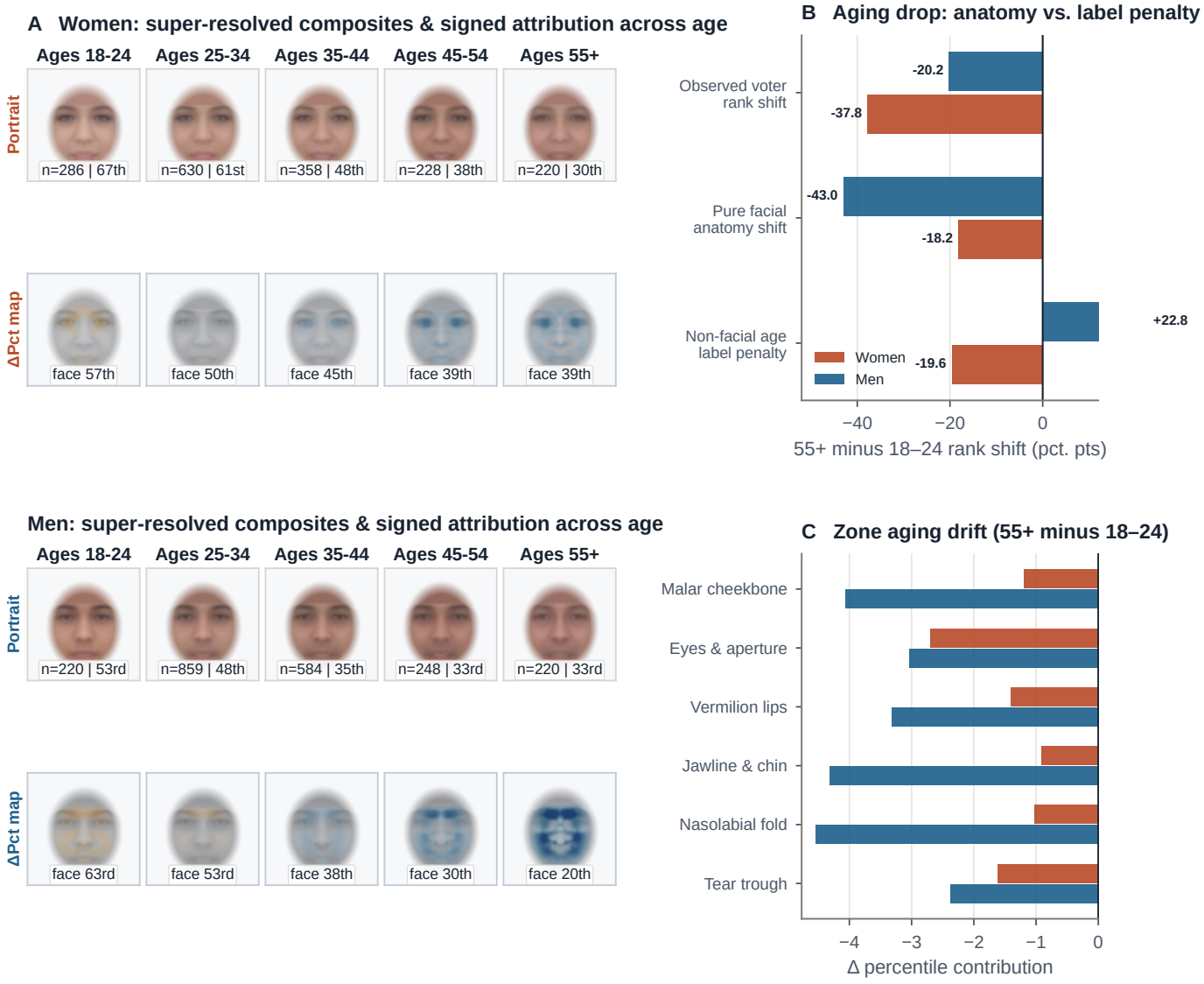


Fig. 13. **Super-resolved canonical facial aging composites, anatomy-versus-label decomposition, and 12-zone drift from ages 18–24 to 55+. (A)** Piecewise-affine Delaunay composites and signed pure-face percentile attribution maps across five adult age brackets (18–24 through 55+, every cell $n \geq 75$). **(B)** 55+ minus 18–24 rank-shift decomposition separating observed voter drop (67.4 → 29.5th pct in women) into zero-metadata pure facial anatomy (57.0 → 38.8th pct) plus the non-facial age-label penalty. **(C)** Anatomical zone drift between 55+ and 18–24, led by periorbital tear-trough hollowing, nasolabial folding, and jawline contour softening.

San Francisco `OkCupid-JSE` (+10.6 pp at 25–29 flipping to −20.0 pp at 45–54, crossover age 38.2 yr [59]; Table V),

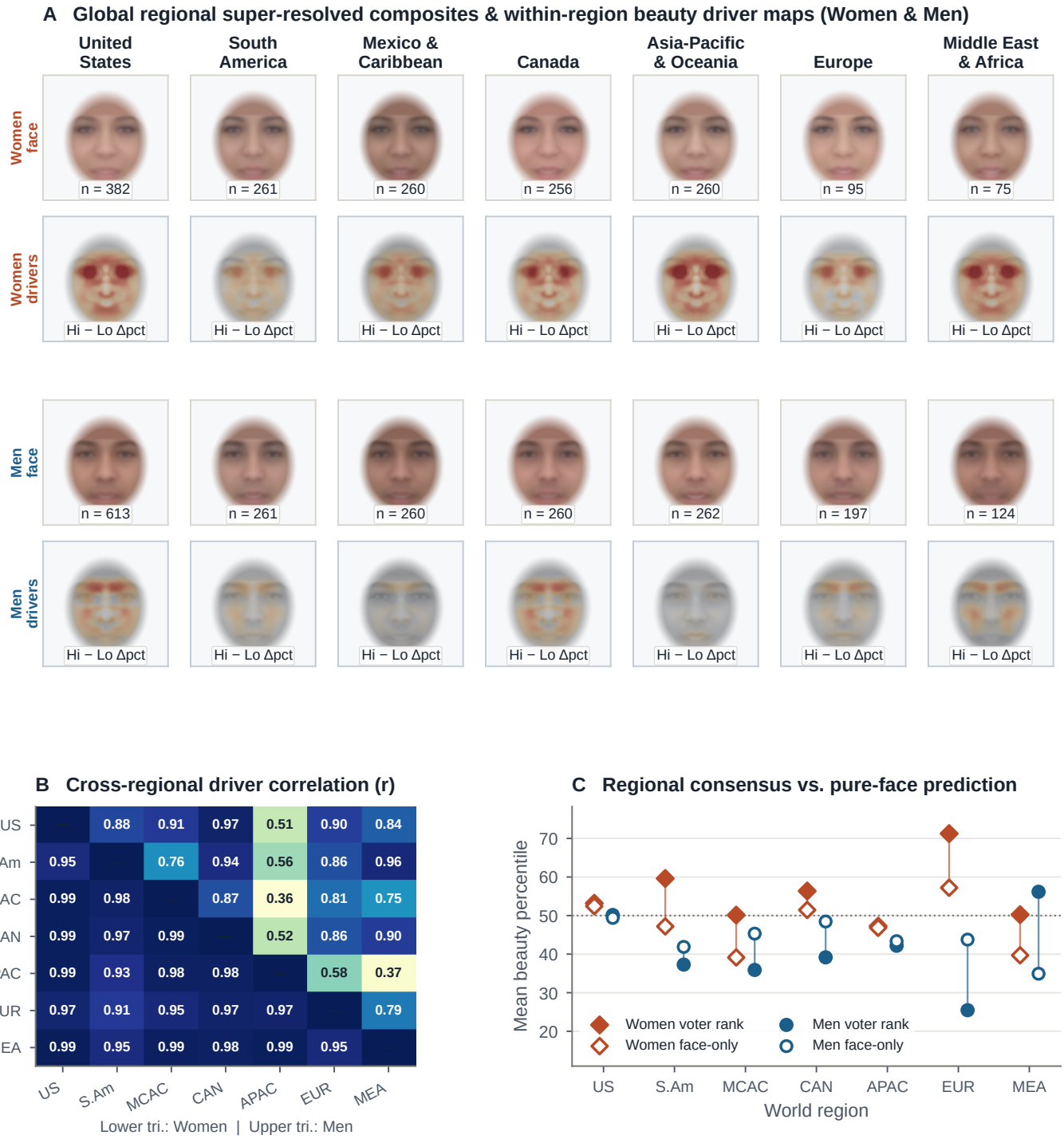


Fig. 14. **Cross-regional invariance of facial beauty drivers across 7 world regions. (A)** Super-resolved regional composites and within-region high-minus-low beauty driver maps across the United States, South America, Mexico/Central America/Caribbean, Canada, Asia-Pacific/Oceania, Europe, and Middle East/Africa (all cells $n \geq 75$). **(B)** Pairwise spatial correlation matrix of regional beauty driver maps (lower triangle: women, mean $r = 0.97$, min 0.91; upper triangle: men, mean $r = 0.76$, min 0.36). **(C)** Observed voter rank versus pure-face predicted rank across regions, separating upload self-selection gaps from invariant anatomical weights.

while postgraduate and "Very Ambitious" women pair lower current parenthood (55.0% vs. 63.9%) with higher forward child desire (31.6% vs. 19.7%) and partnered adults report +6.9 W / +7.9 M higher happiness (0–100, $N = 7{,}060$ [61]; Fig. 15F). Third, stratifying midlife re-entrants by marital history confirms Proposition 23(iv): at ages 40–49, divorced women reduce explicit formal marriage intent by -24.2% relative to never-married single women (5.6% vs. 7.4%) while raising serious pair-bond commitment demand to 80.8%, whereas widowed men—who lost a cooperative hearth without defection—exceed widowed women in explicit remarriage intent (9.9% vs. 8.7% at 40–49; 7.8% vs. 5.4% at 60–75; Fig. 9F).

Finally, applying Proposition 14 to 34,092 completed-fertility respondents (ages 45–79) in the General Social

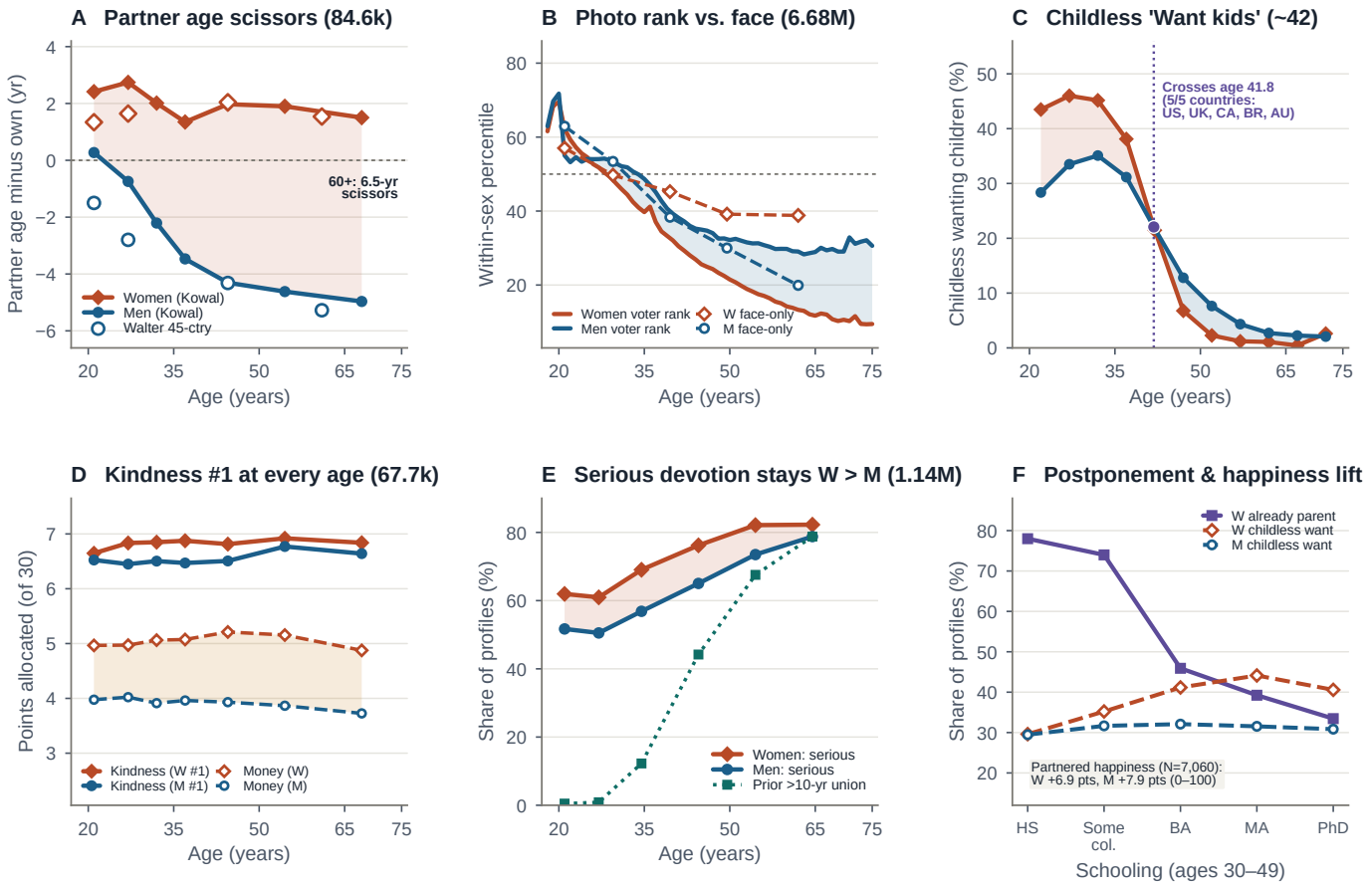


Fig. 15. **How men and women choose across adulthood, ages 18–75: diverging biological clocks (top row) versus unbroken relational truths (bottom row). (A)** Realized partner age-gap scissors (partner age minus own age) across 84,636 partnered adults (37,734 in Kowal et al. [8]; 46,902 in Walter et al. [81]), widening to a 6.5 yr sex divergence at ages 60–79. **(B)** Platform voter rank across 6,677,772 Hot-or-Not profiles versus zero-metadata pure facial anatomy (5,353 aligned faces), isolating steeper female age penalties beyond anatomical drift. **(C)** Explicit "Want children" shares among childless users across 1,137,309 Plenty of Fish profiles, crossing at age 41.8 (exact age 41; 25–29 W−M gap +12.5 pp vs. 45–49 −6.0 pp) and replicating across the U.S., U.K., Canada, Brazil, Australia, and open-science San Francisco `OkCupid-JSE` (age 38.2 yr [59]; Table V). **(D)** Constrained 30-point mate-budget trajectories across 67,733 adults [8]: kindness ranks #1 in *both* women and men in every age bracket from 18–24 to 60–79, outranking financial resources throughout life. **(E)** Serious relationship commitment stays strictly higher in women than men at every adult age (61.0% → 82.3% W vs. 50.5% → 78.8% M) alongside rising > 10-year prior union attainment, proving women's midlife shift reflects childbearing clocks rather than relational exit. **(F)** Completed schooling gradient (ages 30–49, POV) showing fertility postponement without loss of childless desire, alongside the partnered happiness lift (+6.9 points W, +7.9 points M on 0–100; $N = 7{,}060$ [61]).

TABLE V
CROSS-NATIONAL AND OPEN-SCIENCE REPLICATION OF THE CHILDLESS FERTILITY-INTENTION CROSSOVER VERSUS FEMALE RELATIONSHIP COMMITMENT (1,137,309 POV + 59,946 `OKCUPID-JSE` PROFILES).

	Eligible profiles		Childless "Want children" (W − M, pp)		Serious commitment (W − M, pp)	
Country or open-science sub-cohort	Women	Men	Ages 25–29	Ages 45–49	Ages 25–29	Ages 45–49
United States	246,343	559,342	+12.4	−5.4	+9.5	+9.1
United Kingdom	21,420	47,461	+16.7	−3.7	+11.1	+7.1
Canada	19,527	45,913	+16.3	−5.0	+8.8	+12.1
Brazil	34,110	65,346	+4.2	−13.3	+15.1	+16.4
Australia	9,447	24,735	+19.7	−5.7	+14.4	+15.5
San Francisco (`OkCupid-JSE`, open; 45–54 right)	6,817	9,480	+10.6	−20.0	Crossover age: 38.2 yr	

Survey (1972–2024 [11]; Fig. 16; Table VI) locates where selection acts today. Within parenthood ($c > 0$), family-size differentials (Number) are negative across schooling, objective `WORDSUM` vocabulary (g-proxy, −0.054 SD in men), occupational prestige, and income in both sexes. At the marriage pairing gate, however, men face a positive selection gradient across every cognitive and socioeconomic trait—personal income (+0.046 SD; Access +0.047 SD, childlessness falling from 21.9% to 10.5% across income fifths, net +0.013 SD), occupational prestige (+0.023 SD), schooling (+0.015 SD), and objective `WORDSUM` vocabulary (+0.011 SD, 95% CI [+0.001, + 0.022]; Fig. 16A,B)—whereas high-earning women experience higher childlessness (9.5% → 19.5%, Access −0.038 SD; Corollary 8). Across birth cohorts from 1900–1919 to 1960–1979, childlessness rose to account for 46% of men's total opportunity for selection even as non-marital parenthood

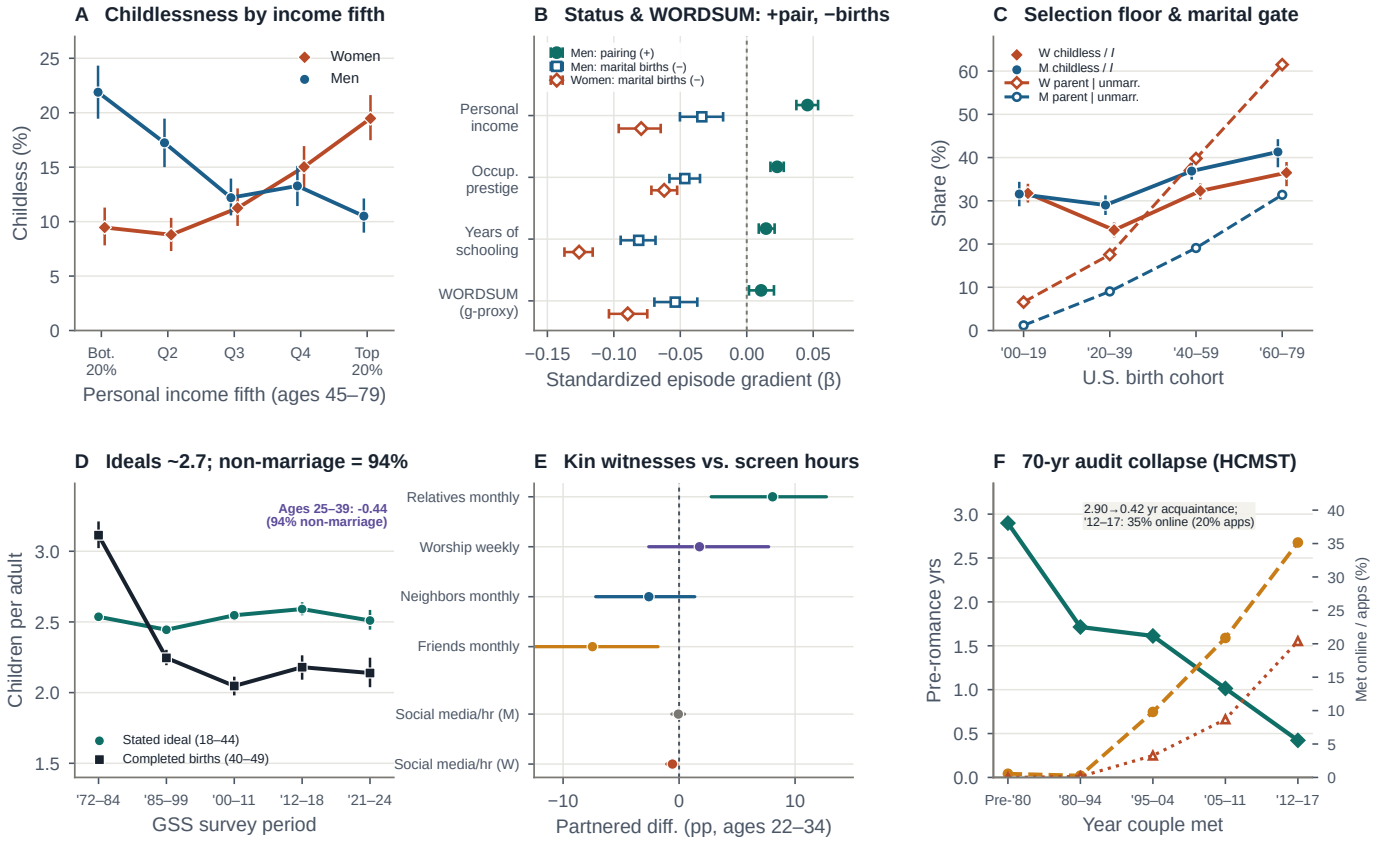


Fig. 16. **Completed-fertility selection floor, `WORDSUM`/status pairing-versus-births inversion, non-marriage shortfall, and 70-year courtship disintermediation (GSS 1972–2024, $n = 34{,}092$; Stanford `HCMST`, $N = 3{,}295$). (A)** Completed childlessness (ages 45–79) across personal-income fifths: low-income men face a sharp pairing floor (21.9% → 10.5% childless), whereas higher-income women experience elevated childlessness (9.5% → 19.5%). **(B)** Standardized episode gradients across personal income, occupational prestige, schooling, and objective `WORDSUM` vocabulary (*g*-proxy): every trait positively predicts male marriage pairing (+0.046, +0.023, +0.015, +0.011 SD) while predicting fewer marital births (−0.054 SD on `WORDSUM`). **(C)** Childlessness share of selection opportunity I (rising to 46% in men) alongside the non-marital parenthood gate. **(D)** Unchanged stated family-size ideals (∼2.7 children) versus the −0.44 drop in births per adult aged 25–39 (94% in non-marriage). **(E)** Partnered probability differences (ages 22–34) for monthly kin contact (+8.1 pp), weekly worship (+1.8 pp), neighbors (−2.6 pp), friends (−7.5 pp), and social-media hours across 70 countries (−0.1 to −0.6 pp/h). **(F)** Stanford `HCMST` 70-year courtship trajectory [10]: mean pre-romance acquaintance collapsed from 2.90 yr (pre-1980) to 0.42 yr (2012–2017) as online meeting rose to 35.2% (20.4% dating apps).

TABLE VI
ACCESS–NUMBER SELECTION DECOMPOSITION IN COMPLETED U.S. COHORTS (GSS 1972–2024, AGES 45–79; TRAIT SD PER GENERATION).

Trait	Sex	n	Access	Number	Total
Years of schooling	Men	15,080	−0.003 [−0.012, +0.005]	−0.081 [−0.094, −0.068]	−0.085 [−0.100, −0.069]
	Women	18,939	−0.038 [−0.044, −0.032]	−0.126 [−0.137, −0.115]	−0.164 [−0.177, −0.151]
Vocabulary (10 words)	Men	7,111	−0.011 [−0.024, +0.003]	−0.054 [−0.070, −0.036]	−0.065 [−0.086, −0.042]
	Women	9,076	−0.027 [−0.036, −0.019]	−0.089 [−0.104, −0.075]	−0.117 [−0.134, −0.099]
Occupational prestige	Men	14,707	+0.016 [+0.008, +0.024]	−0.046 [−0.057, −0.036]	−0.031 [−0.044, −0.017]
	Women	17,606	−0.019 [−0.026, −0.013]	−0.061 [−0.071, −0.051]	−0.081 [−0.093, −0.069]
Personal income (log)	Men	7,380	+0.047 [+0.036, +0.059]	−0.034 [−0.050, −0.017]	+0.013 [−0.008, +0.034]
	Women	7,239	−0.038 [−0.047, −0.028]	−0.078 [−0.094, −0.063]	−0.116 [−0.134, −0.098]

Proposition 14 exact partition: Total = Access + Number; 95% bootstrap CIs (2,000 resamples).

expanded (Fig. 16C; Appendix Table D2). Meanwhile, stated family-size ideals among adults aged 18–44 remained virtually unchanged from 1972 to 2024 (∼2.7 children), yet realized births among adults aged 25–39 fell by −0.44 children per adult, with 94% (−0.41) accounted for by rising non-marriage (16%→47%; Fig. 16D). Consistent with Proposition 19, seeing relatives monthly predicts +8.1 pp higher partnering among young adults (95% CI [+2.8, + 12.7]), whereas monthly friend socializing (−7.5 pp) and daily social-media hours across 70 countries (−0.1 to −0.6 pp/h) do not substitute for kin witnesses (Fig. 16E), paralleling the 70-year Stanford `HCMST` courtship shift where mean pre-romance acquaintance collapsed from 2.90 yr before 1980 to 0.42 yr in 2012–2017 as online meeting climbed to 35.2% (20.4% dating apps; Fig. 16F [10]).

VI. Discussion, AI Evaluation Parallels, Limitations, and Methods

A. Synthesis and Structural Parallel to AI Evaluation

Across fossil paleoclimate records, small-scale societies, laboratory encounters, and 7,996,528 digital profiles, a single structural principle recurs: *an evaluation architecture selects for general capacity and primary devotion only when candidates are observed across multiple common domains and unstaged everyday contexts*. When observation collapses to a single curated channel ($k = 1, d \to 1$), three distortions follow identically in human dating platforms and in artificial-intelligence benchmark leaderboards [65, 93–95]:

1) **Single-context saturation ($R^2_{\infty,1} < 1$):** Just as thousands of Hot-or-Not voters and our 43-wave `HoloFace-L1` ensemble leave 39.9%–48.3% of true human consensus unexplained because one static portrait omits cross-situational conduct (Proposition 20), static AI leaderboards saturate on memorized format cues without bounding out-of-distribution generalization.
2) **Secondary threshold gaming versus breadth dilution:** Just as single-variable height filters generate a $1.20\times$ heap at $6'0''$ on POV and a $+17.1\%$ $5'11'' \to 6'0''$ surge exclusively among straight men on `OkCupid` (-1.3% in gay men), optimizing models against narrow scalar reward proxies induces sycophancy and reward overoptimization [93, 94]; evaluating agents across K diverse, unscripted tasks dilutes domain-specific gaming as $1/K$ (Corollary 4).
3) **Capability tests versus unstaged alignment ($d \to 1$):** Announced benchmarks measure maximum capability ($R_b \to 1$) while carrying zero information about unmonitored default rates ($R_p \to 0$; Proposition 16). Bounding deployment reliability requires either unannounced behavioral auditing ($(1-d)n \gg 1$) or architectural execution mediation ($e \to 1$; Proposition 17) [95].

B. Scope Limits, Failure Regimes, and Falsifiers

Every inference in this monograph is subject to explicit boundaries. First, no ancestral heritability, selection gradient, or Paleolithic marriage parameter is estimated; contemporary survey, ethnographic, and platform datasets test only observable comparative statics of the measurement equations. Second, fossil regional series are not ancestor–descendant lineages, Phase III cranial sampling remains geographically uneven, the 930–813 ka coalescent bottleneck remains actively contested [19, 20, 37], and coincidence between 1.0–0.78 Ma cooking evidence and post-0.7 Ma encephalization is a consistency check rather than causal proof. Third, platform records (HON, POV, `OkCupid-JSE`) are convenience populations of online users rather than national probability censuses; in the 2021 privacy-revised `OkCupid-JSE` release [59], essay rows are randomly permuted relative to structured demographic rows and ages include bounded privacy jitter, so structured orientation/fertility tables and 10-prompt essay portfolios are analyzed strictly within their respective files without cross-file row linking; cross-sectional age, relationship-stage time-use effort, pre-cohabitation timing, retrospective breakup-initiation reports, and marital-status re-entry profiles combine behavioral responses with cohort and union-survival selection, and associations between kin contact, social-media browsing hours, perceived alternatives, and relationship outcomes are observational rather than experimental. Appendix D (Table D3) pre-commits fifteen discriminating empirical designs and explicit falsifying outcomes across paleoanthropology, juvenile metabolic phylogenetics, experimental partner choice, longitudinal dyadic diaries, behavioral genetics, and union-timing survival analysis.

C. Reproducible Pipeline and Privacy Safeguards

All statistical analyses, Monte Carlo simulations, paleoclimate synchronizations, out-of-fold `HoloFace-L1` ensemble solves, and vector figures execute deterministically on CPU from SHA-256-verified inputs (scripts/run.py `-study performance-before-outcomes`, scripts/validate.py `-source`, and scripts/paper.py `-study performance-before-outcomes`). The `LR04` paleoclimate stack (evidence/paleoclimate/lr04_benthic_stack.txt) is public domain; fossil compilations (Gingerich 2022, Will et al. 2021, Du et al. 2018) and modern open microdata (Mehr et al. 2019 `NHS`, Rosenfeld et al. 2019 `HCMST`, Kim & Escobedo-Land 2015 `OkCupid-JSE`, Whyte et al. 2021, Kowal et al. 2025, Joel et al. 2020, Lidborg et al. 2022, Fisman et al. 2006, `D-PLACE`/SCCS, GSS) are analyzed from immutable archival releases. On HON, POV, and `OkCupid-JSE`, minimum cell suppression ($n \geq 100$ in demographic and free-text tables; strict $n \geq 75$ individuals per super-resolved canonical Delaunay facial composite) ensures zero individual-portrait or free-text disclosure.

Acknowledgments and Data Availability

No external funding was received, and the authors declare no competing financial interests. Reproducible CPU analysis scripts, deterministic Monte Carlo simulations, SHA-256 reference manifests, and figure generators are maintained in the project repository (scripts/run.py, scripts/validate.py, and scripts/paper.py with `-study performance-before-outcomes`). Open datasets analyzed include the public-domain Lisiecki & Raymo `LR04` benthic δ^{18}O stack [4], the Gingerich [3], Will et al. [2], and Du et al. [1] hominin endocranial compilations, the Columbia speed-dating dataset [55], Harvard Music Lab `Natural History of Song` discography, naive-listener, and ethnographic corpora [26], Stanford `How Couples Meet and Stay Together` (`HCMST` 2017) public microdata [10], the San Francisco `OkCupid` teaching and research release (`OkCupid-JSE`, 2021 privacy-revised edition) [59], `D-PLACE` releases of the Standard Cross-Cultural Sample (including alloparenting and conjugal dissolution codings [71]), Ethnographic Atlas, and Binford hunter-gatherer files [52, 66, 67], Australian mate-choice microdata [61], 175-country mate-allocation, pair-bond, and triangular-love microdata [8], 43-dataset longitudinal couples machine-learning meta-analyses [9], cross-national survey archives [81], stature/strength meta-analytic tables [62], MHC meta-analytic summaries [85, 86], and the General Social Survey cumulative release (1972–2024) [11]. Platform statistical aggregates enforce cell suppression ($n \geq 100$) and super-resolved facial composites average $n \geq 75$ aligned portraits per cell with zero individual-image disclosure. Large-language-model tooling assisted code refactoring, literature auditing, and manuscript typesetting; every equation, simulation, and empirical table was verified programmatically against primary source files.

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TECHNICAL APPENDIX

Formal Models, Complete Proofs, Paleoclimatic Audit, and Platform Sensitivity

Appendix A: Twenty-Three Propositions, Corollaries, and Complete Proofs · Appendix B: Numerical Verification and the 1.5–0.7 Ma Plateau Audit · Appendix C: Digital-Platform, Stature, Facial-Attribution, and `HoloFace-L1` Audits · Appendix D: Survey Cohort Robustness and Pre-Committed Falsifiers

APPENDIX A
FORMAL MATHEMATICAL MODELS AND PROOFS

A. Setup and the Information–Opportunity Identity

Let $A \sim \mathcal{N}(\mu, V_A)$ denote additive breeding value for a transferable capacity and $C = A + E$ its phenotype, with independent environmental deviation $E \sim \mathcal{N}(0, V_E)$, phenotypic variance $V_P = V_A + V_E$, and heritability $h^2 = V_A/V_P$. Across n observed events, assessors form an averaged score

$$Y = \lambda C + U + \sigma\sqrt{\rho}\, Z_0 + \sigma\sqrt{(1-\rho)/n}\, Z_n, \tag{A1}$$

where $U \sim \mathcal{N}(0, \tau^2)$ is persistent nuisance (training or inherited opportunity), $\rho \in [0,1]$ is the shared error fraction, and $Z_0, Z_n \sim \mathcal{N}(0,1)$ are independent. Target phenotypic validity is

$$R = \mathrm{Corr}(C,Y)^2 = \frac{\lambda^2 V_P}{\lambda^2 V_P + \tau^2 + \sigma^2[\rho + (1-\rho)/n]}. \tag{A2}$$

Let $Z = (Y - \mathrm{E}\,Y)/\,\mathrm{SD}(Y)$ denote standardized *standing*. Under Mendelian transmission in the infinitesimal model [96], the single-generation response is evaluated via the Price equation [97].

Proposition 1 (Information–opportunity identity)**.** *Let within-sex reproductive allocation satisfy $w = W(Y) \geq 0$ with $\mathrm{E}\,W > 0$. If $\mathrm{E}[A \mid Y]$ is linear in Y (exact under joint Gaussianity), the selection differential on breeding value is*

$$S = \frac{\mathrm{Cov}(A,W)}{\mathrm{E}\,W} = \underbrace{\sigma_A}_{\text{genetic}}\ \underbrace{h\sqrt{R}}_{\text{accuracy}}\ \underbrace{\mathrm{Corr}(Y,W)}_{\text{alignment}}\ \underbrace{\mathrm{CV}(W)}_{\text{opportunity}}, \tag{A3}$$

where $h\sqrt{R} = \mathrm{Corr}(A,Y)$ and $\mathrm{CV}(W)^2$ is Crow's opportunity for selection [98]. Without linearity, $|S| \leq \mathrm{SD}(\mathrm{E}[A \mid Y])\ \mathrm{CV}(W)$.

B. Three Routes, Multi-Domain Portfolios, and the Generalist Target

Let individuals of sex s pair by reciprocal rank matching on Y, with expected reproductive contribution

$$w_s = \exp(\beta_{N,s} C + \beta_{D,s} C_p + \beta_{S,s} Z), \tag{A4}$$

separating returns to own capacity ($\beta_{N,s}$), partner's capacity ($\beta_{D,s}$), and standardized social standing ($\beta_{S,s}$).

Proposition 2 (Three-route decomposition)**.** *Under joint Gaussianity and ideal rank matching on Y, the within-sex selection differential is*

$$S_s = V_A\left[\beta_{N,s} + \beta_{D,s} R + \beta_{S,s}\sqrt{R/V_P}\right], \tag{A5}$$

and the offspring response is $\Delta\bar{A} = \frac{1}{2}(S_f + S_m)$.

Corollary 1 (Information bottleneck)**.** *If $R = 0$, choice generates zero selection on capacity ($\beta_D R = \beta_S\sqrt{R/V_P} = 0$) however strong partner preferences are.*

Corollary 2 (Selection follows the score)**.** *If Y also includes an independent heritable display D with additive variance $V_{A,D}$, standing selection shifts display by $\beta_S V_{A,D}/\,\mathrm{SD}(Y)$ and capacity by $\beta_S \lambda V_A/\,\mathrm{SD}(Y)$; display variance dominating Y steers evolution toward presentation rather than capacity.*

Now let performance in domain $k \in \{1,\ldots,K\}$ be $T_k = G + S_k$, where $G \sim \mathcal{N}(0, V_G)$ is general capacity and $S_k \sim \mathcal{N}(0, V_S)$ are independent domain-specific skills. Pooling n events per domain with domain nuisance $U_k \sim \mathcal{N}(0,\tau^2)$ and cross-domain nuisance $\Omega \sim \mathcal{N}(0,\omega^2)$ yields

$$Y = \frac{1}{K}\sum_{k=1}^{K}\left(T_k + U_k + \bar{e}_k\right) + \Omega. \tag{A6}$$

Proposition 3 (Portfolio validity)**.** *Validity for general capacity is*

$$R_G = \frac{V_G}{V_G + \left(V_S + \tau^2 + \sigma^2[\rho + (1-\rho)/n]\right)/K + \omega^2}. \tag{A7}$$

Single-domain repetition ($K = 1, n \to \infty$) stalls at $V_G/(V_G + V_S + \tau^2 + \rho\sigma^2 + \omega^2)$, whereas cross-domain breadth ($K \to \infty$) raises the ceiling to $V_G/(V_G + \omega^2)$.

Proposition 4 (Diversity concentrates selection on general capacity)**.** *Under standing selection $w = \exp(\beta_S Z)$ with heritabilities h^2_G, h^2_S, h^2_Ω, breeding-value differentials satisfy*

$$S_{A_G} = \frac{\beta_S h^2_G V_G}{\mathrm{SD}(Y)}, \quad S_{A_{S_k}} = \frac{\beta_S h^2_S V_S}{K\ \mathrm{SD}(Y)}, \quad S_{A_\Omega} = \frac{\beta_S h^2_\Omega \omega^2}{\mathrm{SD}(Y)}. \tag{A8}$$

Corollary 3 (Generalist aim)**.** *Writing $V_{A,G} = h^2_G V_G$ and $V_{A,S} = h^2_S V_S$, common comparison across K domains yields*

$$\frac{S_{A_G}}{S_{A_{S_k}}} = K\,\frac{V_{A,G}}{V_{A,S}}. \tag{A9}$$

Proposition 5 (Breadth buys time)**.** *For target validity $R^* < V_G/(V_G + \omega^2)$, let $D = V_G(1/R^* - 1) - \omega^2$ and $Q = V_S + \tau^2 + \rho\sigma^2$. Reaching R^* requires*

$$n^*(K) = \frac{(1-\rho)\sigma^2}{KD - Q} \quad \text{for } K > K_{\min} = \frac{Q}{D}, \tag{A10}$$

and is impossible for $K \leq K_{\min}$ regardless of observation duration.

Proposition 6 (Expenditure follows breadth)**.** *Let heritable general investment g and domain-specific investments s_k enter $T_k = G + g + S_k + s_k$ at log-fitness costs $\frac{1}{2}c_G g^2$ and*

$\frac{1}{2}c_S s_k^2$ *under* $w \propto \exp(\beta_S Z)$. *Equilibrium investments and the general-capacity cost share are*

$$g^* = \frac{\beta_S}{c_G\ \mathrm{SD}(Y)}, \quad s_k^* = \frac{\beta_S}{K\, c_S\ \mathrm{SD}(Y)}, \quad \pi_G = \frac{K c_S}{K c_S + c_G}, \tag{A11}$$

passing $\pi_G = \frac{1}{2}$ *at* $K^\dagger = c_G/c_S$.

Corollary 4 (Breadth taxes exaggeration)**.** *If domain-specific showmanship* f_k *inflates perceived performance in domain* k *at cost* $\frac{1}{2}c_F f_k^2$ *without real benefit, its equilibrium is* $f_k^* = \beta_S/(K\, c_F\ \mathrm{SD}(Y))$*, total expenditure on exaggeration falls as* $1/K$*, and selection on heritable domain-specific exaggeration is diluted* K*-fold relative to* G*.*

Proposition 7 (Division of labor restores the specialist)**.** *When each candidate performs and is evaluated solely in their best specialty* $k^* = \arg\max_k S_k$*, within-specialty covariances satisfy*

$$\frac{\mathrm{Cov}(A_G,Y)}{\mathrm{Cov}(A_S,Y)} = \frac{V_{A,G}}{c_K\, V_{A,S}}, \tag{A12}$$

where $c_K = \mathrm{Var}(\max_{1\le k\le K} Z_k)$ *for i.i.d. standard normals (*$c_2 = 0.68$,$c_5 = 0.45$,$c_{10} = 0.34$,$c_{20} = 0.28$*), weakening generalist aim by* Kc_K *relative to Eq.* (A9)*.*

C. Assortment, Positional Persistence, and Energetic Co-Limitation

Proposition 8 (Assortment inflates additive variance)**.** *With phenotypic partner correlation* m *and constant genic variance* V_{LE}*, additive variance evolves as*

$$V_{A,t+1} = \tfrac{1}{2}V_{A,t}(1 + m h_t^2) + \tfrac{1}{2}V_{LE}, \tag{A13}$$

with unique positive equilibrium V_A^* *solving* $(1-m)V^2 + (V_E - V_{LE})V - V_{LE}V_E = 0$ *for* $m < 1$*, amplifying long-run pair-output response over random mating by*

$$F(m) = \frac{V_A^*}{V_{LE}}(1+m). \tag{A14}$$

Let capacity carry quadratic metabolic and developmental cost $c > 0$:

$$w = \exp\Big(\beta_N C + \beta_S Z - \frac{c}{2}C^2\Big). \tag{A15}$$

Proposition 9 (Positional persistence)**.** *Under Gaussianity, the exact single-generation response and evolutionary equilibrium are*

$$\Delta\bar{A} = \frac{V_A(\beta_N + \beta_S\sqrt{R/V_P} - c\mu)}{1 + cV_P}, \qquad \mu^* = \frac{\beta_N + \beta_S\sqrt{R/V_P}}{c}. \tag{A16}$$

Corollary 5 (Positional surplus)**.** *Because* $\mathrm{E}[\beta_S Z] = 0$*,* μ^* *exceeds the mean-log-fitness optimum* β_N/c *by* $\beta_S\sqrt{R/V_P}/c$*.*

Decompose capacity into costly neural tissue B (endocranial volume) and tissue-independent organization K, $C = aB + K + E$ ($a \ge 0$), with additive covariance $\mathbf{G}$ and log fitness

$$\ln w = \beta_{\mathrm{eff}}C - \frac{\kappa_B}{2}B^2 - \frac{\kappa_K}{2}K^2, \qquad \beta_{\mathrm{eff}} = \beta_N + \beta_S\sqrt{R/V_P}. \tag{A17}$$

Proposition 10 (Tissue tracks demand, information, and budget)**.** *Under Lande's multivariate equation* $\Delta\bar{\mathbf{z}} = \mathbf{G}\boldsymbol{\beta}$ *[99] with nonsingular* $\mathbf{G}$*, the unique equilibrium is*

$$B^* = \frac{a\beta_{\mathrm{eff}}}{\kappa_B}, \qquad K^* = \frac{\beta_{\mathrm{eff}}}{\kappa_K}, \qquad \frac{B^*}{K^*} = \frac{a\kappa_K}{\kappa_B}. \tag{A18}$$

If $a = 0$*,* $\bar{B}^* \to 0$ *regardless of genetic correlation between* B *and* C*.*

Corollary 6 (Information and energy co-limit tissue)**.** *Selection enlarges tissue above* $\bar{B}$ *if and only if*

$$\sqrt{R/V_P} > \frac{\kappa_B\bar{B}/a - \beta_N}{\beta_S}, \tag{A19}$$

capping equilibrium volume at $B_{\max} = a(\beta_N + \beta_S\sqrt{R_{\max}/V_P})/\kappa_B$ *until energetic innovations (such as cooking lowering* κ_B*) or broader assessment (raising* $R_{\max}$*) relax the bound.*

D. Contest Aim, Safe Repetition, Leveling, and Episodes

Let physical formidability F and general capacity G load on composite standing across venues k as

$$Y = \sum_k \omega_k(\ell_{Fk}F + \ell_{Gk}G + e_k), \qquad \bar{\ell}_X = \sum_k \omega_k \ell_{Xk}. \tag{A20}$$

Proposition 11 (The score sets the aim)**.** *Under joint Gaussianity, for any allocation* $W(Y)$ *with intensity* $i = \mathrm{Corr}(Y,W)\,\mathrm{CV}(W)$*,*

$$S_{A_F} = \frac{i\, h_F^2 \bar{\ell}_F V_F}{\mathrm{SD}(Y)}, \qquad S_{A_G} = \frac{i\, h_G^2 \bar{\ell}_G V_G}{\mathrm{SD}(Y)}, \tag{A21}$$

so $S_{A_G}/S_{A_F} = (\bar{\ell}_G h_G^2 V_G)/(\bar{\ell}_F h_F^2 V_F)$ *invariant to selection intensity or noise.*

Corollary 7 (A cue to a life stage is not a target)**.** *If a preferred cue* $Q = T + A_Q + e$ *combines transient life stage* T *(such as youth) with heritable* $A_Q \perp G$*, then* $S_{A_Q} = i\, V_{A_Q}/\,\mathrm{SD}(Q)$ *and* $S_{A_G} = 0$*.*

Proposition 12 (Safe contests inform more)**.** *With* $R(n)$ *in Eq.* (A2) *and per-bout risk* $\delta > 0$*, net assessment value* $B\sqrt{R(n)} - \delta n$ *has a unique maximizer* n^* *strictly decreasing in* δ/B*, raising optimal validity* $R(n^*)$ *as contests become non-lethal.*

For coalitional leveling, let a dominant male tolerate k subordinates at reproductive price q each against external coalitions of strength M, prevailing with Lanchester contest probability

$$p(k) = \frac{(1+k)^\theta}{(1+k)^\theta + M^\theta}, \qquad U(k) = (1 - kq)\, p(k). \tag{A22}$$

Proposition 13 (The despot's dilemma)**.** $\ln U(k)$ *is strictly concave on* $[0,1/q)$ *with unique optimum* k^**: (i)* $k^* > 0 \iff \theta M^\theta/(1+M^\theta) > q$*; (ii)* k^* *increases in* M *and decreases in* q*; (iii) as* $M \to \infty$*,* $k^* \to (\theta - q)/[q(1+\theta)]$ *and the dominant's share falls to* $(1+q)/(1+\theta)$*.*

Proposition 14 (Access and number)**.** *For completed family size $c \geq 0$, parenthood rate $p = \Pr(c > 0)$, relative fitness $w = c/\operatorname{E} c$, and mean-zero trait z,*

$$\begin{aligned} S = \operatorname{Cov}(z,w) &= \underbrace{\operatorname{E}[z \mid c > 0]}_{\textit{access}} + \underbrace{\frac{\operatorname{Cov}(z,c \mid c > 0)}{\operatorname{E}[c \mid c > 0]}}_{\textit{number}}, \\ I = \operatorname{Var}(w) &= \frac{1-p}{p} + \frac{I_{c>0}}{p}. \end{aligned} \tag{A23}$$

Corollary 8 (A floor is enough)**.** *Excluding the lowest-scoring fraction q ($z_q = \Phi^{-1}(q)$) with equal reproduction among the remaining $1-q$ yields*

$$S_A = \sigma_A\, h\sqrt{R}\,\frac{\varphi(z_q)}{1-q}, \qquad I = \frac{q}{1-q}. \tag{A24}$$

E. Delivery, Staged Tests, Enforcement, Ancestors, and Two-Way Juries

Let candidate value be $v = pb + c - h$, where $b = \bar{b} + B$ is productive capacity and $p = \bar{p} + P \in [0,1]$ is reliable delivery to partner and offspring, assessed with validities R_b,R_p under $w = 1 + s(\hat{v} - \operatorname{E}\hat{v})$.

Proposition 15 (Capacity pays in proportion to delivery)**.** *Breeding-value differentials are*

$$S_{A_B} = s\,\bar{p}\,R_b\,V_{A,B}, \qquad S_{A_P} = s\,\bar{b}\,R_p\,V_{A,P}. \tag{A25}$$

Under quadratic costs $\frac{1}{2}c_B\bar{b}^2$ and $\frac{1}{2}c_P\bar{p}^2$, $(\bar{b},\bar{p}) = (0,0)$ is unstable if and only if $s^2R_bR_p > c_Bc_P$.

Proposition 16 (Staged tests measure capacity; unstaged observation measures delivery)**.** *Across n occasions with staged fraction $d \in (0,1)$ and unstaged default rate $q = 1-p$ ($\nu = \operatorname{E}[q(1-q)]$),*

$$R_b = \frac{V_B}{V_B + \sigma^2/(dn)}, \qquad R_p = \frac{V_P}{V_P + \nu/[(1-d)n]}, \tag{A26}$$

a candidate with default rate q keeps a clean unstaged record with probability $(1-q)^{(1-d)n}$ (95% upper bound $\approx 3/[(1-d)n]$), and R_bR_p is maximized at

$$\begin{aligned} d^* &= \frac{\sqrt{A(1+C)}}{\sqrt{A(1+C)} + \sqrt{C(1+A)}}, \\ A &= \frac{\sigma^2}{nV_B}, \qquad C = \frac{\nu}{nV_P}. \end{aligned} \tag{A27}$$

Proposition 17 (Enforcement substitutes for verification)**.** *When third parties enforce delivery of share $e \in [0,1]$ ($\delta = e + (1-e)p$),*

$$\begin{aligned} S_{A_B} &= s\,[e + (1-e)\bar{p}]\,R_b\,V_{A,B}, \\ S_{A_P} &= s\,(1-e)\,\bar{b}\,R_p\,V_{A,P}. \end{aligned} \tag{A28}$$

Writing $T = s^2R_bR_p/(c_Bc_P)$, any $e > 0$ eliminates the zero trap; for $T(1-e) < 1$, the unique stable interior equilibrium is

$$\bar{b}^* = \frac{sR_b}{c_B}\cdot\frac{e}{1 - T(1-e)^2}, \qquad \bar{p}^* = \frac{T\,e\,(1-e)}{1 - T(1-e)^2}, \tag{A29}$$

with $\bar{p}^*$ maximized at $e^* = 1 - (1-\sqrt{1-T})/T$ and crowded out as $e \to 1$.

If a fraction κ of matings is settled by coercion rather than assessment, relative fitness is a mixture $w = (1-F)w_a + Fw_f$ ($\operatorname{E} F = \kappa$), giving

$$S_A = (1-\kappa)\operatorname{Cov}(A,w_a) + \kappa\operatorname{Cov}(A,w_f). \tag{A30}$$

Proposition 18 (Who becomes an ancestor)**.** *If share q of a sex leaves zero offspring and parents have mean m and variance $m + \alpha m^2$, variance effective population size satisfies $N_e/N = (1-q)/(1+\alpha)$, independent of m, and for equal sex counts $N_{e,f}/N_{e,m} = [(1-q_f)/(1-q_m)][(1+\alpha_m)/(1+\alpha_f)]$.*

For m observers with single-observer reliability r_1 and shared rater error ρ_e, pooled Spearman–Brown agreement and its shared-error ceiling are

$$\begin{aligned} R_m &= \frac{mr_1}{1 + (m-1)r_1}, \\ R_m^{(\rho_e)} &= \frac{mr_1}{1 + (m-1)(r_1 + \rho_e)} \xrightarrow{m\to\infty} \frac{r_1}{r_1 + \rho_e}. \end{aligned} \tag{A31}$$

Proposition 19 (Thin evidence delays commitment)**.** *Reaching delivery reliability R^* when k witnesses each observe λ occasions per period (staged share d, single-occasion reliability r_1) requires waiting time*

$$T = \frac{R^*(1-r_1)}{r_1(1-R^*)}\cdot\frac{1}{k\,\lambda\,(1-d)}. \tag{A32}$$

Corollary 9 (Safe contests buy reliability by repetition)**.** *With dangerous-trial reliability ρ_d and safe-trial reliability $\rho_s < \rho_d$, n safe trials match capacity reliability iff $n \geq \rho_d(1-\rho_s)/[\rho_s(1-\rho_d)]$, while delivery validity remains zero if staged ($d = 1$).*

Corollary 10 (A narrow test makes a deficit of a difference)**.** *A trait x with domain loadings $\sum_{k=1}^{K} a_k = 0$ is neutral under full K-domain comparison but penalized in proportion to $|\bar{a}_T|$ when evaluated on a narrow subset $T \subset \{1,\ldots,K\}$ where $\bar{a}_T < 0$.*

Proposition 20 (Two-way observer and context validity)**.** *Let candidate i's long-run trait be $T_i \sim (0,\sigma_T^2)$ and observer $j \in \{1,\ldots,m\}$ in context $c \in \{1,\ldots,k\}$ record*

$$X_{ijc} = T_i + H_i + C_{ic} + O_{ij} + \varepsilon_{ijc}, \tag{A33}$$

with mutually orthogonal target halo $H_i \sim (0,\sigma_H^2)$, person-by-context state $C_{ic} \sim (0,\sigma_C^2)$, observer idiosyncrasy $O_{ij} \sim (0,\sigma_O^2)$, and residual $\varepsilon_{ijc} \sim (0,\sigma_\varepsilon^2)$. For the two-way jury mean $\bar{X}_{i,m,k} = (mk)^{-1}\sum_{j,c} X_{ijc}$,

$$R^2_{m,k} = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_H^2 + \sigma_C^2/k + \sigma_O^2/m + \sigma_\varepsilon^2/(mk)}. \tag{A34}$$

In a single context ($k = 1$), adding arbitrarily many voters ($m \to \infty$) stalls at the static-presentation ceiling

$$R^2_{\infty,1} = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_H^2 + \sigma_C^2} < 1, \tag{A35}$$

whereas multi-witness, multi-context observation ($m,k \to \infty$) eliminates both σ_O^2/m and σ_C^2/k, reaching $\sigma_T^2/(\sigma_T^2 + \sigma_H^2)$.

Proposition 21 (Entry filtering versus pair-bond welfare)**.** *Let candidate j carry an entry-legible vertical attribute*

$X_j \sim \mathcal{N}(0,1)$ *with encounter reliability* $\rho_X \in (0,1]$ *and a pair-bond character trait* $K_j \sim \mathcal{N}(0,1)$ *(kindness, responsiveness, unstaged devotion) observed at entry with low reliability* $\rho_{K,0} \ll \rho_X$ *via noisy signals* $\hat{X}_{j,0} = \sqrt{\rho_X} X_j + \sqrt{1-\rho_X}\eta_j$ *and* $\hat{K}_{j,0} = \sqrt{\rho_{K,0}} K_j + \sqrt{1-\rho_{K,0}}\xi_j$*, while pair* (i,j) *realizes unique dyadic chemistry* $D_{ij} \sim \mathcal{N}(0,\sigma_D^2)$ *orthogonal to* (X_j,K_j) *and observed only upon live interaction. Suppose true long-run pair-bond welfare is*

$$U_{ij} = \alpha_X X_j + \alpha_K K_j + \alpha_D D_{ij}, \qquad \alpha_K,\alpha_D \gg |\alpha_X| \geq 0. \tag{A36}$$

(i) Stage-weight reversal. *Optimal Bayesian entry screening on* $(\hat{X}_{j,0},\hat{K}_{j,0})$ *prior to live interaction places standardized weight* $w_{X,0} = \alpha_X\sqrt{\rho_X}$ *versus* $w_{K,0} = \alpha_K\sqrt{\rho_{K,0}}$*, so entry filtering weights* X_j *above* K_j *whenever* $\rho_{K,0}/\rho_X < (\alpha_X/\alpha_K)^2$*, whereas inside established unions (repeated joint life* $\rho_{K,T} \to 1$*) realized bond quality has standardized slopes* $\beta_{U,K} = \alpha_K \gg \beta_{U,X} = \alpha_X$*.*

(ii) Consensus filtering loss. *Pre-filtering to the top* $q \in (0,1)$ *quantile on consensus* $\hat{X}_{j,0}$ *alone captures expected welfare gain* $\alpha_X\sqrt{\rho_X}\,\varphi(\Phi^{-1}(1-q))/q$ *while forfeiting all unscreened dyadic and character surplus* $(\alpha_K^2+\alpha_D^2\sigma_D^2)^{1/2}$ *available across excluded pairs.*

(iii) Perceived outside options and commitment. *Under investment-commitment dynamics* $C_{ij,t} = U_{ij} + I_{ij,t} - \gamma\widehat{O}_{i,t}$ *(*$\gamma > 0$*), an exogenous platform shock* $\Delta\widehat{O}_{i,t} > 0$ *that inflates perceived consensus alternatives without raising realized post-entry* $U_{ij'}$ *strictly lowers commitment (*$\partial C_{ij,t}/\partial\widehat{O}_{i,t} = -\gamma < 0$*) and raises dissolution/defection hazard* $\Pr(C_{ij,t} < \theta)$*.*

Proposition 22 (Childhood metabolic bottleneck and pair-bond encephalization ceiling)**.** *Suppose adult endocranial volume* $B > B_0$ *imposes peak juvenile brain metabolic expenditure* $\mathcal{C}_{\text{juv}}(B) = \kappa_1(B-B_0)^2$ *(*$\kappa_1 > 0$*) across childhood dependence* T_J*, funded by maternal surplus* $M_{\text{mom}} > 0$*, kin/allomaternal transfers* $\Delta_{\text{allo}} \geq 0$*, and biparental provisioning* $\Delta_{\text{pair}} \geq 0$ *delivered whenever a partner with baseline stay utility* u_0 *and emotional pair-bond commitment* $L \geq 0$ *resists juvenile-period outside temptation* $\widehat{O} \sim \mathcal{N}(\mu_O,\sigma_O^2)$ *(*$u_0 + L \geq \widehat{O}$*), so* $p_{\text{bond}}(L) = \Phi(z_L)$ *with* $z_L = (u_0 + L - \mu_O)/\sigma_O$ *and* $M_{\text{tot}}(L) = M_{\text{mom}} + \Delta_{\text{allo}} + p_{\text{bond}}(L)\Delta_{\text{pair}}$*. Let juvenile feasibility cap* $B_{\text{cap}}(L,\kappa_1) = B_0 + \sqrt{M_{\text{tot}}(L)/\kappa_1}$ *anchor adult log-fitness* $\ln w(B) = \beta_S B - \frac{\lambda_V}{2}\big(B - B_{\text{cap}}(L,\kappa_1)\big)^2$ *(*$\beta_S,\lambda_V > 0$*). Then the unique evolutionary Lande equilibrium is*

$$B^*(L,\kappa_1) = B_0 + \sqrt{\frac{M_{\text{mom}} + \Delta_{\text{allo}} + \Phi(z_L)\Delta_{\text{pair}}}{\kappa_1}} + \frac{\beta_S}{\lambda_V}. \tag{A37}$$

For $\Delta_{\text{pair}} > 0$*, equilibrium brain volume increases strictly in pair-bond commitment (*$\partial B^*/\partial L > 0$*) and exhibits strict supermodular complementarity between dietary/cooking efficiency* $\tau = \kappa_1^{-1/2}$ *and commitment love* L*:*

$$\frac{\partial^2 B^*}{\partial\tau\,\partial L} = \frac{\Delta_{\text{pair}}\,\varphi(z_L)}{2\,\sigma_O\,\sqrt{M_{\text{tot}}(L)}} > 0. \tag{A38}$$

Proposition 23 (Lifecycle staging and marital exit)**.** *Consider courtship, union lock-in, dissolution, and remarriage across stages:*

(i) Courtship staging drop. *If daily secondary display effort* $e \geq 0$ *yields market return* $\gamma_{\text{mkt}}(1-\lambda_{\text{stage}})e - \frac{c_e}{2}e^2$ *with courtship* $\lambda_{\text{date}} = 0$ *and marital commitment shield* $\lambda_{\text{mar}} \in (0,1)$*, optimal effort falls from* $e^*_{\text{date}} = \gamma_{\text{mkt}}/c_e$ *to* $e^*_{\text{mar}} = (1-\lambda_{\text{mar}})e^*_{\text{date}}$ *(fractional drop* $-\lambda_{\text{mar}}$*).*

(ii) Pre-cohabitation rush hazard. *Suppose latent primary reliability* $p \sim \mathcal{N}(\mu_p,\sigma_p^2)$ *is audited across* $m > 0$ *unstaged periods via* $\bar{Y}_m = p + \bar{\varepsilon}_m$ *(*$\bar{\varepsilon}_m \sim \mathcal{N}(0,\sigma_\varepsilon^2/m)$*), with posterior mean* $\hat{p}_m \sim \mathcal{N}(\mu_p,\sigma_p^2-\tau_m^2)$ *and residual variance* $\tau_m^2 = (\sigma_p^{-2} + m\sigma_\varepsilon^{-2})^{-1}$*. Entering lock-in when* $\hat{p}_m \geq \theta_{\text{in}}$ *yields post-entry failure rate* $H(m) = \Pr(p < \theta_{\text{stay}} \mid \hat{p}_m \geq \theta_{\text{in}})$*, which decreases strictly with* m *whenever* $\theta_{\text{in}} \geq \mu_p > \theta_{\text{stay}}$*.*

(iii) Marriage-specific unilateral exit flip. *Let spouses* $s \in \{f,m\}$ *draw independent baseline continuation utilities* $U_{s,0} \sim \mathcal{N}(\mu_u,\sigma_u^2)$ *(*$\mu_u > 0$*). Without lock-in or defection (*$q = 0$*), unilateral exit* $\Pr(U_f < 0 < U_m \mid U_f < 0 \oplus U_m < 0) = \frac{1}{2}$*. Under marital lock-in where unscreened primary defection occurs with probability* $q \in (0,1)$ *and imposes asymmetric domestic/care loss* $\delta_f > \delta_m > 0$ *(*$U_s = U_{s,0} - D\delta_s$*,* $D \sim \text{Bernoulli}(q)$*) with* $\mu_u \geq \frac{1}{2}\delta_f$*, women's share of unilateral divorces strictly exceeds* $\frac{1}{2}$*.*

(iv) Second-act re-entry divergence. *Let formal marriage demand be* $P_{s,h} = \Phi(z_{s,h})$ *for sex* s *and marital history* $h \in \{\text{sin},\text{div},\text{wid}\}$*. Updating legal lock-in cost after divorce (*$z_{f,\text{div}} = z_{f,\text{sin}} - \kappa_{\text{lock}}$*,* $\kappa_{\text{lock}} > 0$*) lowers* $P_{f,\text{div}} < P_{f,\text{sin}}$ *without lowering intrinsic pair-bond valuation, whereas losing a cooperative hearth via widowhood raises male replacement demand above female (*$P_{m,\text{wid}} > P_{f,\text{wid}}$*) whenever male solitary domestic loss exceeds female (*$z_{m,\text{wid}} > z_{f,\text{wid}}$*).*

F. Complete Proofs

Gaussian tilting lemma. For jointly Gaussian $(A,\mathbf{X})$ and $w = \exp(\mathbf{h}^\top\mathbf{X})$, differentiating the joint moment-generating function gives $\mathrm{E}[Aw]/\mathrm{E}[w] = \mathrm{E}[A] + \mathrm{Cov}(A,\mathbf{h}^\top\mathbf{X})$. For $x \sim \mathcal{N}(0,V)$ and $w = \exp(kx - \frac{c}{2}x^2)$, completing the square yields tilted law $\mathcal{N}(kV/(1+cV),V/(1+cV))$.

Proposition 1. Write $\mathrm{E}[A \mid Y] = \mu + \kappa(Y - \mathrm{E}\,Y)$ with $\kappa = \mathrm{Cov}(A,Y)/V_Y$. Since $A - \mathrm{E}[A \mid Y] \perp W(Y)$, $\mathrm{Cov}(A,W) = \kappa\,\mathrm{SD}(Y)\,\mathrm{Corr}(Y,W)\,\mathrm{SD}(W) = \sigma_A\,\mathrm{Corr}(A,Y)\,\mathrm{Corr}(Y,W)\,\mathrm{SD}(W)$. Under Eq. (A1), $\mathrm{Corr}(A,Y) = \lambda V_A/(\sigma_A\,\mathrm{SD}(Y)) = h\sqrt{R}$; dividing by $\mathrm{E}\,W$ gives Eq. (A3). Without linearity, Cauchy–Schwarz on $\mathrm{Cov}(\mathrm{E}[A \mid Y],W)$ gives the bound. □

Proposition 2 and Corollaries 1–2. Ideal rank matching equates standardized partner scores ($Y_p = Y$, $Z_p = Z$), so $C_p = \mu + (\lambda V_P/V_Y)(Y - \mathrm{E}\,Y) + \varepsilon$ with ε independent of focal variables. By the tilting lemma, $S_s = \mathrm{Cov}(A,\beta_{N,s}C + \beta_{D,s}C_p + \beta_{S,s}Z)$ with $\mathrm{Cov}(A,C) = V_A$, $\mathrm{Cov}(A,C_p) = \lambda^2 V_P V_A/V_Y = V_A R$, and $\mathrm{Cov}(A,Z) = \lambda V_A/\,\mathrm{SD}(Y) = V_A\sqrt{R/V_P}$. Setting $R = 0$ gives Corollary 1; adding independent D gives $\mathrm{Cov}(A_D,Z) = V_{A,D}/\,\mathrm{SD}(Y)$ (Corollary 2). □

Propositions 3–7 and Corollaries 3, 4. In Eq. (A6), $\mathrm{Cov}(G,Y) = V_G$ and $\mathrm{Var}(Y) = V_G + (V_S + \tau^2 + \sigma^2[\rho + (1-\rho)/n])/K + \omega^2$, yielding Eq. (A7). Writing $A_X = h_X^2 X + \eta_X$ ($\eta_X \perp Y$) gives $\mathrm{Cov}(A_G,Z) = h_G^2 V_G/\,\mathrm{SD}(Y)$, $\mathrm{Cov}(A_{S_k},Z) = h_S^2 V_S/(K\,\mathrm{SD}(Y))$, and $\mathrm{Cov}(A_\Omega,Z) = h_\Omega^2\omega^2/\,\mathrm{SD}(Y)$ (Proposition 4); dividing the first two gives Eq. (A9). Setting $R_G = R^*$ and solving for n yields Eq. (A10) for $KD > Q$ (Proposition 5). In Proposition 6, $\partial Y/\partial g = 1$ and $\partial Y/\partial s_k = 1/K$; setting Lande gradients $\beta_S/\,\mathrm{SD}(Y) - c_G g = 0$ and $\beta_S/(K\,\mathrm{SD}(Y)) - c_S s_k = 0$ yields Eq. (A11), with cost ratio $[\frac{1}{2}c_G g^{*2}]/[K\frac{1}{2}c_S s_k^{*2}] = Kc_S/c_G$. Domain-specific exaggeration f_k enters identically with $\partial Y/\partial f_k = 1/K$ (Corollary 4). Under specialization $k^* = \arg\max_k S_k$ (Proposition 7), $G \perp k^*$ gives $\mathrm{Cov}(A_G,Y \mid k^*) = V_{A,G}$, whereas S_{k^*} has order-statistic variance $c_K V_S$, giving $\mathrm{Cov}(A_{S_{k^*}},Y \mid k^*) = c_K V_{A,S}$ and Eq. (A12). □

Proposition 8. Under the infinitesimal model, $A_{\text{off}} = (A_f + A_m)/2 + \delta$ with $\mathrm{Var}(\delta) = V_{LE}/2$ and $\mathrm{Cov}(A_f,A_m) = h^4\,\mathrm{Cov}(C_f,C_m) = mh^2V_A$, giving Eq. (A13). Stationarity $V_{A,t+1} = V_{A,t} = V$ yields $(1-m)V^2 + (V_E - V_{LE})V - V_{LE}V_E = 0$, which has one positive root for $m < 1$, and response $\frac{b}{2}V_A^*(1+m)$ gives Eq. (A14). □

Propositions 9–10 and Corollaries 5, 6. Centering $C = \mu + x$ and $Y - \mathrm{E}\,Y = \lambda x + \nu$ ($\nu \perp x,A$), $\ln w = kx - \frac{c}{2}x^2 + (\beta_S/\,\mathrm{SD}(Y))\nu + \text{const}$ with $k = \beta_N + \beta_S\sqrt{R/V_P} - c\mu$. Tilting gives $\mathrm{E}_{\text{tilt}}[x] = kV_P/(1+cV_P)$ and $\Delta\bar{A} = h^2\,\mathrm{E}_{\text{tilt}}[x] = V_A k/(1+cV_P)$, vanishing at μ^* in Eq. (A16); $\mathrm{E}[\ln w] = \beta_N\mu - \frac{c}{2}(\mu^2 + V_P)$ peaks at β_N/c (Corollary 5). For $C = aB + K + E$, $\boldsymbol{\beta} = (a\beta_{\text{eff}} - \kappa_B\bar{B}, \beta_{\text{eff}} - \kappa_K\bar{K})^\top$; nonsingular $\mathbf{G}$ implies $\mathbf{G}\boldsymbol{\beta} = \mathbf{0} \iff \boldsymbol{\beta} = \mathbf{0}$, yielding Eq. (A18) and threshold (A19). □

Propositions 11–14 and Corollaries 7, 8. Joint Gaussianity of (A_F,A_G,Y) gives $S_{A_X} = i\,\mathrm{Cov}(A_X,Y)/\,\mathrm{SD}(Y)$ with $\mathrm{Cov}(A_X,Y) = h_X^2\bar{\ell}_X V_X$ (Proposition 11); for $Q = T + A_Q + e$, $\mathrm{Cov}(A_G,Q) = 0$ (Corollary 7). In Proposition 12, $g(n) = \mathrm{d}\sqrt{R}/\mathrm{d}n > 0$ has $\mathrm{d}\ln g/\mathrm{d}n < 0$ with $g(0^+) = \infty, g(\infty) = 0$, so $g(n^*) = \delta/B$ has a unique root decreasing in δ/B. In Proposition 13, $\phi(k) = \mathrm{d}\ln U/\mathrm{d}k = -q/(1-kq) + \theta[1-p(k)]/(1+k)$ is strictly decreasing on $[0,1/q)$, with $\phi(0) > 0 \iff \theta M^\theta/(1+M^\theta) > q$ and $\lim_{M\to\infty}\phi(k) = 0$ at $k^* = (\theta - q)/[q(1+\theta)]$. In Proposition 14, $\mathrm{E}\,z = 0$ and $\bar{c}_+ = \mathrm{E}[c \mid c > 0]$ give $\mathrm{Cov}(z,w) = \mathrm{E}[zc \mid c > 0]/\bar{c}_+ = \mathrm{E}[z \mid c > 0] + \mathrm{Cov}(z,c \mid c > 0)/\bar{c}_+$ and $\mathrm{Var}(w) = (1 + I_{c>0})/p - 1$; truncating below z_q gives $\mathrm{E}[Z \mid Z > z_q] = \varphi(z_q)/(1-q)$ (Corollary 8). □

Propositions 15–21 and Corollaries 9–10. With independent posterior estimates $\hat{b},\hat{p}$, $\mathrm{Cov}(A_B,\hat{p}\hat{b}) = \bar{p}\,\mathrm{Cov}(A_B,\hat{b}) = \bar{p}R_b V_{A,B}$ and symmetrically for S_{A_P}; Lande dynamics $V_{A,B}(sR_b\bar{p} - c_B\bar{b})$ and $V_{A,P}(sR_p\bar{b} - c_P\bar{p})$ destabilize (0,0) iff $s^2R_bR_p > c_Bc_P$ (Proposition 15). With dn staged tests and $m = (1-d)n$ unstaged occasions, $F \mid q \sim \mathrm{Binomial}(m,q)$ has $\mathrm{Cov}(F,q) = mV_P$ and $\mathrm{Var}(F) = m^2V_P + m\nu$, giving Eq. (A26), clean-record probability $(1-q)^m$, and unique interior maximizer d^* of $[(1 + A/d)(1 + C/(1-d))]^{-1}$ in Eq. (A27) (Proposition 16). Replacing $\hat{p}$ by $e + (1-e)\hat{p}$ gives Eq. (A28); stationarity yields $\bar{p}^* = T(1-e)[e + (1-e)\bar{p}^*]$ and Eq. (A29), with $\partial\bar{p}^*/\partial(1-e) = 0$ at $1 - e = (1 - \sqrt{1-T})/T$ (Proposition 17). Conditioning on coercion indicator F gives Eq. (A30). In Proposition 18, $\mathrm{E}[k^2] = (1-q)(m + (1+\alpha)m^2)$ with $m = \bar{k}/(1-q)$ yields $\bar{k} - 1 + V/\bar{k} = \bar{k}(1+\alpha)/(1-q)$, hence $N_e/N = (1-q)/(1+\alpha)$ [100]. Setting Spearman–Brown reliability $nr_1/(1 + (n-1)r_1) = R^*$ with $n = k\lambda(1-d)T$ gives Eq. (A32) (Proposition 19) and Corollary 9; linear covariance of $Y_T = G + x\bar{a}_T + \bar{\varepsilon}_T$ gives Corollary 10. Averaging Eq. (A33) over m observers and k contexts yields $\mathrm{Var}(\bar{X}_{i,m,k}) = \sigma_T^2 + \sigma_H^2 + \sigma_C^2/k + \sigma_O^2/m + \sigma_\varepsilon^2/(mk)$ and $\mathrm{Cov}(T_i,\bar{X}_{i,m,k}) = \sigma_T^2$, establishing Eqs. (A34)–(A35) (Proposition 20). Finally, in Proposition 21, standard normal signal extraction gives $\mathrm{E}[X_j \mid \hat{X}_{j,0}] = \sqrt{\rho_X}\hat{X}_{j,0}$, $\mathrm{E}[K_j \mid \hat{K}_{j,0}] = \sqrt{\rho_{K,0}}\hat{K}_{j,0}$, and $\mathrm{E}[D_{ij} \mid \hat{X}_{j,0},\hat{K}_{j,0}] = 0$, so $\mathrm{E}[U_{ij} \mid \hat{X}_{j,0},\hat{K}_{j,0}] = \alpha_X\sqrt{\rho_X}\hat{X}_{j,0} + \alpha_K\sqrt{\rho_{K,0}}\hat{K}_{j,0}$, yielding $|w_{X,0}| > |w_{K,0}| \iff \rho_{K,0}/\rho_X < (\alpha_X/\alpha_K)^2$ while post-entry observability ($\rho_{K,T} \to 1$) restores structural slopes (α_X,α_K) (part (i)); truncating $\hat{X}_{j,0} \geq \Phi^{-1}(1-q)$ gives $\mathrm{E}[U_{ij} \mid \hat{X}_{j,0} \geq z_{1-q}] = \alpha_X\sqrt{\rho_X}\varphi(z_{1-q})/q$ by orthogonality of (K_j,D_{ij}) to $\hat{X}_{j,0}$ (part (ii)); and differentiating $C_{ij,t} = U_{ij} + I_{ij,t} - \gamma\widehat{O}_{i,t}$ with respect to $\widehat{O}_{i,t}$ gives $-\gamma < 0$, shifting $C_{ij,t}$ downward stochastically and strictly increasing $\Pr(C_{ij,t} < \theta)$ whenever $C_{ij,t}$ has positive density at θ (part (iii)). □

Proposition 22. Under $\widehat{O} \sim \mathcal{N}(\mu_O,\sigma_O^2)$, bond survival through T_J has probability $\Pr(\widehat{O} \leq u_0 + L) = \Phi\big((u_0 + L - \mu_O)/\sigma_O\big) = p_{\text{bond}}(L)$. Equating $\kappa_1(B_{\text{cap}} - B_0)^2 = M_{\text{tot}}(L)$ on $B_{\text{cap}} > B_0$ gives $B_{\text{cap}}(L,\kappa_1) = B_0 + \tau\sqrt{M_{\text{tot}}(L)}$ with $\tau = \kappa_1^{-1/2}$. For $B \sim \mathcal{N}(\bar{B},V_P)$ under $\ln w(B) = \beta_S B - \frac{\lambda_V}{2}(B - B_{\text{cap}})^2$, Lande's selection gradient $\partial\,\mathrm{E}[\ln w]/\partial\bar{B} = \beta_S - \lambda_V(\bar{B} - B_{\text{cap}})$ has unique globally stable zero $B^*(L,\kappa_1) = B_{\text{cap}}(L,\kappa_1) + \beta_S/\lambda_V$, establishing Eq. (A37) (reducing to the solitary-mother ceiling $B_0 + \sqrt{M_{\text{mom}}/\kappa_1} + \beta_S/\lambda_V$ when $\Delta_{\text{allo}} = \Delta_{\text{pair}} = 0$). Differentiating $B^* = B_0 + \tau\sqrt{M_{\text{tot}}(L)} + \beta_S/\lambda_V$ yields $\partial B^*/\partial L = \tau\Delta_{\text{pair}}\varphi(z)/(2\sigma_O\sqrt{M_{\text{tot}}(L)}) > 0$ for $z = (u_0 + L - \mu_O)/\sigma_O$ and $\Delta_{\text{pair}} > 0$, and differentiating with respect to τ yields Eq. (A38). □

Proposition 23. (i) Strict concavity in e gives first-order condition $\gamma_{\text{mkt}}(1 - \lambda_{\text{stage}}) - c_e e^* = 0$, hence $e^*_{\text{mar}}/e^*_{\text{date}} = 1 - \lambda_{\text{mar}}$. (ii) Conditional on $\hat{p}_m = z \geq \theta_{\text{in}}$, $p \mid (\hat{p}_m = z) \sim \mathcal{N}(z,\tau_m^2)$, so $\Pr(p < \theta_{\text{stay}} \mid \hat{p}_m = z) = \Phi\big((\theta_{\text{stay}} - z)/\tau_m\big)$, which decreases strictly in both z and τ_m^{-1} since $z \geq \theta_{\text{in}} > \theta_{\text{stay}}$; moreover, $\hat{p}_m = \mu_p + s_m Z$ ($Z \sim \mathcal{N}(0,1)$, $s_m = \sqrt{\sigma_p^2 - \tau_m^2}$ strictly increasing in m), and for $\theta_{\text{in}} \geq \mu_p$ the truncated standardized variable $Z \mid (Z \geq (\theta_{\text{in}} - \mu_p)/s_m)$ induces $\hat{p}_m(m) = \mu_p + s_m Z$ with $\partial[\theta_{\text{stay}} - (\mu_p + s_m z)]/\partial m < 0$ at every quantile above truncation, strictly reducing $H(m)$. (iii) At $q = 0$, $U_{f,0},U_{m,0}$ are i.i.d., so $\Pr(U_f < 0 < U_m) = \Pr(U_m < 0 < U_f)$. For $q > 0$, let $p_{s,1} = \Phi(-(\mu_u - \delta_s)/\sigma_u)$; since $\delta_f > \delta_m > 0$, $p_{f,1} > p_{m,1}$, so $p_{f,1}(1 - p_{m,1}) - p_{m,1}(1 - p_{f,1}) = p_{f,1} - p_{m,1} > 0$, strictly lifting women's unilateral share $[(1-q)p_0(1-p_0) + qp_{f,1}(1 - p_{m,1})]/[(1-q)2p_0(1-p_0) + q(p_{f,1} + p_{m,1} - 2p_{f,1}p_{m,1})] > \frac{1}{2}$. (iv) Monotonicity of $\Phi(\cdot)$ yields $\Phi(z_{f,\text{sin}} - \kappa_{\text{lock}}) < \Phi(z_{f,\text{sin}})$ and $\Phi(z_{m,\text{wid}}) > \Phi(z_{f,\text{wid}})$. □

TABLE B1
SUMMARY OF NUMERICAL VERIFICATION SUITE (E1–E23, A1–A17).

Suite	Models Verified	Agreement / Audit Bound
E1–E3, E9	Props. 1–4 (10^5–2×10^5 agents)	Max $\lvert\text{MC} - \text{exact}\rvert = 2.7 \times 10^{-4}$; max $\lvert z\rvert = 1.2$; bound never violated
E4–E6	Props. 8–10, deferred acceptance	Infinitesimal V_A within 2%; Lande tissue eq. 2.007 ± 0.016 vs. 2.001
E10–E12	Props. 11–13 (contest aim, risk, skew)	Aim $\lvert z\rvert = 1.1$; optimal bout count $\lvert z\rvert = 2.4$; log-concavity exact
E15–E16	Props. 5–7, 16–17	Aim ratio within 0.3%; staging ≤ 0.0005; IBM enforcement ≤ 0.006
E21–E23, 2-way	Props. 18–23, Cor. 9–10	Drift N_e/N within 4%; 2-way $\lvert\Delta R^2\rvert \leq 0.003$; E22 $\lvert\Delta B^*\rvert \leq 0.75\,\text{cm}^3$; E23 $\lvert\Delta e^*\rvert \leq 0.04$ min, $\lvert\Delta\,\text{Pr}\rvert \leq 0.006$
A1–A17	Independent re-derivations & raw-file recounts	Quadrature 4×10^{-17}; Lande 1×10^{-12}; raw survey/culture recounts 0 diff

APPENDIX B
SIMULATION VERIFICATION AND PALEOCLIMATIC AUDIT OF THE 1.5–0.7 MA PLATEAU

Every analytical proposition in Appendix A was verified in Python with fixed `SeedSequence` seeds against independent Monte Carlo and individual-based evolutionary simulations (E1–E23, plus zero-shared-code audit scripts A1–A17; Table B1). Across all closed-form checks, simulated selection differentials, equilibrium trait means, and variance recursions agree with analytical formulas within Monte Carlo sampling precision (maximum standardized deviation $\lvert z\rvert = 1.2$; exact algebraic and quadrature checks $\leq 1 \times 10^{-12}$).

Table B2 presents the complete chronological and epistemic audit of the 1.5–0.7 Ma hominin endocranial volume plateau (Phase III, mean 908 cm^3), synchronizing the three open fossil compilations (233 crania in Gingerich [3], 166 in Will et al. [2], 94 in Du et al. [1]) against the public-domain 57-core `LR04` benthic δ^{18}O stack [4] (`SHA-256: ec3c4af8...`). Two constraints govern causal inference. First, stasis began near 1.50 Ma, preceding the 1.12 Ma European cooling crisis by $\sim$380 kyr, the proposed 930–813 ka coalescent bottleneck by $\sim$570 kyr, and the 789 ka Australasian impact by $\sim$711 kyr. Second, evidence tiers separate measured paleoclimatic and archaeological records from contested coalescent parameterizations and dated physical events lacking demonstrated hominin mortality.

APPENDIX C
DIGITAL-PLATFORM, STATURE, FACIAL-ATTRIBUTION, AND HOLOFACE-L1 AUDITS

A. *HoloFace-L1 Multimodal Architecture and Psychometric Reliability Ceiling*

To bound what static portraits and profile metadata reveal about human consensus ratings (Hot-or-Not: 10,000 profiles, 20,000 images, stratified 8,000/1,000/1,000 train/val/test split, zero identity leakage), we synthesized 43 experimental waves (H1–H43, 61 representations) into `HoloFace-L1` (Table C1):

1) **`HoloFace-Pure` (zero metadata):** Pools four visual families strictly from aligned pixels via OOF Cholesky/Woodbury ridge and validation simplex weights: (F1) 3D Kendall Riemannian shape (`1k3d68` + 106-landmark Delaunay log-Jacobians, zero pixels); (F2) skin hemoglobin/melanin chromophores and $1/f^\beta$ microrelief; (F3) pose, smile, eye aperture, and framing; and (F4) 12-space foundation/cortical backbones (`DINOv2`, `CLIP`, `SigLIP`, `ConvNeXt`, `EVA-02`, `Swin`, `ArcFace`, `FARL`), with exact $2^4 = 16$ Shapley ρ^2 attribution.
2) **`HoloFace-Full` ($L_1 + L_2$ stacker):** Combines 29 OOF visual, tabular (`LightGBM`/`CatBoost`), and cross-modal FiLM/OT/RKHS predictors with empirical-Bayes shrinkage, 5-fold exact-L_1 LP (`HiGHS`, Test MAE 0.3361), and ridge L_2 calibration eliminating male right-skew bias (-0.016).
3) **Psychometric ceiling:** Regressing training OOF squared residuals $(y_i - \hat{f}_{\text{oof}}(x_i))^2$ on $1/n_i$ ($n_i \geq 20$) isolates per-voter rating variance ($\hat{\sigma}^2_{\text{voter}} = 8.23$), yielding finite-voter Oracle Spearman and MAE bounds across vote tiers (Table C1).

B. *Platform Eligibility, Vote-Depth, and Response-Coding Sensitivity*

Tables C2–C4 audit the 6.68M Hot-or-Not and 1.14M Plenty of Fish registries (complemented by 59,946 `OkCupid-JSE` profiles [59]) across vote thresholds ($n_{\text{votes}} \geq 1$ to 100), image inventory, non-response coding, orientation, and marital status; both the female-steeper HON age slope and the POV childless crossover hold throughout.

APPENDIX D
SURVEY COHORT ROBUSTNESS AND PRE-COMMITTED FALSIFIERS

Tables D1 and D2 verify that the sex-asymmetric access differential on men's personal income ($+0.047$ SD) versus negative family-size differentials in both sexes (General Social Survey, 1972–2024, $n = 34{,}092$) holds across unweighted estimation, narrower completed-fertility ages (50–74), survey-period splits, and birth cohorts (1920–1979) even as nonmarital childbearing rose. Finally, Table D3 pre-commits fifteen discriminating empirical tests and explicit observations that would falsify or bound each link of the multi-context jury framework.

TABLE B2
CHRONOLOGICAL AND EPISTEMIC AUDIT OF GLOBAL EVENTS OVERLAPPING THE 1.5–0.7 MA HOMININ ENDOCRANIAL STASIS PLATEAU.

Window	Lag After Onset	Event or Hypothesis	Evidence Tier	Supported Empirical Record	Scope Limit / Why It Cannot Alone Explain Full Plateau
1.25–0.70 Ma	+250 kyr	MPT ice-cycle shift (41→100 kyr) [4, 29, 30, 32]	Measured global climate proxy	Glacial P90-P10 swing widens from 0.80 to 0.93‰ across MPT (1.21 post-MPT); CO_2 falls and deep-Atlantic circulation reorganizes.	Fragments habitats & amplifies ecological variability in late plateau; begins ∼250 kyr AFTER ∼1.5 Ma stasis onset.
∼1.12 Ma	+380 kyr	MIS 34 extreme cooling (Eur. depop.) [18, 33]	Measured regional climate + habitat model	Severe North Atlantic/Mediterranean cooling renders western Europe climatically unsuitable after ∼1.4 Ma occupation.	Supports regional European depopulation/turnover, NOT global extinction; occurs ∼380 kyr after plateau onset.
∼1.07 Ma	+430 kyr	Bosumtwi crater (10.5 km, Ghana) [43]	Dated impact — no hominin mortality link	Confirmed 10.5 km West African impact structure dated near 1.07 Ma.	Regional blast hazard only; zero paleontological or demographic evidence of continental hominin die-off.
∼1.00 Ma	+500 kyr	Wonderwerk Cave fire (∼1.0 Ma) [5]	Dated archaeological energetic marker	In-situ combustion features ∼1.0 Ma inside deep cave context.	Early thermal processing candidate; precedes Phase IV expansion as fire use generalizes.
930–813 ka	+570 kyr	Proposed N_e ≈1,280 bottleneck (contested) [19, 20, 36–39]	Contested coalescent genomic inference	Hu et al. (2023) & Zhou et al. (2026) infer N_e ≈1,280; Deng et al. (2025), Cousins & Durvasula (2025), and Terhorst (2025) find SFS artifact / structured-stem alternatives fit without collapse.	Coalescent N_e is NOT a census headcount; occurs ∼570 kyr after plateau onset; potentially reflects metapopulation fragmentation across MIS 24–22.
815±11 ka	+685 kyr	Pantasma crater (14 km, Nicaragua) [44]	Dated impact — no hominin mortality link	Confirmed 14 km Central American impact crater in uninhabited Americas.	No hominins in Americas; impact scale below global winter threshold.
∼800 ka	+700 kyr	Oldest Toba Tuff eruption (∼800 ka) [45]	Dated super-eruption — no hominin die-off shown	Major Quaternary silicic eruption in Sumatra near 0.80 Ma.	Occurs near end of plateau (∼700 kyr after onset); Southeast Asian hominin occupation persists across interval.
789±3 ka	+711 kyr	Australasian Bolaven tektite impact (∼789 ka) [21, 46–48, 50]	Dated impact glass — crater debated, no kill link	Largest Quaternary tektite field; Bolaven source proposed (Sieh 2020, 2024) and disputed (Mizera 2022, 2024); Bose/Vietnam bifaces co-occur.	Occurs ∼710 kyr after plateau onset right before Phase IV rise; bifacial toolmakers persist in Bose & Vietnam.
∼780 ka	+720 kyr	Gesher Benot Ya'aqov cooking (∼780 ka) [6, 51]	Dated archaeological cooking evidence	Earliest well-dated controlled thermal cooking of aquatic/animal tissue.	Relaxes gut/metabolic budget ceiling right at the ∼0.7 Ma Phase IV parallel brain-expansion onset.
∼773 ka	+727 kyr	Matuyama–Brunhes magnetic reversal (∼773 ka) [49]	Dated geomagnetic horizon — no biological kill	Last full geomagnetic polarity reversal; key chronostratigraphic anchor for Middle Pleistocene.	Chronological marker only; no demonstrated hominin or mammalian extinction.

Lag after onset is measured from the 1.50 Ma plateau start (fitted stasis–step breakpoint 1.45 Ma). Benthic $\delta^{18}O$ statistics are computed directly from the public-domain Lisiecki & Raymo `LR04` stack [4] (`evidence/paleoclimate/lr04_benthic_stack.txt`).

TABLE C1
HOLOFACE-L1 43-WAVE MULTIMODAL FACIAL BENCHMARK, EXACT 4-FAMILY SHAPLEY DECOMPOSITION, AND FINITE-VOTER PSYCHOMETRIC CEILING (10,000 PROFILES, 20,000 PORTRAITS).

Model / Tier	Input Regime	Val MAE ↓	Test MAE ↓	RMSE ↓	Test ρ ↑	Rank ρ^2	Key Structural or Psychometric Finding
Panel A: 43-wave benchmark progression and `HoloFace-L1` *synthesis ($N = 1{,}000$ held-out test profiles, $n_{votes} \geq 20$)*							
H1 Baseline	Raw UV pixels	0.4915	0.4831	0.6082	0.234	5.5%	Unmasked 40% blue background matte confounds ResNet-18
H4 Domain Face	Aligned face pixels	0.4125	0.3911	0.4962	0.554	30.7%	SFace 112px + RegNetY/ResNet50-V2 human-face backbones
H8 Pre-UV + Pose	Visual + 72-pt mesh	0.3874	0.3664	0.4875	0.580	33.7%	Pre-UV landmark geometry without demographic metadata
HoloFace-Pure	Zero metadata (F1–F4)	0.3980	0.3776	0.4867	0.583	34.0%	Strict zero-metadata visual convex synthesis ($\rho = 0.704$ at 500+ votes)
H10 FiLM + GBDT	Multimodal	0.3689	0.3447	0.4663	0.633	40.1%	Cross-modal FiLM residual + 5-fold grouped GBDT/CatBoost
H20 Exact-L_1 LP	Multimodal	0.3576	0.3362	0.4541	0.662	43.8%	DINOv3 + ConvNeXtV2 + GlintR100 + CORAL/FiLM exact-L_1 LP
H29 Kendall 3D	Multimodal + 3D mesh	0.3609	0.3378	0.4522	0.660	43.5%	Riemannian shape manifold ($\Sigma_2^{72} \times \Sigma_2^{106} \times \Sigma_3^{68}$); best single-wave RMSE
H30 Cortical Router	Multimodal + 12-space	0.3576	0.3362	0.4542	0.662	43.9%	Retinotopic cortical fovea + dynamic 12-alignment router; best single-wave ρ
H40 Set-Transformer	Multimodal 20k views	0.3650	0.3429	0.4617	0.644	41.5%	Permutation-invariant multi-photo set attention (single-image MAE 0.3422)
HoloFace-Full (L_1)	29-OOF exact-L_1 LP	0.3569	**0.3361**	0.4558	0.660	43.6%	Best Val/Test/Single-Image MAE (0.3420); gender Pct MAE 16.6
HoloFace-Full (L_2)	Skew-unbiased ridge	–	0.3531	**0.4524**	0.658	43.3%	Eliminates male right-skew rating bias ($+0.088 \rightarrow -0.016$)
Panel B: Standalone performance and exact $2^4 = 16$ Shapley share of zero-metadata `HoloFace-Pure` *rank variance*							
F4 Foundation 12-space	6,520-dim visual	0.3974	0.3767	0.4868	0.583	34.0%	Shapley share **62.4%** ($+\Delta\rho^2 = 21.2\%$, across 8 foundation/face backbones)
F1 3D Kendall shape	582-dim mesh/Jacobian	0.4452	0.4256	0.5374	0.413	17.1%	Shapley share **18.3%** ($+\Delta\rho^2 = 6.2\%$, pure 3D/2D landmarks, zero pixel intensities)
F2 Skin chromophores	288-dim albedo/$1/f^\beta$	0.4455	0.4336	0.5435	0.386	14.9%	Shapley share **14.8%** ($+\Delta\rho^2 = 5.0\%$, hemoglobin/melanin & spectral micro-relief)
F3 Staging & pose	106-dim pose/framing	0.4650	0.4586	0.5714	0.241	5.8%	Shapley share **4.5%** ($+\Delta\rho^2 = 1.5\%$, changeable yaw/pitch, smile & framing only)
Panel C: Finite-voter psychometric Oracle ceiling ($\hat{\sigma}^2_{voter} = 8.23$) across held-out vote-depth tiers							
Oracle: 20–49 votes ($n = 87$)	Median 33 raters	–	0.3969	–	0.643	41.4%	`HoloFace-Full` $\rho = 0.603$ ($\rho^2 = 36.4\%$); irreducible single-context gap 12.1%
Oracle: 50–199 votes ($n = 304$)	Median 106 raters	–	0.2277	–	0.832	69.2%	`HoloFace-Full` $\rho = 0.537$ ($\rho^2 = 28.9\%$); irreducible single-context gap 58.2%
Oracle: 200+ votes ($n = 609$)	Median 1,941 raters	–	0.0681	–	0.984	96.8%	`HoloFace-Full` $\rho = 0.763$ ($\rho^2 = 58.2\%$); irreducible single-context gap 39.9%
Oracle: 500+ votes ($n = 451$)	Median 3,526 raters	–	0.0463	–	0.993	98.7%	`HoloFace-Full` $\rho = 0.770$ ($\rho^2 = 59.3\%$); irreducible single-context gap 39.9%
Oracle: All test ($n = 1{,}000$)	Median 367 raters	–	0.1457	–	0.918	84.3%	`HoloFace-Full` $\rho = 0.660$ ($\rho^2 = 43.6\%$); irreducible single-context gap 48.3%

All models are trained strictly on the 8,000-profile training split (H1–H43 archive) and evaluated once on the held-out 1,000-profile test set ($n_{\text{votes}} \geq 20$). Finite-voter Oracle ceilings use out-of-fold heteroscedastic variance components ($\hat{\sigma}^2_{\text{voter}} = 8.23$) estimated strictly on training profiles without test-set leakage.

TABLE C2
HON SENSITIVITY ACROSS MINIMUM VOTE THRESHOLDS AND IMAGE INVENTORY.

		All eligible			Image inventoried		
Minimum votes	Gender	Profiles	Mean	Pooled	Profiles	Mean	Pooled
1	Women	3,589,427	59.3%	60.3%	366,379	54.9%	57.3%
1	Men	3,129,114	60.0%	63.1%	531,211	59.8%	62.7%
20	Women	3,589,397	59.3%	60.3%	366,375	54.9%	57.3%
20	Men	3,088,375	60.0%	63.1%	529,317	59.8%	62.7%
100	Women	3,471,920	59.4%	60.3%	364,063	54.9%	57.3%
100	Men	1,597,216	61.6%	63.5%	340,942	61.5%	63.1%
20, recorded U.S.	Women	2,353,568	58.0%	59.8%	363,798	54.8%	57.3%
20, recorded U.S.	Men	2,874,289	60.0%	63.2%	524,984	59.8%	62.8%

TABLE C3
POV FERTILITY-INTENTION RESPONSE AUDIT BY AGE BAND, SEX, AND PARENTHOOD STATUS (1,137,309 ELIGIBLE PROFILES).

Age	Recorded gender	Profiles	Want	Do not want	Undecided/Open	Nonresponse
25–29	Women	45,792	46.0%	5.1%	26.5%	22.5%
25–29	Men	173,520	33.5%	4.1%	32.7%	29.7%
45–49	Women	4,608	6.8%	33.3%	29.3%	30.6%
45–49	Men	11,668	12.8%	12.8%	43.8%	30.5%

TABLE C4
POV CROSSOVER ROBUSTNESS ACROSS SAMPLE RESTRICTIONS AND OUTCOME DEFINITIONS (AGES 25–29 VS. 45–49).

Cohort or weighting	Ages 25–29			Ages 45–49		
	Women	Men	W − M	Women	Men	W − M
Main cohort	46.0%	33.5%	+12.5	6.8%	12.8%	−6.0
Standardized by exact age	46.0%	33.5%	+12.5	6.8%	12.8%	−6.0
Marked single	46.0%	33.5%	+12.5	7.8%	13.6%	−5.8
All seeking records	43.7%	33.2%	+10.5	6.2%	12.3%	−6.0
Answered only	59.3%	47.6%	+11.7	9.8%	18.4%	−8.7
Single, answered, age-standardized	59.4%	47.6%	+11.7	11.3%	19.6%	−8.3

TABLE D1
ROBUSTNESS OF THE ACCESS–NUMBER SELECTION DECOMPOSITION ACROSS SURVEY WEIGHTS, AGE WINDOWS, AND PERIODS (GSS 1972–2024).

Variant	Trait, sex	Access	Number	Total
Main analysis	Schooling, men	−0.003 [−0.011, +0.005]	−0.081 [−0.095, −0.068]	−0.085 [−0.100, −0.069]
	Schooling, women	−0.038 [−0.044, −0.033]	−0.126 [−0.137, −0.114]	−0.164 [−0.176, −0.151]
	Income, men	+0.047 [+0.035, +0.059]	−0.034 [−0.050, −0.016]	+0.013 [−0.006, +0.033]
	Income, women	−0.038 [−0.047, −0.028]	−0.078 [−0.093, −0.063]	−0.116 [−0.135, −0.098]
Unweighted	Schooling, men	−0.011 [−0.018, −0.003]	−0.082 [−0.093, −0.070]	−0.093 [−0.106, −0.078]
	Schooling, women	−0.053 [−0.059, −0.047]	−0.121 [−0.130, −0.112]	−0.174 [−0.185, −0.164]
	Income, men	+0.049 [+0.038, +0.059]	−0.026 [−0.039, −0.012]	+0.023 [+0.002, +0.039]
	Income, women	−0.053 [−0.062, −0.043]	−0.078 [−0.093, −0.065]	−0.131 [−0.149, −0.115]
Ages 50–74 only	Schooling, men	−0.002 [−0.012, +0.007]	−0.087 [−0.102, −0.071]	−0.089 [−0.108, −0.071]
	Schooling, women	−0.037 [−0.044, −0.031]	−0.121 [−0.132, −0.111]	−0.159 [−0.172, −0.145]
	Income, men	+0.044 [+0.032, +0.058]	−0.036 [−0.055, −0.018]	+0.008 [−0.015, +0.030]
	Income, women	−0.034 [−0.044, −0.024]	−0.079 [−0.096, −0.060]	−0.113 [−0.133, −0.093]
Surveys 1990–2024	Schooling, men	−0.008 [−0.018, +0.003]	−0.071 [−0.086, −0.056]	−0.079 [−0.095, −0.062]
	Schooling, women	−0.045 [−0.052, −0.038]	−0.127 [−0.140, −0.113]	−0.172 [−0.187, −0.157]
	Income, men	+0.050 [+0.037, +0.065]	−0.034 [−0.055, −0.015]	+0.017 [−0.009, +0.042]
	Income, women	−0.036 [−0.048, −0.024]	−0.082 [−0.100, −0.065]	−0.118 [−0.139, −0.098]
Surveys 1972–1989	Schooling, men	+0.007 [−0.006, +0.018]	−0.105 [−0.126, −0.083]	−0.098 [−0.121, −0.072]
	Schooling, women	−0.022 [−0.033, −0.012]	−0.121 [−0.140, −0.102]	−0.144 [−0.165, −0.120]
	Income, men	+0.040 [+0.021, +0.057]	−0.034 [−0.062, −0.007]	+0.006 [−0.029, +0.037]
	Income, women	−0.040 [−0.057, −0.024]	−0.067 [−0.097, −0.037]	−0.107 [−0.141, −0.075]

TABLE D2
THE PAIRING GATE ACROSS BIRTH COHORTS: NEVER-MARRIED SHARE, NON-MARITAL PARENTHOOD, AND MALE INCOME PAIRING DIFFERENTIAL.

Sex	Born	Never married (%)	Parents among the never married (%)	Children of the never married (% of all children)	Pairing differential, income
Men	1920–1939	4.5 [4.0, 5.1]	9.0 [5.3, 13.4]	0.4 [0.2, 0.6]	+0.027 [+0.016, +0.039]
	1940–1959	7.9 [7.2, 8.7]	19.1 [15.5, 22.8]	1.7 [1.3, 2.2]	+0.042 [+0.030, +0.053]
	1960–1979	13.3 [11.8, 14.8]	31.4 [25.6, 37.2]	5.1 [3.6, 6.7]	+0.078 [+0.056, +0.102]
Women	1920–1939	2.9 [2.4, 3.3]	17.6 [11.8, 24.2]	0.6 [0.4, 0.9]	−0.009 [−0.018, 0.000]
	1940–1959	6.8 [6.1, 7.4]	39.8 [35.1, 44.6]	2.8 [2.2, 3.4]	−0.003 [−0.013, +0.009]
	1960–1979	11.4 [10.1, 12.7]	61.5 [55.7, 66.5]	8.3 [6.8, 9.8]	+0.020 [+0.003, +0.038]

TABLE D3
FIFTEEN PRE-COMMITTED DISCRIMINATING PREDICTIONS AND EXPLICIT FALSIFYING OUTCOMES.

ID	Mechanism Link	Identifying Empirical Design	Observation That Refutes or Bounds Claim
F1	Energetic–information co-limitation (Prop. 10)	High-resolution dated fossil crania vs. regional archaeological fire/cooking and diet proxies	Sustained brain-volume steps precede energetic/foraging shifts or track population size alone
F2	Causal tissue coupling ($a > 0$)	Sibling / Mendelian-randomization designs linking brain volume to g	Within-family causal effect $a \approx 0$ (stratification only)
F3	Multi-domain breadth (Prop. 3)	Fixed budget randomized across K distinct tasks vs. K repeats	Single-domain repetition predicts held-out g as well as breadth
F4	Breadth taxes gaming (Cor. 4)	Coached single-domain inflation under K-domain vs. specialist choice	Single-domain coaching shifts K-domain ranks without $1/K$ dilution
F5	Division of labor (Prop. 7)	Embodied generalist foraging bands vs. niche-specialized economies	Specialist evaluation weights g as strongly as common domains
F6	Two-way context ceiling (Prop. 20)	Scaling $m \to 10^3$ portrait raters vs. pooling $k \geq 4$ live contexts	Infinite single-context raters reach multi-context $R^2_{\infty,1} \approx R^2_{m,k}$
F7	Legibility governs stranger choice (Eq. (A31))	Randomizing encounter length (4 vs. 60 min) and multi-witness panel reports before partner choice	Weights on intelligence and sincerity fail to rise as inter-rater agreement increases
F8	Staged capacity vs. unstaged delivery (Prop. 16)	Comparing announced test performance vs. unannounced everyday conduct as predictors of later partner defaults	Announced staged assessments predict long-run reliability/default rates as well as unstaged conduct
F9	Partial enforcement crowds in/out commitment (Prop. 17)	Behavioral games and cross-cultural sharing norms varying third-party enforced transfer share $e \in [0,1]$	Voluntary generosity rises monotonically at $e \to 1$ or partner-choice weights ignore e
F10	Male access floor vs. podium (Prop. 14)	Decomposing male reproductive variance into childlessness ($c = 0$) vs. father family size ($c > 0$)	Selection differential on male standing sits primarily in high-parity fathers rather than pairing access
F11	Cue vs. trait response (Cor. 7)	Separating preference slopes on transient life-stage cues (youth) or thresholded stature ($6'0''$) from completed fertility	Steep online/stated preference slopes translate into proportional completed-fertility differentials
F12	Witness density and union timing (Prop. 19)	Survival analysis of time-to-cohabitation/marriage by mutual-acquaintance and kin-witness density at first meeting	Thin witness networks show equal or faster progression to durable fertility-producing unions, net of selection
F13	Entry filtering vs. pair-bond welfare (Prop. 21)	Longitudinal tracking of unions formed via vertical filters vs. social networks, plus social-media abstinence trials	Entry consensus attributes (money, stature, photo rank) predict multi-year satisfaction as strongly as kindness
F14	Childhood metabolic & pair-bond ceiling (Prop. 22)	Phylogenetic regression of adult endocranial volume on juvenile brain RMR, biparental care, and diet across mammals	Lineages cross the ~ 650–$700\,\mathrm{cm}^3$ juvenile ceiling under solo maternal care without pair-bond/dietary release
F15	Lifecycle staging, rush hazard & exit (Prop. 23)	Prospective dyadic time-use diaries from courtship into marriage, pre-cohabitation witness audits, and registry remarriage panels	Within-couple staging effort stays flat after marriage, ≤ 3 mo rushed cohabitation shows equal survival, or marital exit stays sex-symmetric